Current Opinion in Solid State and Materials Science 27 (2023) 101106

Contents lists available at ScienceDirect

Current Opinion in Solid State & Materials Science

journal homepage: www.elsevier.com/locate/cossms

Recent research progress in hydrogen embrittlement of additively manufactured metals – A review

Ju Yao, Qiyang Tan *, Jeffrey Venezuela *, Andrej Atrens, Ming-Xing Zhang *

School of Mechanical and Mining Engineering, The University of Queensland, St. Lucia, QLD 4072, Australia

ARTICLE INFO

Keywords:
Additive manufacturing
Hydrogen embrittlement
Metals
Microstructure
Mechanical properties

ABSTRACT

Hydrogen is considered as a primary energy carrier for the hydrogen economy. However, hydrogen embrittlement (HE) is an inescapable problem that needs to be solved because metals, particularly steels, are commonly used in the transportation and storage of hydrogen, and because HE occurs in high-performance structural components in contact with moisture or hydrogen. In particular, HE concerns of additively produced alloys should be addressed, because additive manufacturing (AM) can provide significant advantages in the manufacturing of such structural components. This review overviews the recent research progress in HE of metals fabricated using AM. This review introduces AM and HE and summarises and discusses (i) the factors that influence the HE of AM metals, (ii) possible mechanisms of HE, (iii) the differences and similarities of HE behaviour between metals processed by AM and those produced through conventional manufacturing processes, and (iv) the current challenges and research gaps of HE in AM metals. The review covers structural steels, titanium alloys, tool steels, nickel-based superalloys, stainless steels and high-entropy alloys.

1. Introduction

Hydrogen embrittlement (HE) [1,2] is associated with high performance metallic materials widely used in industry, including steels, titanium alloys and superalloys. HE manifests as the deterioration in structural properties [3], like tensile and fatigue properties [4], and can lead to unexpected catastrophic failure [5]. The ubiquitous nature of hydrogen means that the ingress of hydrogen atoms is typically difficult to avoid. HE has been extensively studied for over 100 years since it was first documented by Johnson in 1875 [5,6]. It is now widely agreed that HE of high-performance structural components is an unavoidable issue, particularly for components exposed to environments that can liberate hydrogen, such as sea water, hydrogen gas, or moist air. Thus, it is a long-term challenge to select the right material or to develop new materials for hydrogen production, storage, and transportation.

Additive manufacturing (AM), also known as 3D printing, is a revolutionary manufacturing technology due to its advantages over traditional subtractive processes, including faster part delivery from concept to market, the ability to create objects with high geometric complexity, and reduced waste generation during fabrication [7,8]. In the last decade, there has been considerable research in the AM of metallic materials with two frontier AM technologies: direct energy deposition (DED) and powder bed fusion (PBF). The latter can be further divided into the major techniques: selective laser melting (SLM) or sintering (SLS) and electron beam melting (EBM) [8]. These AM processes use either powder or wire as the feedstock and use layer-by-layer fabrication to shape and consolidate the feedstock into arbitrary configurations with complex geometries possible [9]. At present, AM can fabricate highly dense metal parts with acceptable mechanical properties for industrial applications from steels, aluminium (Al) alloys, titanium (Ti) alloys, nickel-based (Ni-based) superalloys, and cobalt-chrome (CoCr) alloys[10,11]. In addition, modelling and simulation

Abbreviations: AM, Additive manufacturing; ASS, Austenitic stainless steel; BCC, Body centred cubic; BCT, Body centred tetragonal; DMD, Direct metal deposition; DED, Directed energy deposition; EBM, Electron beam melting; FCC, Face centred cubic; FEA, Finite element analysis; GB, Grain boundary; HAC, Hydrogen-assisted cracking; HCP, Hexagonal close-packed; HEA, High-entropy alloys; HE, Hydrogen embrittlement; HEDE, Hydrogen enhanced decohesion; HELP, Hydrogen enhanced localised plasticity; HEE, Hydrogen environment embrittlement; HRE, Hydrogen reaction embrittlement; IHE, Internal hydrogen embrittlement; LAM, Laser additive manufacturing; LENS, Laser engineered net shaping; LMD, Laser metal deposition; PBF, Powder bed fusion; SEM, Scanning electron microscope; SLM, Selective laser melting; SS, Stainless steel; SSRT, Slow strain rate tensile; TDS, Thermal desorption spectroscopy; TEM, Transmission electron microscopy; UTS, Ultimate tensile strength; XRD, X-ray diffraction; YS, Yield strength.

* Corresponding authors.
E-mail addresses: q.tan@uq.edu.au (Q. Tan), j.venezuela@uq.edu.au (J. Venezuela), mingxing.zhang@uq.edu.au (M.-X. Zhang).

https://doi.org/10.1016/j.cossms.2023.101106
Received 25 July 2023; Received in revised form 13 September 2023; Accepted 13 September 2023
1359-0286/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

can indicate how process variables affect the quality of AM components. Modelling and simulation can assist with a fundamental understanding of the underlying physical processes and rapid design can shorten the qualification cycle of the AM part [12]. In summary, AM is already widely used to create high-quality products and there is much AM has focused on generating products with superior mechanical properties.

Hydrogen embrittlement (HE) of AM products has received relatively less attention. The earliest report by Baek et al. in 2017 [13] compared the HE behaviour of 304L stainless steel (SS) fabricated by direct metal tooling (DMT), one of the AM methods, with that of the cold rolled. The DMT 304L SS had higher HE resistance than the rolled specimen in gaseous hydrogen, and the strain-induced martensitic transformation did not occur in the DMT specimen. This study concluded that the DMT AM technology was capable of creating products for use in hydrogen [13]. Subsequently, HE research of AM metallic materials has received increasing attention, including research on 304L SS [13–16], 316L SS [17–20], Inconel 718 [21–28] and Ti-6Al-4 V [29–37]. These studies confirmed that the HE behaviour of metals fabricated with AM differs from those made through conventional processes.

Reviews on hydrogen embrittlement (HE) and on metal additive manufacturing (AM) have been published in the past few years [8,9,11,38–44]. The reviews on HE emphasised mechanisms and microstructure [38–40,45–47]. Those on metal AM emphasised processes, microstructure and properties [8,9,11,48,49]. These reviews have provided the background of HE and AM. However, the HE knowledge generated from conventional alloys cannot be fully applied to AM metals because HE is sensitive to microstructure, and generally microstructure of AM alloys differs from their counterparts produced through conventional processes. Thus, understanding the HE behaviour of the AM metals is essential for extending the AM technology to hydrogen related industries. However, there is a lack of an integrated review on the intersection area of the HE susceptibility of AM metals.

Thus, this review will provide valuable insights into the current state of knowledge regarding HE of metals fabricated using AM, effectively building critical reviews and discussions on previous research results. The present work aims to overview the current understanding of HE of AM metals, providing the most up-to-date knowledge in this area. The review starts with the characteristics and applications of AM metals in Section 2, followed by briefing the two typical mechanisms of HE and the influence of material microstructures on HE in Section 3. Section 4, the major part of this review, focuses on the mechanisms and primary influencing factors of HE on the properties of various AM metals, including Ti alloys, Ni-based superalloys, stainless steels, high-entropy alloys, and ultrahigh strength steels. This review summarises the reported HE mechanisms in the literature. The evaluation of those HE mechanisms is beyond the scope of this paper. But, comparisons of the HE susceptibility of AM metals with conventionally produced metals are provided, together with suggestions for future studies.

2. Metal additive manufacturing

2.1. Introduction

The ASTM defined additive manufacturing (AM) as “a process of joining materials to make objects from 3D model data, usually layers upon layer, as opposed to subtractive manufacturing methodologies” [50]. Currently, there are large number of AM technologies for metals [8,51,52], such as selective laser melting (SLM) [15,22,26,30,34,35,53–57], electron beam melting (EBM) [29,31–34,58–60], and directed energy deposition (DED) [14,15,18]. These technologies have strong potential to revolutionise the production paradigm in a variety of industries, including aerospace, automotive, energy, medicine, tooling, and consumer goods [51]. Applications for common metals fabricated by AM are listed in Table 1.

Most of these metals are sensitive to hydrogen or are used as critical components in environments where hydrogen may be produced. Thus, it is necessary to investigate their HE susceptibility.

Table 1
Applications for common AM metals [10,11].

Alloys	Applications
Aluminium	Aerospace
Stainless steel	Corrosion resistance; Aerospace; Energy, oil and gas; Medical; Automotive; Marine; Machinability and weldability; Tools and moulds; Consumer products; High temperature
Structural steel	Tools and moulds
Titanium	Corrosion resistance; Aerospace; Automotive; Marine; Machinability and weldability; High temperature; Medical
Cobalt chrome	Corrosion resistance; Aerospace; Medical
Nickel-Based superalloys	Corrosion resistance; Aerospace; Marine; Machinability and weldability; High temperature
Precious metals	Medical; Consumer products

2.2. AM processes

The ASTM F2792-12a standard defines two major methods of AM, powder bed fusion (PBF) (as shown in Fig. 1 (a)), and direct energy deposition (DED) (as shown in Fig. 1 (b)). PBF uses a high-energy beam (laser or electron beam) to sinter or melt the pre-spread powder layer on the bed into the desired profile, followed by spreading and sintering/melting subsequent layers in a repeat fashion until the whole part is built-up. In contrast, DED builds up layer profiles utilizing a focused high-energy beam to simultaneously melt and deposit the feedstock metal that can be either powder (injected by single or multiple nozzles) or wire (directly feed in) on a substrate plate or on existing layers [8,9].

All metal AM methods basically operate in a similar manner although different terminologies have been used over the last two decades [8]. All AM methods feature fast heating, rapid cooling, a high thermal gradient within a small melt pool and complicated thermal history [8,11,61,62] which leads to the different microstructure and properties of parts processed by AM compared with those produced with traditional manufacturing processes. In addition, AM has many more processing variables or parameters that influence the microstructure and properties of the products [8,11]. Table 2 lists the most prevalent parameters.

Lambrakos and Cooper [69,70] identified the most important process parameters in metal AM as (i) powder or wire feed rate, (ii) laser or electron-beam power density, (iii) traverse and scanning speed, (iv) hatch spacing, and (v) layer thickness. Considerable effort has been devoted to investigate the influences on the microstructure and properties of AM alloys [71–78], and there have been several review papers [8,11,43,79–84].

Generally, DED requires higher energy density because coarser powders or wires are used, which results in larger melt pools (around 1 mm in diameter), a faster build rate, and a relatively slower cooling rate. Hence, DED is also associated with poorer surface finish and lower geometric accuracy. Subsequent machining is required [48,85] to achieve the final shape and size. Many DED techniques have been developed. Commercially available systems include laser direct metal forming (LDMF) [23,86], laser engineered net shaping (LENS) [87–90], direct light fabrication (DLF) [91,92], and direct metal deposition (DMD) [93,94].

In comparison, PBF uses lower beam energy density with a smaller spot size, resulting in smaller melt pools, (generally ~100 μm in diameter), greater resolution, better surface finish and much higher dimensional accuracy [8,9,95]. Hence, PBF is more suitable for manufacturing components with highly complex geometry at a small production volume, such as those in the medical and aerospace sectors [9]. In addition, PBF is related to much faster heating and cooling and a larger thermal gradient within the melt pools than DED. Hence, for the same alloy, the microstructure and properties, including HE, of parts fabricated with PBF differ from those made by DED. Based on the type of power source, PBF can be further divided into two principal processes: (i) selective

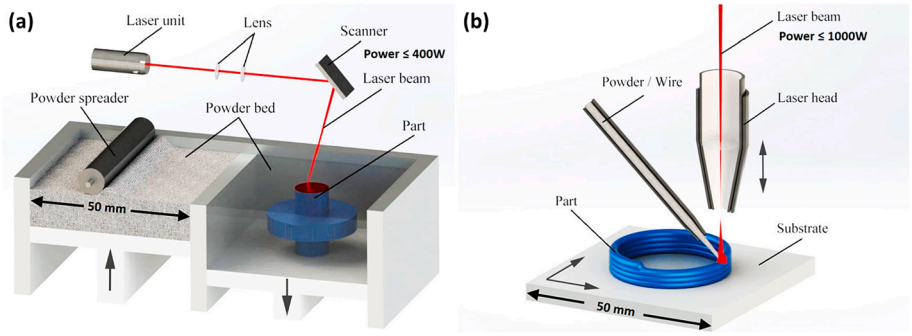

Fig. 1. Schematic illustration of the (a) LPBF process and (b) LDED process. Reproduced from Yin [8] with permission from Taylor & Francis, and Tan is the author of the paper.

Table 2
Summary of key processing parameters in AM processes [8,11,63–68].

Parameters	DED		PBF	
Feedstock	Powder	Wire	Powder	
Nomenclature	LDED	EDED	LPBF	EPBF
Heat source	Laser	Electron beam	Laser	Electron beam
Power	100 ~ 3000 W	500 ~ 2000 W	50 ~ 1000 W	
Beam spot size	up to millimetres	N/A	≤ 100 μm	
Scan speed	5 ~ 20 mm/s	1 ~ 10 mm/s	10 ~ 1000 mm/s	
Powder particle size	50 ~ 150 μm	N/A	10 ~ 60 μm	60 ~ 105 μm
Hatch distance	≤ 1 mm	N/A	≤ 400 μm	200 mm
Layer thickness	≤ 500 μm	N/A	≤ 100 μm	50 ~ 100 μm
Atmosphere	Protective, e.g., Air, Argon and nitrogen	Vacuum	Protective, e.g., Argon and nitrogen	Vacuum

* Typical value.

laser melting (SLM) or sintering (SLS) and (ii) electron beam melting (EBM). Both processes require a platform to hold the powders [96]. However, EBM also requires high vacuum in order to focus the electron beam.

2.3. Microstructure characteristics

As mentioned above, the AM processes are associated with a high cooling rate and directional solidification within small melt pools. Thus, microstructures and properties of metals processed using AM differ from those produced by conventional methods [8,51]. For example, Fig. 2 shows that the as-built SLM 316L stainless steel has cellular dendritic structures and columnar grain structures that are not present in the wrought 316L [97].

The distinguishing microstructural features of AM metals include (i) the finer microstructure due to the fast cooling, (ii) metallurgical defects resulting from improper processing settings, (iii) anisotropic microstructures with elongated grains owing to the directional solidification characteristics, and (iv) non-equilibrium microstructures including metastable phases, solute trapping, and non-equilibrium compositions as a result of fast cooling [98]. Common defects include (i) unmelted or partially melted particles of powders, (ii) lack of fusion, (iii) keyholes, (iv) cracking, (v) deposited layers, (vi) delamination, (vii) ejecting molten droplets, and (viii) porosity [51,99]. Fig. 3 shows examples of defects that form in SLM alloys, including the porosity that resulted from either insufficient energy input or excessive energy use, the balling effect due to excess energy, and hot cracking caused by stress at elevated temperatures [51]. Remelting is a way to improve the microstructure and mechanical properties of AM parts by reheating and resolidifying the material, which can help to achieve higher densification, improve surface finish, and increase strength and toughness [100,101].

AM parts often have an epitaxial columnar grain morphology because of the fast directional solidification with an extremely high thermal gradient (approximately 10^6 K/m). This corresponds to microstructural texture and inhomogeneity, leading to undesired anisotropic properties. Additionally, due to their elongated shape, columnar structures are susceptible to intergranular hot-cracking when exposed to high thermal shocks during the AM process [11,102–104]. An example of the epitaxial columnar grain morphology in SLM Co-Cr-Mo alloy is presented in Fig. 4 [105]. Inoculation treatment is an effective method for refining and homogenizing the microstructure, converting the columnar structure into equiaxed grains. This is achieved by introducing nucleants, which can be either externally added or in-situ formed, to promote heterogeneous nucleation [106]. In the past years, inoculation treatments to promote heterogeneous nucleation has been widely utilised in metal AM to process high-performance Ti alloys [107–109], Al alloys [102,104,110–112], and steels [106,113–116].

Different materials produced with AM exhibit varying microstructural characteristics. For example, retained γ austenitic films along the building direction is common in most AM steels [117]. Fig. 5 (a) and (b)

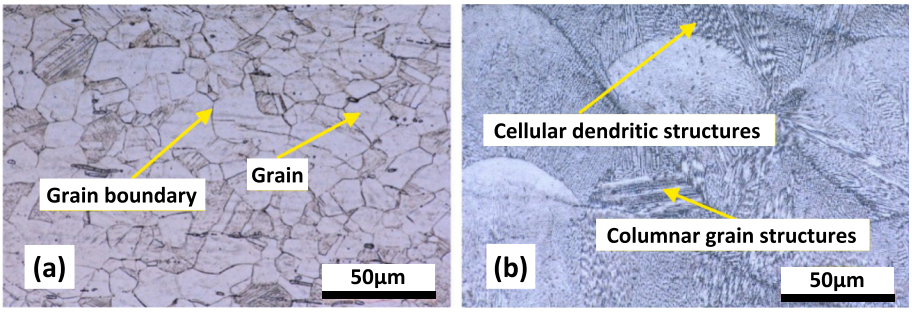

Fig. 2. Optical microscope microstructure of (a) wrought 316L and (b) as-built SLM 316L. Reproduced from Tascioglu [97] with permission from Springer.

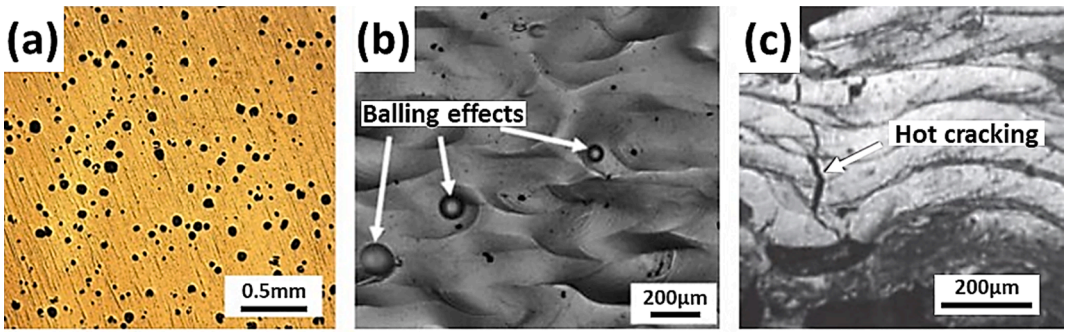

Fig. 3. Defects in SLM Ti-6Al-4 V: (a) porosity, (b) balling, and (c) hot-tearing cracks. [51].

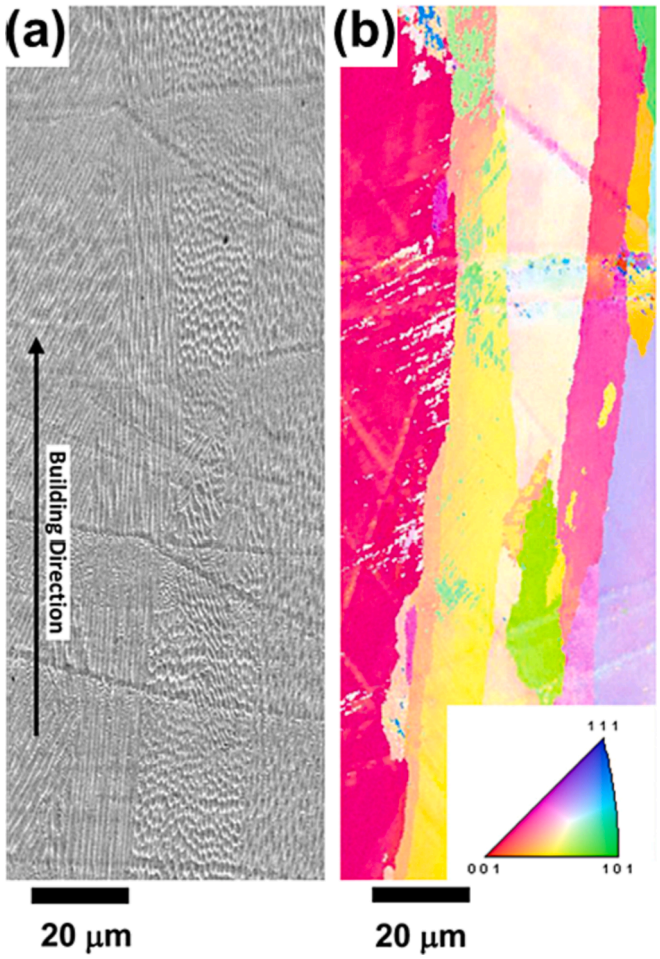

Fig. 4. (a) SEM micrograph of the cross section of a Co-29Cr-6Mo alloy fabricated via the SLM process. (b) The corresponding crystal orientation map of the micrograph showing the epitaxial columnar grain morphology in the metal AM part. Reproduced from Takaichi [105] with permission from Elsevier.

show the ultrafine cellular-dendritic substructure in the SLM H13 steel. Most of the cells present a pancake-like morphology along the building direction, and there are extremely thin retained austenite (γ) films in-between the cells. This microstructure is formed from directional solidification during AM and solute segregation [118–121]. The inter-dendritic segregation of solutes, carbon in particular, stabilised the local austenite by decreasing its martensitic transformation point below room temperature and suppressed the subsequent martensitic transformation in these areas, forming austenitic films. Fig. 5 (d) presents the retained austenitic films amongst the martensite blocks [117].

Moreover, Fig. 6 shows that the microstructure in the AM steels is sensitive to the cooling rate and the temperature gradient, which are controllable via the energy input by changing the feeding mechanism such as laser power or/and scanning speed. Therefore, the mechanical characteristics of steels produced through AM can be deliberately customised by considering the specific correlation between the microstructure and the desired property [8].

The microstructural features of various metals fabricated by AM have been comprehensively summarised in previous reviews on metal AM [8,9,11,48] and are not detailed herein since the present work focuses on HE.

2.4. Mechanical properties

Metals produced with different AM methods have different microstructures that result in distinct properties. For example, the microstructure of metals produced with the laser PBF process is much finer as a result of the fast-cooling, resulting in better tensile strength but lower ductility when compared to other processing methods [66]. Table 3 lists the tensile properties, yield strength (YS), ultimate tensile strength (UTS) and elongation, of the mostly studied metals fabricated with AM. For comparison, properties of the counterpart metals produced with the typical conventional manufacturing processes are also listed. Other mechanical properties such as impact toughness, fatigue resistance, and creep resistance, are not included in this review since they have been investigated and reviewed previously [8,103,122–125].

Fig. 7 compares the mechanical properties of AM alloys and the conventionally produced alloys with processes such as casting, forging, and other forging processes for the properties (UTS, YS and elongation). The X-axis representing data obtained from the traditional methods, and the Y-axis representing data obtained from AM metals.

The reported property data for the alloys fabricated by AM is more scattered due to the obvious anisotropy in the mechanical properties, especially elongation [8,51,117,175]. Nevertheless, the tensile properties of AM metals are comparable with those made using conventional methods. This is attributed to the relatively high cooling rates achieved in AM, which reduces the partitioning and favours finer microstructure than conventional production [176]. This implies that AM has strong and practical potential for industry applications in producing high-quality parts with complex geometries.

3. Hydrogen embrittlement

Hydrogen embrittlement (HE) is the deterioration of mechanical properties of solid metals due to the presence of hydrogen [3], which can degrade tensile and fatigue properties [4], and can even cause catastrophic failure [5]. The pervasive nature of hydrogen makes the entry of hydrogen atoms into a substance typically difficult to avoid. Thus, HE is an inevitable problem, particularly for high-performance structural components.

The causes of HE can be explained from different aspects, including

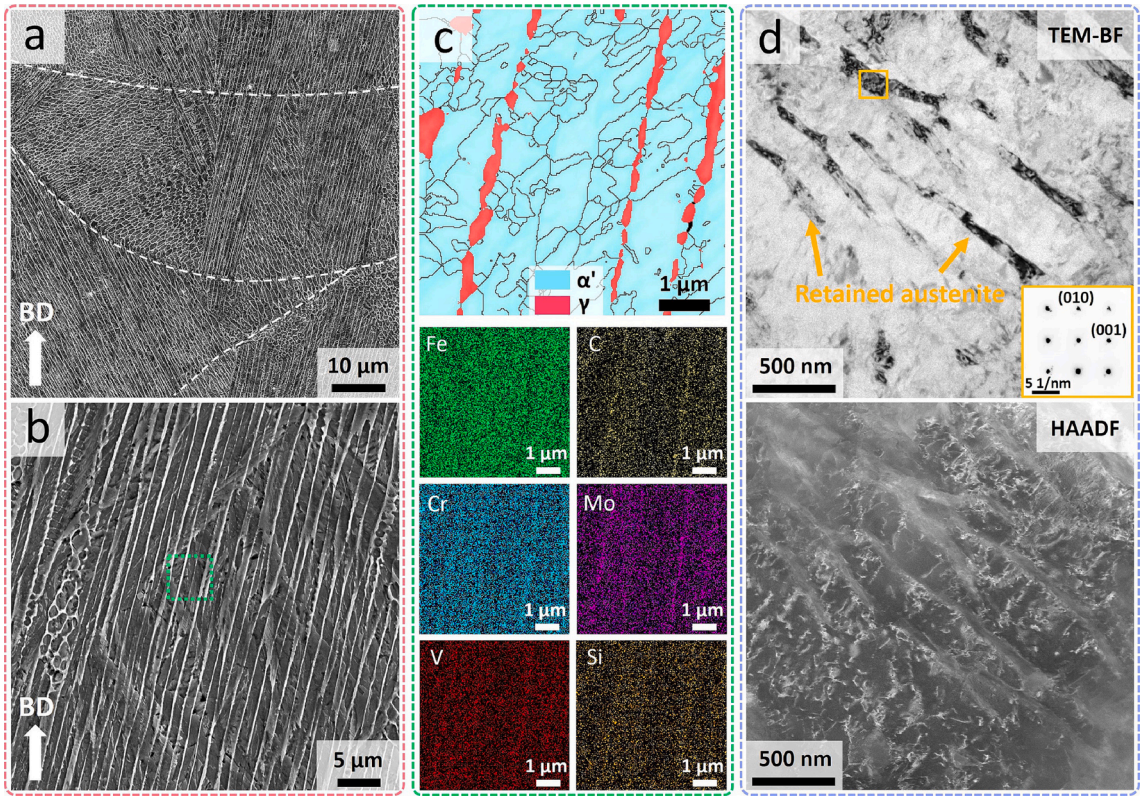

Fig. 5. (a) SEM micrograph of the SLM H13 steel, with the dashed lines representing the melt pool boundaries; (b) high magnification SEM micrograph, showing extremely fine cell-like substructures in the sample. (c) EBSD phase map and the corresponding EDS maps of the main constituent elements of the area marked in (b). (d) TEM bright-field (upper) and high-angle annular dark field scanning-TEM (lower) micrographs viewed along the $[100]_{\gamma}$, with the inset showing the SAED pattern of the marked area. Reproduced from Tan [117] with permission from Elsevier, and Tan is the author of the paper.

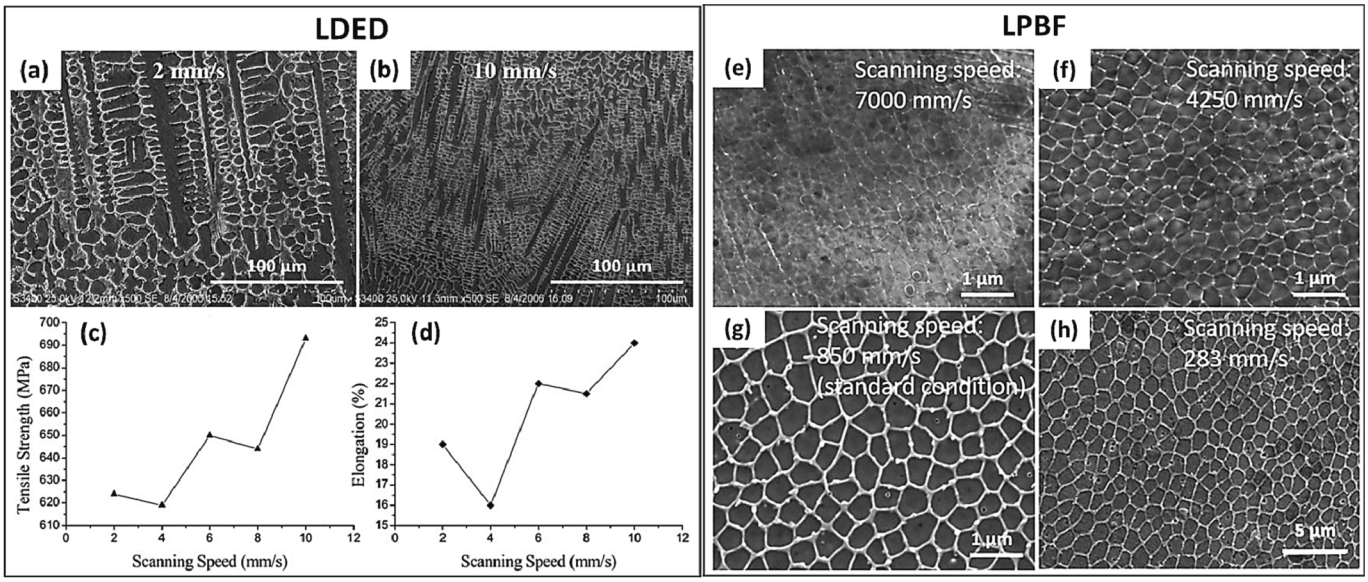

Fig. 6. (a & b) SEM microstructure, (c) tensile strength & (d) elongation of LDED deposited 316L alloys with different scan speeds; (e-h) SEM images from the etched surfaces of the LPBF built 316L samples built with different laser scanning speeds. Reproduced from Yin [8] with permission from Taylor & Francis, and Tan is the author of the paper.

the sources of hydrogen and the types of embrittlement. Generally, HE is divided in the literature into three categories: internal hydrogen embrittlement (IHE), hydrogen reaction embrittlement (HRE), and hydrogen environment embrittlement (HEE) [3,177–180]. IHE is caused by hydrogen in the metal lattice, typically introduced during prior processing or fabrication. The severity of hydrogen embrittlement increases with hydrogen concentration, and depends on temperature and the state of stress [178,181]. HRE is caused by the high pressures resulting from the chemical reaction of hydrogen inside the metal, for example the production of methane by the reaction of hydrogen and

Table 3
Mechanical properties of metals by additive manufacturing (AM) and traditional (TR) methods.

Materials	AM	UTS (MPa)	YS (MPa)	Elongation (%)	Hardness (HV)	Ref.	TR	UTS (MPa)	YS (MPa)	Elongation (%)	Hardness (HV)	Ref.
316L SS	LPBF	480 ~ 760	350 ~ 640	10 ~ 30	220 ~ 279	[66]	Cast	596 ± 16	365 ± 22	69 ± 9		[126]
		670 ± 30	520 ± 70	47.5 ± 11.5		[127]	Forged	450 ~ 818	150 ~ 230	50 ~ 62	133 ~ 140	[66]
		653 ± 3.4	534 ± 5.7	16.2 ± 0.8		[128]		450	170	40		[129]
		704.1 ± 2.5	581.1 ± 7.3	54.8 ± 2.3	239.6 ± 6.6	[130]		480	170	40		[129]
	E-PBF	437 ~ 580	253 ~ 396	10 ~ 59	165 ~ 173	[66]		620	310	30		[129]
	DED	533 ~ 685	235 ~ 485	12 ~ 43	180 ~ 210	[66]		485	170	40	218	[129]
		900	580	6.5		[131]	Wrought	887	382			[132]
		770	415	4		[131]		935	390			[132]
		430 ~ 510	400 ~ 440	14 ~ 20		[89]						
		639	558	21		[133]						
		536	352	46		[133]						
304L SS	LPBF	706.7 ± 6.2	454.7 ± 11.5	50.3 ± 1.4	217.5 ± 3.7	[130]	Cast	295	125	23		[129]
		530 ± 24	430 ± 12	32 ± 11		[134]	Forged	450	170	40		[129]
		660 ± 20	540 ± 15	36 ± 12		[134]		480	170	40		[129]
	DED	609 ± 18	337 ± 29	48.2 ± 2.5		[78]		620	310	30		[129]
		606 ± 13	314 ± 6	56.4 ± 5.8		[78]						
		685 ± 5	490 ± 8	51 ± 2		[135]						
		580 ± 10	280 ± 6	62 ± 5		[135]						
Ti-6Al-4V	LPBF	973 ~ 1407	885 ~ 1333	5 ~ 19	381 ~ 479	[66]	Cast	950	940	23.1		[66]
		1000.4	803.6	7.9		[136]	Forged	1380	1192	19.1		[66]
		1013	836.6	9.5		[136]		926 ± 2	878 ± 4	20 ± 1		[137]
		911.1	738.6	4.2		[136]		1030	970	16		[138]
		1011.5	783.2	5.2		[136]		980	865	13.5		[139]
		1219 ± 20	1143 ± 30	4.89 ± 0.6		[140]	Wrought	1008	962	19		[141]
		1269 ± 9	1195 ± 19	5 ± 0.5		[140]		870 ± 10	790 ± 20	18.1 ± 0.8		[142]
		1246 ± 134	1150 ± 67	1.4 ± 0.5		[143]		933 ± 7	832 ± 10	13 ± 1.5		[143]
		1273 ± 53	1421 ± 120	3.2 ± 0.5		[143]		942 ± 8	836 ± 9	12.5 ± 1.2		[143]
	E-PBF	915 ~ 1200	830 ~ 1150	13 ~ 25	360 ~ 460	[66]						
		979	905	10		[144]						
		971	910	10		[144]						
		1066	1006	15		[138]						
		1073	1001	11		[138]						
		1032	973	12		[138]						
		1032.9 ± 12.9	984.1 ± 8.5	9 ± 2.9		[145]						
		1008.6 ± 15.2	961 ± 7.1	7.1 ± 3.4		[145]						
		1029.7 ± 7	982.9 ± 5.7	12.2 ± 0.8		[145]						
		966.5 ± 5.3	1017.4 ± 4.9	12.2 ± 2		[145]						
	DED	920 ~ 1163	850 ~ 1105	4 ~ 17		[66]						
		1099 ± 2	976 ± 24	4.9 ± 0.1		[146]						
		1025 ± 2	950 ± 2	5 ± 1		[137]						
		1063 ± 20	960 ± 26	10.9 ± 1.4		[147]						
IN 718	LPBF	1400	1180	16.9		[148]	Cast	950	940	23.1		[148]
		1440	1186	18.5		[148]		1090	915	11		[129]
		1450	1190	20.4		[148]	Forged	1380	1192	19.1		[148]
		1126	849	22.8		[149]	Wrought	1350	1105			[129]
		1143 ± 5	995 ± 10	23 ± 3		[150]		1435	1185			[129]
		1085 ± 11	816 ± 24	19.1 ± 0.7		[151]		1530	1365			[129]
		1010 ± 10	737 ± 4	20.6 ± 2.1		[151]		1407	1172	21		[151]
	E-PBF	1207	669	21		[152]						
		1060 ± 26	822 ± 25	22		[153]						
		929 ± 20	744 ± 44	5.5		[153]						
		1138 ± 24	925 ± 20	15.7 ± 4.3		[154]						
		1061 ± 83	894 ± 24	11.5 ± 6.9		[154]						
AISI 4340	LPBF	1207 ~ 1227	1076 ~ 1110	16 ~ 18		[155]	Forged	1315	1250	8.2	393	[129]
		1386 ~ 1434	1276 ~ 1311	19		[155]		1315	1240	14.6	393	[129]

(*continued on next page*)

Table 3 (*continued*)

Materials	AM	UTS (MPa)	YS (MPa)	Elongation (%)	Hardness (HV)	Ref.	TR	UTS (MPa)	YS (MPa)	Elongation (%)	Hardness (HV)	Ref.
	DED				480 ~ 681	[94]	Wrought	1862	1517	11		[155]
H13	LPBF	1712 ± 103	1236 ± 178	4.1 ± 1.2		[156]	Forged	1580	1365	14.4	458	[129]
		1620 ± 215	835 ± 23	4.1 ± 2.3		[156]		1960	1570	13.0	549	[129]
		1965 ± 145	1073 ± 72	3.7 ± 1.7		[156]	Wrought	1999	1569	7.5		[157]
		2005 ± 68	1190 ± 50	3.8 ~ 6.9		[117]						
		1370 ± 175.1	1003 ± 8.5	1.7 ± 0.6		[158]						
		1440 ± 34.6	892 ± 55.2	1.5 ± 0.2		[159]						
		1704 ± 30	1342 ± 67	1.55		[160]						
		1321 ± 11	1167 ± 73	0.35		[160]						
		1227 ± 46	1133 ± 42	0.3		[160]						
	DED	1820	1505	6		[93]						
		2064 ± 51	1288 ± 54	6 ± 2.2		[157]						
		2033 ± 38	1564 ± 24	5.0 ± 2.8		[157]						
17—4 PH	LPBF	1103.8 ~ 1105.8	633.8 ~ 666.2	20.5 ~ 21.3	259	[161]	Forged	795	515	18	252	[129]
		980	500	7	340	[162]		930	725	16	271	[129]
		1100	500	12	340	[162]		965	795	14	292	[129]
		1370	1190	8.3	380	[163]		1000	860	13	303	[129]
		1060	650	14.5		[164]		1070	1000	12	332	[129]
		940	580	5.8		[164]		1170	1070	10	361	[129]
		1188	1160	4.8		[165]		1310	1170	10	388	[129]
		1203	1170	4.8		[165]	Wrought	1018	992	13.4	430	[166]
	DED	900 ± 200	400 ± 100	5 ± 3	441	[167]		1450	1370	9		[168]
18Ni 300	LPBF	1290 ± 114	1214 ± 99	13.3 ± 1.9	388	[169]	Cast	1750	1650	8		[129]
		1050	930	5	350	[170]	Wrought	1000 ~ 1170	760 ~ 895	6 ~ 15	332	[169]
		1187.6 ± 10.4	914.9 ± 12.5	6.14 ± 1.33	371	[171]						
		1050 ± 45	903 ± 98	8 ± 1.2	371	[172]						
		1165 ± 7	915 ± 7	12.44 ± 0.14	332 ~ 342	[173]						
		1260.1 ± 79	768.0 ± 29	13.9 ± 2		[174]						
		825.9 ± 96	1324.7 ± 51	14.0 ± 1.5		[174]						

carbon in the steel [178,182]. HEE is caused by hydrogen that can enter the material (initially hydrogen free) during service or during mechanical testing, such as during mechanical testing in gaseous hydrogen [3]. HEE takes place in hydrogen-containing atmospheres. During testing in hydrogen gas, some hydrogen molecules are adsorbed on the surface, the hydrogen molecules dissociate into hydrogen atoms, and some hydrogen atoms enter the metal [178,183]. Alternatively, if hydrogen is released by a chemical reaction at the steel surface, as in acid pickling to remove mill scale after hot working of an ingot, the hydrogen activity can be extremely large, so large that the hydrogen can deposit at internal defects like inclusions and produce sufficient pressures to exceed the ultimate strength of the steel and cause blisters. Hydrogen in excess of the hydrogen solubility causes the precipitation of hydrides in hydride forming metals such as Ti and Al (but typically not in steels).

The HE sensitivity depends on the microstructure and properties of the metal. Consequently, it is expected that the HE sensitivities of metals manufactured by AM differ from those produced by conventional methods because microstructures and properties of parts fabricated with AM are different from those produced with conventional methods. Before reviewing the HE of AM alloys, major HE mechanisms are summarised in the following.

3.1. HE mechanisms

The following mechanisms have been usually used to explain the observed manifestations of HE in metals: (i) hydrogen-enhanced decohesion (HEDE) [184–186], (ii) hydrogen-enhanced localised plasticity (HELP) [187,188], (iii) adsorption-induced dislocation emission (AIDE) [40], and (iv) hydride formation and cleavage (HFC) [189,190]. HEDE and HELP are the mechanisms most often considered [13] to be the operative mechanism in AM steels. HEDE assumes that hydrogen causes brittle cleavage of metal bonds, whereas hydrogen is associated with plastic deformation in HELP [23]. In addition, some studies suggested that there may be synergistic effects between the HE mechanisms and that the details of the HE fracture process are more complicated than assumed in these mechanisms [23,44,191,192].

3.1.1. Hydrogen enhanced decohesion (HEDE)

The HEDE mechanism proposes that hydrogen reduces the cohesive strength of an alloy [184–186]. The decohesion mechanism was originally proposed in 1926 by Pfeil, who stated that “hydrogen decreased the cohesion across cubic cleavage planes and grain boundaries” [193]. A refined version proposed that the presence of hydrogen reduces the cohesive strength of interface boundaries or lattice planes [194,195]. Troiano suggested that the underlying mechanism of HEDE is that the electron from a hydrogen atom dissolved in steel enters the unfilled 3D shell of the atom comprising the steel (e.g. Fe atoms as suggested by Troiano in his study of pure Fe), and that these extra electrons in the d-shell increase the interatomic repulsive forces, and result in decreasing cohesive strength [5,195]. The schematic of the HEDE mechanism is shown in Fig. 8 [196], which shows another way to explain the HEDE model. The entry of hydrogen causes the crystal lattice to expand, weakening the atomic bonds at the tip of a crack. As a result, the energy required to enable crack propagation is lowered, resulting in a macroscopic brittle fracture [197,198].

Few studies have provided evidence that HEDE is the sole HE mechanism in AM metals, but often HEDE is assumed in combination with the HELP mechanism. For example, Li et al. [23] reported that the HE of IN 718 was caused by the synergistic effect of the HELP and HEDE mechanism, while Wan et al. [199] proposed that the HE of the

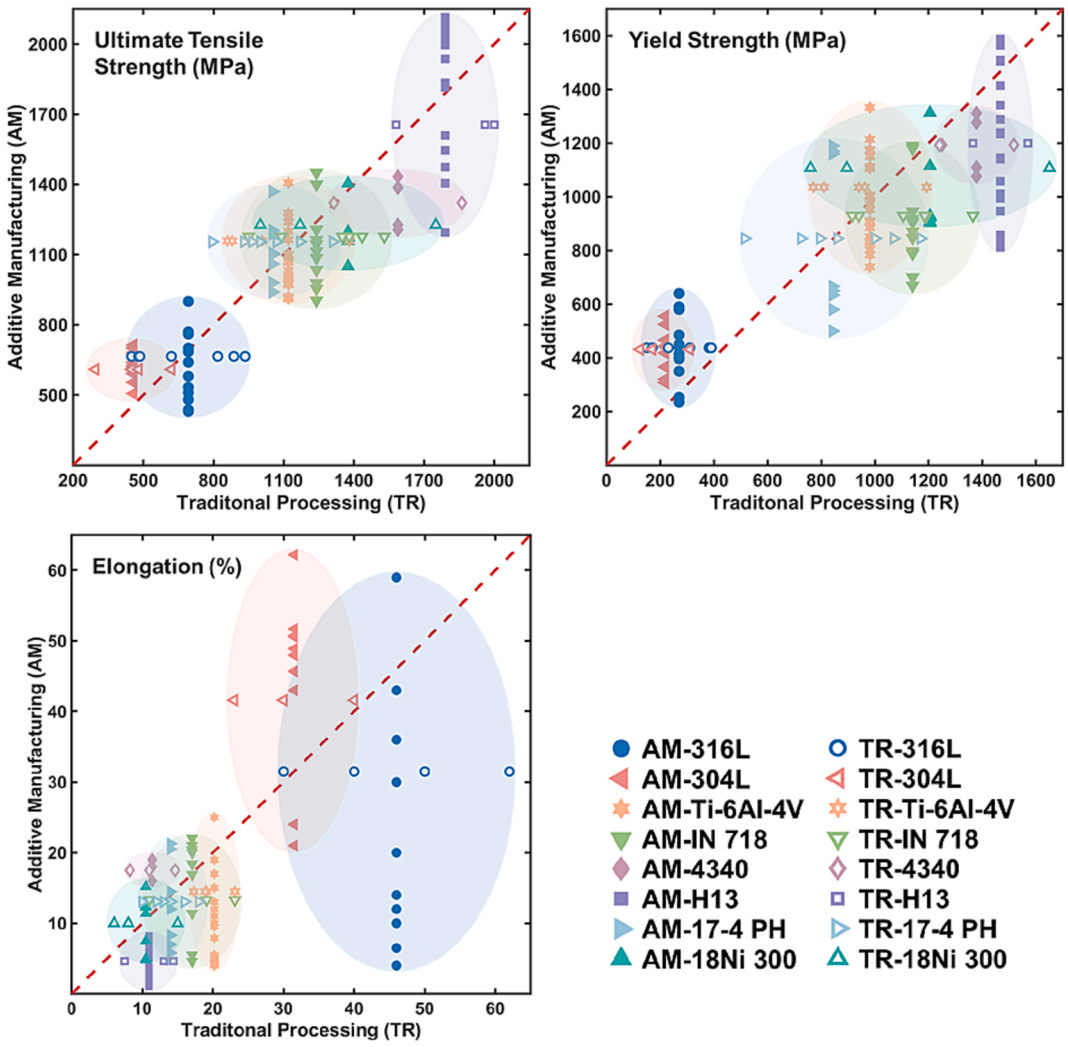

Fig. 7. Comparison between the UTS, YS & elongation of traditional methods (x-axis) and AM methods (y-axis).

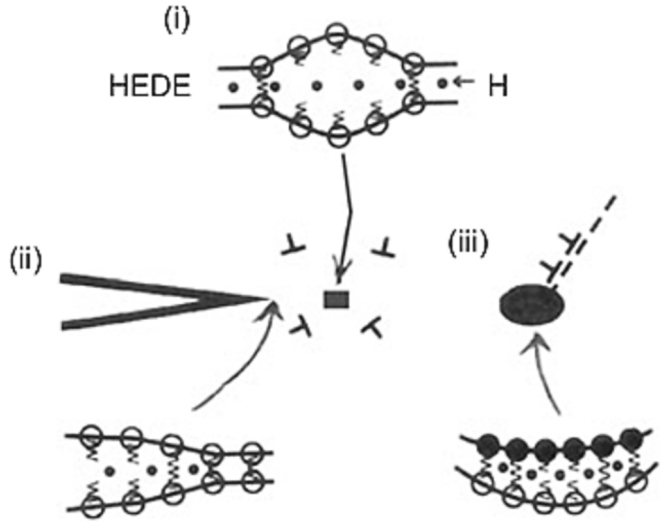

Fig. 8. Schematic of the HEDE mechanism, including tensile separation of atoms owing to weakening of interatomic bonds by (i) hydrogen in the lattice, (ii) adsorbed hydrogen, and (iii) hydrogen at particle–matrix interfaces. Reproduced from Lynch [196] with permission from Elsevier.

$AlCoCrFeNi_{2.1}$ eutectic high-entropy alloy (HEA) was due to the activation of the HEDE and HELP mechanisms.

3.1.2. Hydrogen enhanced localised plasticity (HELP)

The HELP mechanism considers that hydrogen facilitates local dislocation nucleation and motion at the crack tip, resulting in local dislocation pile-ups near slip band intersections, causing premature failure [187,188]. The schematic diagram of the HELP mechanism is shown in Fig. 9 [196]. In 1971, Beachem first proposed the mechanism of hydrogen-enhanced localised plasticity by analysing fracture surfaces in quenched and tempered steels. HE was assessed as a function of the stress intensity factor, and compared flow stresses with and without hydrogen [200]. Later, Birnbaum, Robertson and other researchers refined the model, using elasticity theory combined with observations using in-situ transmission electron microscopy (TEM) [201–203].

The HELP model assumes that the diffusible hydrogen increases the mobility of dislocations, causing a reduction in the material's shear strength. Such reduction in the shear strength enables the formation of micro voids that connect along preferential slip planes, resulting in highly localised fracture [197,198].

Some research on materials processed using AM has identified HELP as the primary mechanism of HE [21,23,34,199,204–206]. More details are presented in the subsequent sections.

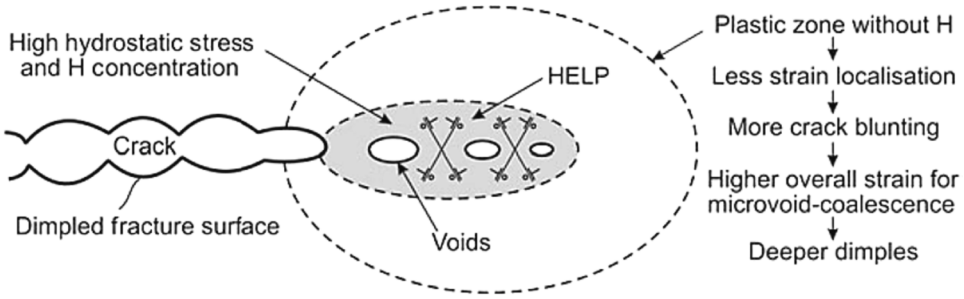

Fig. 9. Schematic of the HELP mechanism, including a micro void-coalescence process, with plasticity localised and facilitated in regions of high hydrogen concentration. Reproduced from Lynch [196] with permission from Elsevier.

3.1.3. Hydride formation and cleavage (HFC)

Embrittlement of hydride-forming metals is primarily attributed to the brittle nature of the hydride phase and/or the failure along the metal/hydride phase, known as the HFC mechanism [207]. This mechanism was first proposed by Westlake in 1969 [208] and is important in hydride forming metals such as Ti and Al [190,207,209–211]. The schematic diagram of the HFC mechanism is shown in Fig. 10 [196], which involves repetitive processes of (i) hydrogen diffusion towards areas with elevated hydrostatic stress in advance of cracks, (ii) nucleation and growth of a hydride phase, (iii) the fracturing of the hydride upon reaching a critical size, and (iv) arrest of crack propagation at the interface between the hydride and the matrix [40].

The HFC has been identified as the main mechanism of HE in Ti alloys processed using various AM techniques, including SLM and EBM [31,32,34]. Additional information regarding this mechanism is presented in the subsequent sections.

3.1.4. The synergistic action of HE mechanisms

Numerous investigations of HE primarily focused on the dominance of only one particular mechanism. However, recent advancements in modelling and experimental studies have shed light on the collaborative interplay of multiple HE mechanisms [44,212–214]. Djukic et al. [44] indicated that the cooperative action of multiple HE mechanisms in steels and iron could be classified into two categories (i) HELP-mediated HEDE model and (ii) HELP + HEDE model. The HELP-mediated HEDE model focuses on the activation of the HEDE mechanism through previous activity of the HELP mechanism, while the HELP + HEDE model involves the synergistic effect of both plasticity-mediated HE mechanisms and the HEDE mechanism. The summary of the unified model for synergistic interplay of HE mechanisms in steels is illustrated in Fig. 11. The HELP-mediated HEDE model assumes that the governing mechanism of HE in steels and other materials is primarily the HELP mechanism. In this case, HE is initiated by HELP and is a prerequisite for the eventual activation of the decohesion-based (HEDE) mechanism at specific microstructural locations. On the other hand, the HELP + HEDE describes the simultaneous action of hydrogen-enhanced plasticity and decohesion. According to this model, some or multiple plasticity-mediated HE mechanisms (e.g., HELP, AIDE or HESIV) together with the decohesion-based (HEDE) mechanism are responsible for the HE. The dominance of a specific mechanism is contingent upon factors such as hydrogen concentration, material stress state, and environmental conditions.

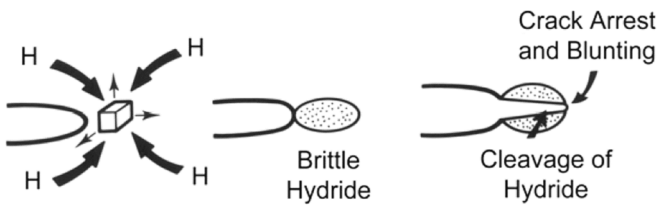

Fig. 10. Schematic presenting subcritical crack growth involving hydrogen diffusion to hydrostatically stressed regions, then formation and fracture of a brittle hydride at a crack tip. Reproduced from Lynch [196] with permission from Elsevier.

The concurrent operation of both HELP and HEDE mechanisms, in accordance with the HELP + HEDE model, has recently been substantiated across a range of materials, including various types of steels [212,215], nickel-based alloys [216,217], aluminium alloys [218–220], and high-entropy alloys [221,222]. These mechanisms also apply to some metals fabricated through AM processes [13,16,23,28,199,223]. Further discussion of the synergistic action of HE mechanisms in AM metals will be provided in the subsequent sections.

3.2. Microstructure influence

HE is sensitive to microstructure, even for the same alloy. Microstructural alteration may result in an improvement in the resistance to HE [34]. For example, Jeong et al. [224] considered that the lath-type of retained austenite in steels has a more positive effect in relieving hydrogen susceptibility than the equiaxed retained austenite at a similar volume fraction and mechanical stability. It has been verified that lath-type retained austenite slows hydrogen diffusion as the lath boundaries inhibit hydrogen migration. Venezuela et al. [225] considered that martensite is known to be the most susceptible to hydrogen compared with other ferrous microstructures such as pearlite and bainite. Martensite possesses a highly strained nano-structure because of lattice distortions induced by the interstitial carbon atoms [226]. The presence of a high dislocation density and a high density of interface defects introduces higher residual stress in the lattice. Hydrogen can interact with these inherent stresses in martensite, causing embrittlement [225]. The order of HE susceptibility of microstructure is ranked as follows from low to high: fine-grained massive ferrite and acicular ferrite < fine-grained granular bainite and massive ferrite < coarse-grained granular bainite and bainite ferrite embedded with martensite-austenite constitute < untempered martensite [227]. Hydrogen trapping and diffusion are closely related to the microstructure of materials and are critical to assessing HE susceptibility. This is further discussed in the following sections [39].

3.2.1. Hydrogen trapping

Lynch [40] proposed that solute hydrogen occupies and diffuses across interstitial lattice sites in metals and occupies to varying degrees the lower potential-energy sites than normal interstitial sites. Schematic illustrations of sites and traps for hydrogen in materials are shown in Fig. 12. These sites are generally listed in terms of hydrogen trapping strength from low to high: solute atoms < locations between the first several atomic layers beneath surfaces and free surfaces < vacancies < strain fields and dislocation cores < grain boundaries < inclusion or matrix interfaces < internal cracks and voids.

Traps influence the solubility and diffusivity of hydrogen. Thus,

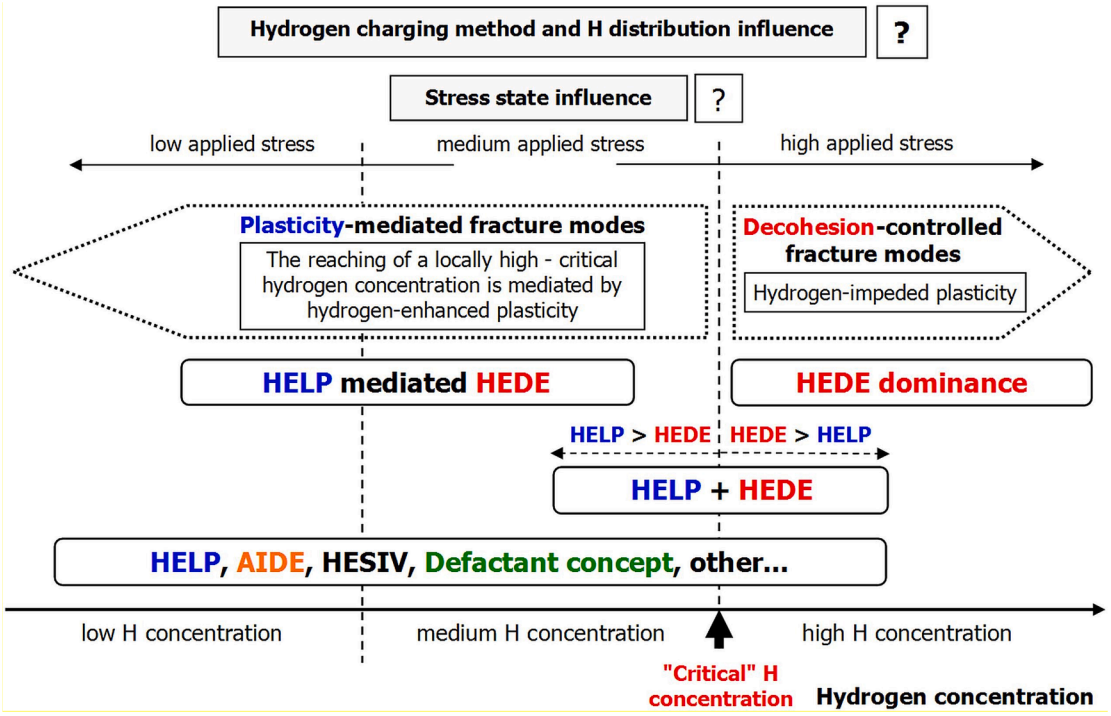

Fig. 11. Summary of the unified model for synergistic interplay of HE mechanisms in steels: localised plasticity and decohesion (HELP + HEDE model). Reproduced from Djukic [44] with permission from Elsevier.

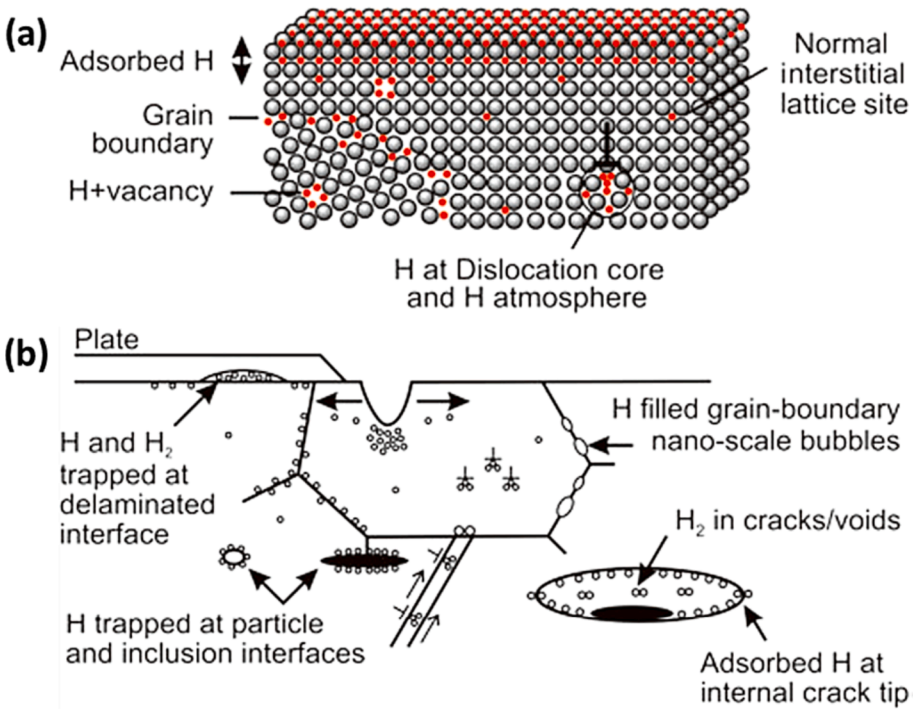

Fig. 12. Schematic illustrations of sites and traps for hydrogen in materials (a) on the atomic scale, and (b) on a microscopic scale. Reproduced from Lynch [196] with permission from Elsevier.

trapping can either enhance or weaken the material's HE susceptibility [39,228]. A thorough investigation of trapping mechanisms in metals is necessary because it is important for understanding the complicated interactions between metals and hydrogen and could aid in the design of hydrogen-induced failure prevention methods [39].

The location of hydrogen trapping influences the dynamics of HE. In HEE, hydrogen gas dissociates and is adsorbed at crack tips. In contrast, in IHE, solute hydrogen diffuses to interior fracture-tip surfaces and is adsorbed there. Solute hydrogen can also precipitate as hydrogen gas in voids and at delaminated non-porous plating/substrate interfaces, sometimes resulting in high pressures, which can then be reabsorbed and then diffuse under applied stresses. In normal interstitial lattice

sites, hydrogen concentration occurs ahead of notches and fractures, where strong hydrostatic stresses lead to a slightly enlarged lattice [40].

Furthermore, crystalline defects such as grain boundaries (GBs) and dislocations play a significant role in HE of metals [215,229]. A GB is a planar defect that can trap hydrogen. Similarly, the dislocation is a line defect with a stress field surrounding the dislocation core that can interact with hydrogen [39]. Wan and co-workers [215] have shown that the reaction of GBs with dislocations is a key component in HE mechanism for polycrystalline metals. The dislocation-GB reaction can cause a locally activated state of the GB with a more disordered atomistic structure, introducing a local dilative stress concentration. When GB is segregated with hydrogen atoms, the cohesion of the GB is significantly weakened when it is activated. At present, numerous studies demonstrated the importance of GBs in HE, especially in the following aspects:

(i) Hydrogen atoms tend to segregate on GBs [230];
(ii) The quasi-cleavage-like fracture predominates in materials with moderate hydrogen concentrations, and the micro-scale facets on the quasi-cleavage fracture surface, as well as the sites of microcracks within the material, usually correlate to GBs [231–233]; while the intergranular fracture is common in materials charged with relatively high hydrogen content [234–236];
(iii) HE suppression can be achieved by grain structure refinement [237–239].

3.2.2. Hydrogen diffusion

Previous studies have indicated that HE is generally caused by local hydrogen enrichment due to hydrogen trapping at dislocations, GBs, and other defects. A high density of hydrogen traps would reduce the hydrogen diffusion coefficient [240,241]. The principal mechanism for hydrogen diffusion in steel is lattice diffusion via interstitial jumps [242]. Metals with relatively low hydrogen diffusion coefficients have been found to be more susceptible to HE [229,240,243].

Due to hydrogen diffusion is sensitive to trapping in lattice imperfections, diffusion models have been proposed to quantify trapping parameters, which could be utilised to determine HE susceptibility [39,244]. This information may be acquired using several approaches, with the most common being the hydrogen permeation test [39]. According to Oriani [245], the McNabb-Foster model solutions can be resolved by assuming a dynamic local hydrogen equilibrium between traps and lattice sites. Several constitutive equations linking the diffusion coefficient (D_{eff}) to trap parameters result from this assumption. For instance, by assuming low trap occupancy, the effective diffusion coefficient can be obtained using the equation [246,247]:

$$D_{eff} = \frac{D_L}{1 + \frac{N_T}{N_L}\exp\left(\frac{E_b}{RT}\right)} \tag{1}$$

where N_{T} is the number of trap sites (cm^3), N_{L} is the number of lattice sites (cm^3), R is the gas constant, and T is the absolute temperature. Another method to assess D_{eff} from permeation experiments is to use either the time lag or the breakthrough method [248]:

$$\ln\left(\frac{N_T}{N_L}\right) = \ln\left(\frac{D_L}{D_{eff}} - 1\right) - \frac{E_b}{RT} \tag{2}$$

Aside from vacancies, different microstructural features can influence hydrogen diffusion in metals, including dislocations, GBs and crystal structure. For example, for pure metals at ambient temperatures, hydrogen diffusion mainly depends on the crystal structure [249]. A body-centred cubic (BCC) structured ferrite allows a high diffusion rate and a low solubility because of its open lattice structure. Conversely, the face-centred cubic (FCC) structured austenite has a lower diffusion rate and a higher solubility because of the close-packed lattice. Martensite is primarily a body-centred tetragonal structures (BCT), with high density dislocations for low carbon martensite, which reduces hydrogen diffusivity. As increasing the carbon content, hexagonal close packed (HCP) structure martensite may form with lower stacking fault energy [242,250]. HCP structure is closer packed than BCC. Thus, the diffusion rate of hydrogen in HCP martensite is lower than in ferrite [242]. In general, D_{eff} of hydrogen is 4 to 5 orders of magnitude greater in BCC metals than in FCC and HCP metals at 20 °C [249]. However, there are exceptions, for instance, palladium (FCC) and cobalt (HCP) which have D_{eff} of several orders of magnitude higher than most other FCC and HCP metals, likely due to quantum effects overriding lattice-packing effects [40]. Besides, D_{eff} quoted in the literature span 4 to 5 orders of magnitude at 20 °C for some metals such as steels and Al, presumably because of experimental difficulties related to controlling surface conditions (e.g. oxide films) that influence hydrogen ingress and egress [40,251]. A decrease in D_{eff} indicates an increase in hydrogen trapping in metals. In further, some studies have found that the reduction in D_{eff} increases with the strength of metals, which means the higher-strength metals are more susceptible to HE [244].

4. HE characteristics of AM metals

This section reviews major reported results on HE characteristics of different AM metals, including titanium (Ti) alloys (4.1), nickel (Ni)-based superalloys (4.2), steels (4.3), and high-entropy alloys (4.4). In addition, some metals are also known to be sensitive to hydrogen. But, the HE susceptibility of their counterparts fabricated by AM has not been studied, such as AISI 4340 and H13. These metals will also be briefed in this section.

Table 4 lists the metals processed by AM, for which HE was studied. Among the extensively studied materials are Ti-6Al-4 V, IN 718, AISI 304L stainless steel, and AISI 316L stainless steel. It is noteworthy that the bulk of the research on HE of AM-fabricated metals is currently concentrated on powder-based AM, while research in wire-arc AM remains relatively limited [252]. In light of this, the forthcoming sections of this review will primarily focus on the HE characteristics of metals processed by powder-based AM.

4.1. Titanium alloys

Titanium (Ti) alloys are widely utilised as structural and functional materials in automotive, aerospace, medical and chemical sectors because of their high corrosion resistance, high strength, good biocompatibility and lightweight. The problem of hydrogen penetration

Table 4
AM metals for which HE was studied.

Materials	Material type	AM processes	Ref.
AISI 304L	Austenitic stainless steel	DED (LENS and DMT), PBF (SLM)	[13–16]
AISI 316L	Austenitic stainless steel	DED (LENS), PBF (SLM)	[17–20]
Nb-alloyed	Austenitic steel	PBF (EBM)	[60]
17-4PH	Precipitation hardening stainless steel	PBF (SLM)	[55]
AerMet100	Low-alloy high strength martensite steel	PBF (SLM)	[204,253]
18Ni 300	Maraging steel	PBF (SLM)	[57,254–256]
Ti-6Al-4 V	Ti & Ti alloy	PBF (SLM and EBM)	[29–37]
Ti-6.5Al-3.5Mo-1.5Zr-0.3Si	Ti & Ti alloy	PBF (EBM)	[59]
IN 718	Nickel super-alloy	DED (DMF), PBF (SLM)	[21–28]
$AlCoCrFeNi_{2.1}$	High-entropy alloy	DED (LENS)	[199]
CoCrFeNiMn	High-entropy alloy	PBF (SLM)	[223,257,258]
Pure copper	Copper	PBF (EBM)	[58]

and accumulation in Ti alloys arises at different stages of manufacturing (e.g. pickling treatment) and service (e.g. in a highly corrosive media). Hydrogen dissolves in the Ti lattice through penetration, causing significant changes in physical–chemical properties, microstructure [259], mechanical properties [211,260], and defect structure [261]. In many cases, the influence of hydrogen on Ti alloys can lead to a disastrous accident. The risk of hydrogen-induced failure is more hazardous due to the fact that it is difficult to detect the defects at the beginning of the process [262].

The production of hydrides is the most common cause of embrittlement in Ti alloys [210,211]. Hardie and Ouyang [263] reported that hydrogen absorption causes embrittlement, as evidenced by a decrease in elongation. In α + β Ti alloys with a high fraction of β phase, hydrogen tends to migrate within the β lattice and react with the α-phase along the α/β boundaries, leading to the formation of hydrides. This scenario exacerbates degradation, and the severity of embrittlement depends on the hydrogen pressure and the hydrogen mobility in the β phase [211]. The interaction of hydrogen with metals and the HE is influenced by a variety of parameters, such as environment, microstructure, chemical composition and phase composition [211,264–267].

In some cases, Ti alloys may benefit from hydrogen interaction [59]. A combination of reversible hydrogenation and thermal effects on hydrogenated Ti alloys is known as thermo-hydrogen processing (THP). The use of THP allows the microstructure to be refined, the processability to be improved and the mechanical properties of Ti alloys to be enhanced. [268–270]. Furthermore, Ti-H systems can be utilised as promising hydrogen storage materials [271,272].

The microstructure has a strong influence on the HE susceptibility of Ti alloys. Ti alloys with more α phase have better resistance to HE due to the close packed HCP lattice. In the α phase, the maximum solubility of hydrogen is 0.18 wt% at around 300 °C. Hydrogen solubility reduces with temperature, leading to the formation of hydrides. In the β phase of Ti, the hydrogen solubility is approximately 5 times higher than that in the α phase at the eutectoid temperature and achieves a maximum of 1.98 wt% at 640 °C. However, the hydrogen solubility at room temperature is still unknow because the embrittlement mechanism that operates in highly β stabilised alloys has not been determined [273,274].

Various AM technologies have been used to produce Ti alloy samples or/and engineering parts [59,275,276]. Actually, Ti alloy is the mostly studied metal for AM. Dutta and Froes [276] reviewed Ti alloys produced by different AM methods, including DED (e.g., DMD and LENS) and PBF (e.g., SLM/DMLS and EBM). The AM products exhibited comparable performance with the cast and wrought products, as presented in Fig. 13. The HE behaviours of various AM Ti alloys have also been studied and reported [29–32,34,205,206,262]. The following section discusses HE of the most typical AM Ti alloys.

4.1.1. Ti-6Al-4V

Ti-6Al-4 V alloy is the working horse of Ti alloys. But, Ti-6Al-4 V alloy has low machinability because of poor thermal conductivity [277], high chemical reactivity to oxygen [278], and proclivity for strain hardening [279,280]. Traditional methods for producing Ti-6Al-4 V parts are forging, casting and rolling, followed by machining to final shape and dimension. These methods invariably result in significant material waste, expensive manufacturing costs, and long lead times. [281]. In such cases, AM is beneficial to production of Ti-6Al-4 V components with geometrical complexity [83,281,282].

Due to the fast cooling, SLM Ti-6Al-4 V alloy generally has an acicular α'-martensite phase, but EBM Ti-6Al-4 V, which is associated with a relatively slow cooling, is characterised by prior β grains with a transformed α/β microstructure, as shown in Fig. 14 [34,83]. Moreover, Fig. 15 presents micrographs of hydrogenated SLM and EBM Ti-6Al-4 V [34]. The interaction of hydrogen with defects of Ti-6Al-4 V manufactured by different AM methods is expected to be varying. A few papers have been published [29–32,34].

Metalnikov et al. [31,34] observed hydrogen-assisted cracking (HAC) in EBM Ti-6Al-4 V with fine duplex α + β phases. They suggested that EBM Ti-6Al-4 V had better resistance to HAC in electrochemical and gaseous environments than the conventionally manufactured wrought alloys because EBM Ti-6Al-4 V has a lower amount of discontinuous β-phase, absorbing less hydrogen than the wrought alloy. They also reported that, compared with EBM Ti-6Al-4 V, the SLM alloy demonstrated a higher trend of hydrogen absorption and hydride precipitation because of the α'-martensite microstructure. This means that Ti-6Al-4 V processed by SLM is more susceptible to hydrogen than that processed by EBM [34]. Fig. 16 shows a series of thermal desorption spectroscopy (TDS) results, showing the desorption peaks in hydrogenated SLM and EBM Ti-6Al-4 V. There were three desorption peaks in the spectra for the SLM samples, which implied the presence of three different hydrogen traps. The first desorption peak appeared at low temperatures. The

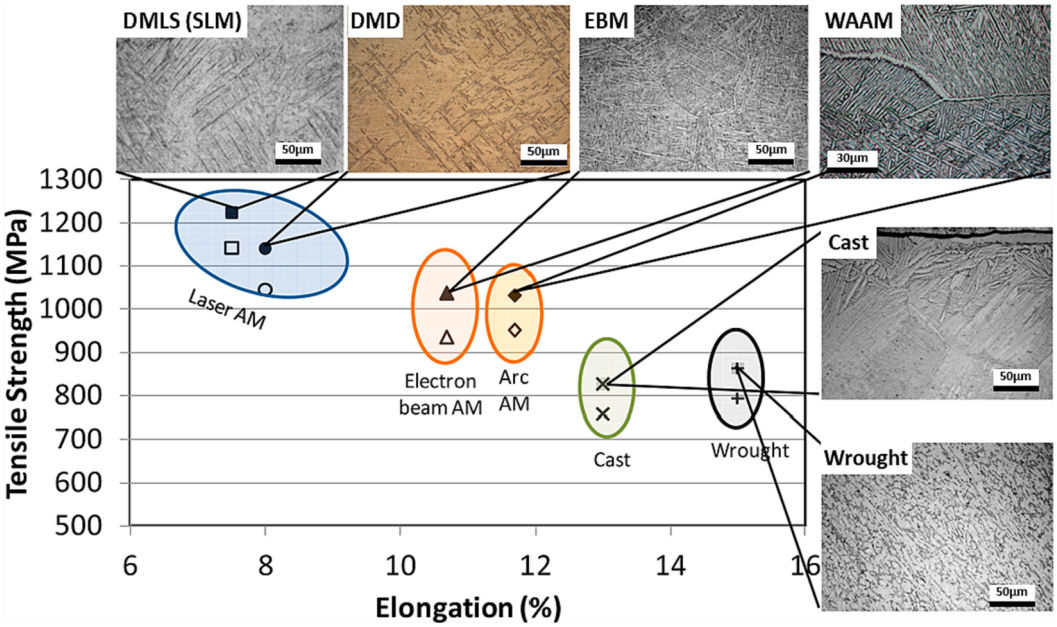

Fig. 13. Tensile behaviour comparison of AM, cast and wrought Ti-6Al-4 V at room temperature. Reproduced from Dutta [276] with permission from Elsevier.

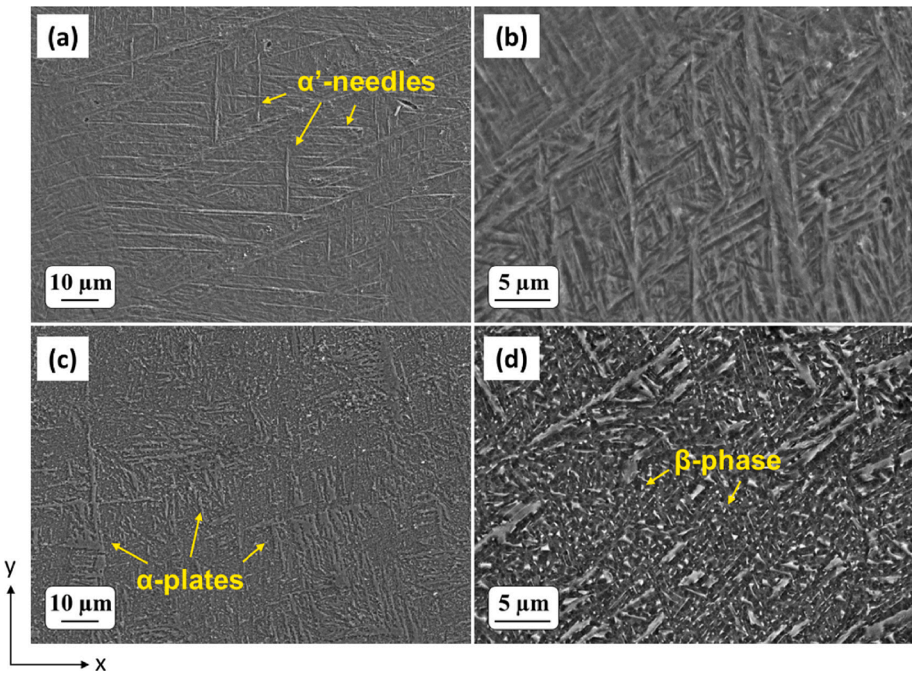

Fig. 14. SEM images showing the microstructure of Ti–6Al–4 V alloy processed by (a & c) SLM and (b & d) EBM. Reproduced from Metalnikov [34] with permission from Elsevier.

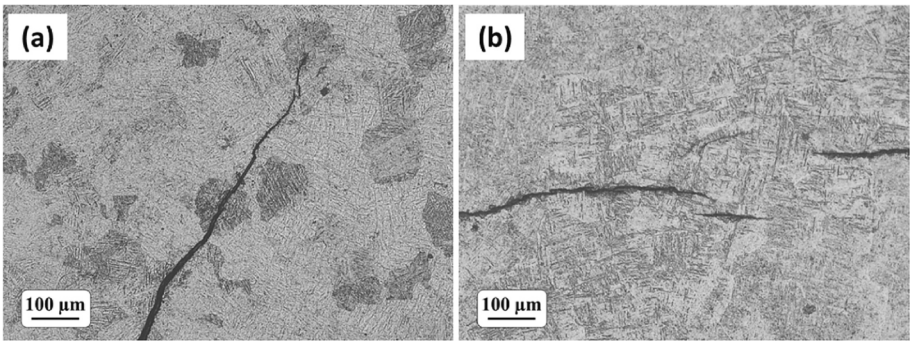

Fig. 15. Micrographs of electrochemical hydrogen charged (a) SLM and (b) EBM Ti–6Al–4 V. Reproduced from Metalnikov [34] with permission from Elsevier.

activation energy was 31 ± 5 kJ/mol, which indicated a reversible trap that could be assigned to dislocations (~30 kJ/mol) [32,283]. The second hydrogen trap was also reversible and could be assigned to GBs (~50 kJ/mol) [31,32]. The third peak appeared at high temperatures and was characterised by the activation energy of 85 ± 9 kJ/mol, which indicated a strong irreversible trap with Ti hydrides (approx. 90 ~ 125 kJ/mol) [283,284]. In contrast, the EBM sample had only two peaks, which were associated with two hydrogen traps: (i) a relatively low activation energy (53 ± 3 kJ/mol) at low temperatures, which corresponded to GB reversible trapping, and (ii) a high activation energy (88 ± 5 kJ/mol) at higher temperatures, which indicates irreversible trapping by Ti hydrides. Thus, based on the TDS spectra in Fig. 14, the main trap was irreversible in the SLM Ti-6Al-4 V, whereas in the EBM Ti–6Al–4 V, it was reversible [34].

In contrast, Navi et al. [32] reported that the HE susceptibility of EBM Ti-6Al-4 V was greater than that of the wrought alloy because of the larger amount of α/β interfaces, and the discontinuous β-phase particles along the short-horizontal direction in the EBM Ti-6Al-4 V. Both factors promoted the formation of hydrides and microvoids, inducing HAC along the interphase boundaries. Fig. 17 presents the time-of-flight secondary ion mass spectrometry image mapping of the total hydrogen intensity and the Ti-hydride intensity in wrought and EBM samples. The brighter areas indicate a higher concentration of hydrogen, and the dark areas were related to the less hydrogenated α phase. The hydrides were aligned parallel to the interphase boundaries, and the higher structural complexity with more Ti hydrides was present in the EBM alloy. In addition, the discontinuous β-phase also resulted in faster hydrogen desorption in the EBM Ti-6Al-4 V than the wrought alloy because the BCC β-phase was associated with the higher solute oxygen content (see Fig. 18).

The contradiatory conclusion between Matalninkov et al. and Navi et al.'s might be attributed to Metalnikov et al.'s [34] underestimation of the influence of hydrides and microvoid formation on the HE susceptibility. Metalnikov et al. only compared the microstructure difference between wrought and EBM Ti-6Al-4 V, without other direct comparative experimental results. In addition, the TDS spectra in these two studies were different. Only one H-trap was identified in the TPD/TDS spectra for the EBM sample in Navi et al.'s [32] work. This was probably attributed to the difference in heating rate or/and different instruments used because Navi and co-workers used their home made system rather than a commercially available machine. Thus, further studies are necessary to clarify the accuracy of these data and conclusions.

Additionally, Silverstein and Eliezer [30] reported that the susceptibility of AM Ti-6Al-4 V to HE strongly depended on the printing

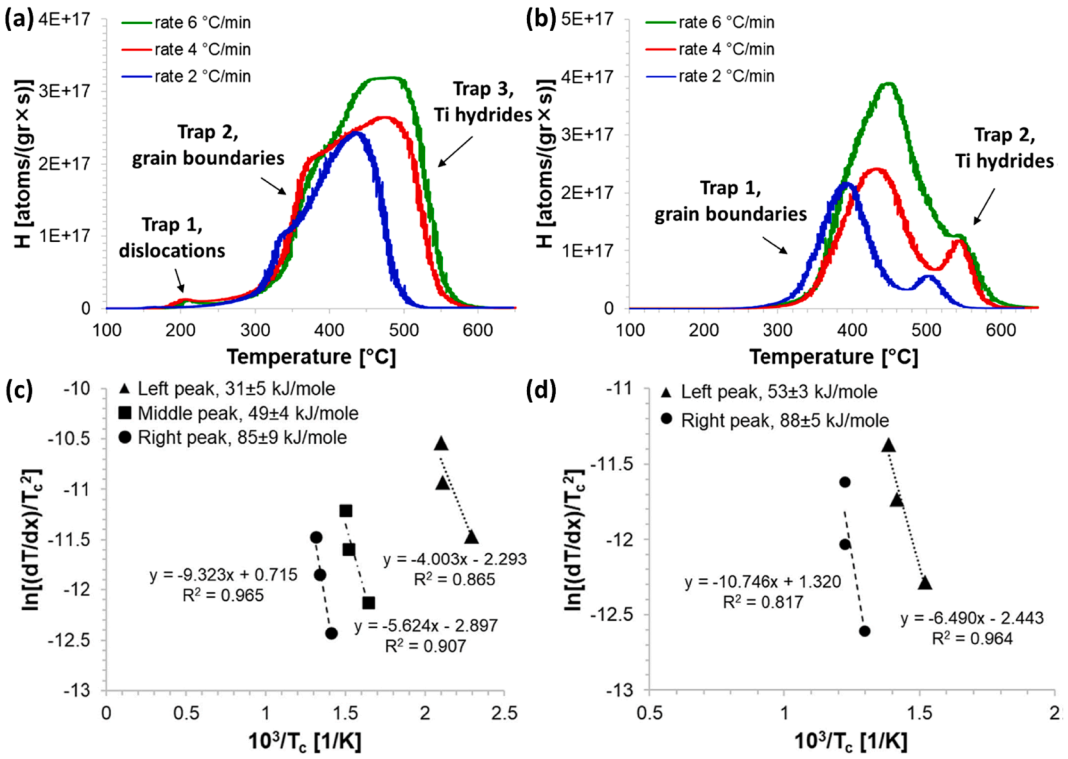

Fig. 16. (a & b) TDS spectra of electrochemical hydrogen charged and (c & d) determination of the activation energies of each hydrogen desorption peaks (a & c) SLM and (b & d) EBM Ti–6Al–4 V. Reproduced from Metalnikov [34] with permission from Elsevier.

direction. The vertical direction (SLM1) had a lower resistance to HE compared with the horizontal direction (SLM2), as shown in Fig. 19. The higher HE susceptibility of the SLM1 sample was attributed to the higher amount of Ti-hydrides as indicated in Fig. 20 and a lower trapping energy of TiH (Fig. 21). In contrast, the SLM2 printing direction may have created a high-stress intensity environment that reduced the probability of the stress-induced hydride mechanism (see Fig. 22, Fig. 23, Fig. 24).

4.1.2. Ti-6.5Al-3.5Mo-1.5Zr-0.3Si

Ti-6.5Al-3.5Mo-1.5Zr-0.3Si is another typical α + β Ti alloy that is used in compressor disks and blades of jet engines because of high creep resistance, and thermal stability at high temperatures [59,285]. There are two studies on the hydrogenation of this alloy manufactured using AM [59,286].

Stepanova et al. [59] investigated the HE of Ti-6.5Al-3.5Mo-1.5Zr-0.3Si processed by EBM and found that the microstructure has a substantial impact on the hydrogen sorption kinetics during gas-phase hydrogenation at 650 °C. The average rate of hydrogen sorption was higher in the EBM samples produced with lower electron beam current (3 mA), which corresponded to smaller melt pools [287], faster cooling and therefore finer secondary α precipices, than those produced with higher electron beam current (4 mA), which was associated with slower cooling and coarser α. Further increasing the electron beam current to 6 mA caused GB cracking due to high internal stresses in the material during solidification, which increased the rate of hydrogen sorption and the hydrogen [59]. In addition, no other research has been done to directly measure the HE of AM Ti-6.5Al-3.5Mo-1.5Zr-0.3Si.

Up to now, the research on HE of AM Ti alloys is limited. Although a few papers were published on Ti-6Al-4 V, inconsistent results were reported. Due to the more processing variables in AM compared with traditional manufacturing routes, the microstructure of AM alloys is more complicated, which leads to the complexity of HE behaviour of AM Ti alloys. Hence, to enable the wider applications of AM of Ti alloys, more and in-depth study on HE of AM Ti alloys needs to be done.

4.2. Nickel-based superalloys

Nickel-based (Ni-based) superalloys are commonly used at temperatures above 500 °C when high creep and/or fatigue resistance is required. These alloys usually contain up to 10 alloying elements, including light elements such as carbon or boron, aluminium and heavy refractory elements such as tantalum, rhenium or tungsten [288]. Although the alloys were designed for turbine engine application, they are also increasingly applied to other sectors, such as ultra-supercritical power plants, nuclear power plants and fossil fuel-fired diesel engines, and fuel cells. The applications are most prevalent over 750 °C because the characteristics of ferritic steels deteriorate dramatically above this temperature [289].

Due to the geometrical complexity of energy generation and aerospace components, AM of Ni-based superalloys has attracted increasing interest from scientists and engineers. At present, different AM techniques have been used to process both solid solution and precipitate-strengthened Ni-based superalloys [290]. Comprehensive details of Ni-based superalloys processed by AM, including SLM, EBM and DED, can be found in recently published review papers [81,84,291]. Table 5 summarises the reported commercial Ni-based superalloys processed through the powder-based fusion approach. Inconel 718 (also called Ni-based 718 or IN 718) is the mostly studied alloy compared with others as it is the most widely used [290].

Chemical, microstructure and property inhomogeneities are major concerns in AM Ni-based superalloys. Such inhomogeneities result from the rapid and repeated thermal gyrations during AM, which are related to non-equilibrium phase formation, spatially dependent microstructure gradients and multi-length scale microstructure inhomogeneities [292]. Although superalloys have good high-temperature oxidation and creep resistance and good corrosion resistance at room temperature, it does not mean that the alloys are immune to HE and/or water vapour effects in high-pressure hydrogen environments [293]. It has been found that HE occurs over a much wider temperature range in many superalloys. More significant HE has been evidenced at substantially higher

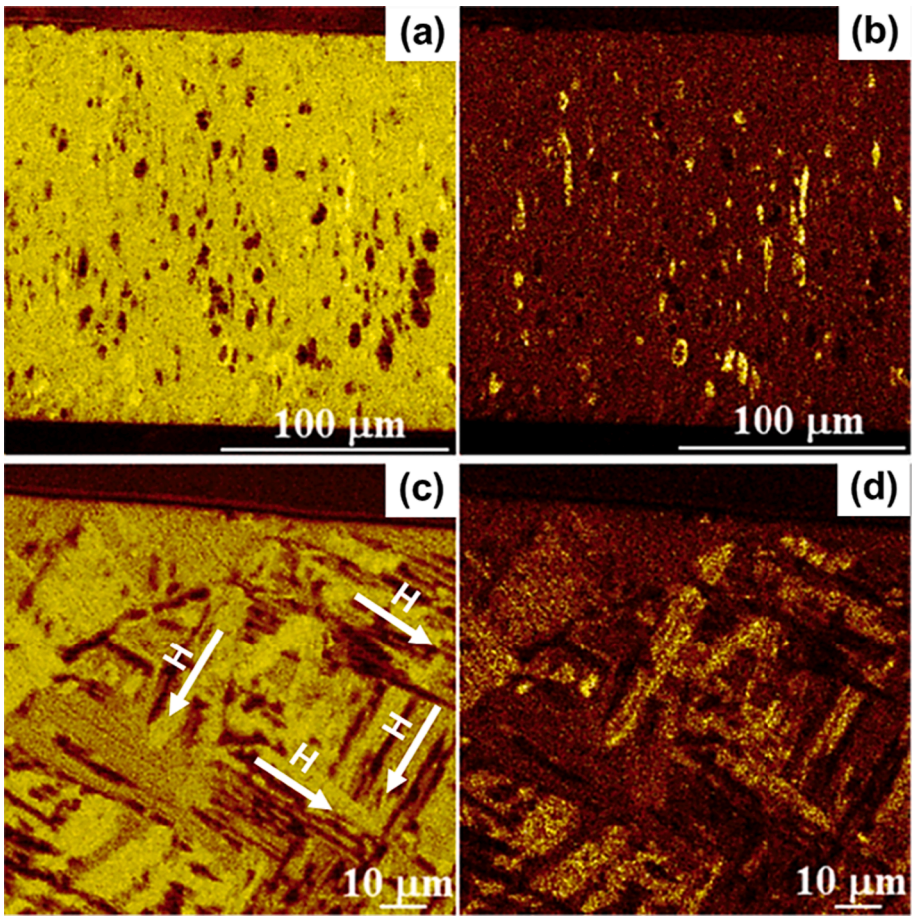

Fig. 17. Time-of-flight secondary ion mass spectrometry image mapping of (a & c) the total hydrogen intensity and (b & d) the Ti-hydride intensity in (a & b) wrought and (c & d) EBM samples. (a & b) Reveal the whole cross-section, whereas (c & d) reveal the upper side of the cross-section. Brighter areas indicate higher concentration of hydrogen. Arrows mark the hydrides aligning parallel to the interphase boundaries. Reproduced from Navi [32] with permission from Elsevier.

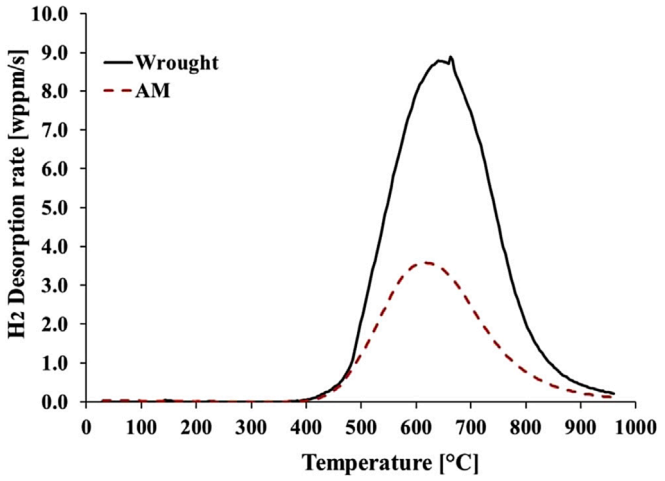

Fig. 18. TDS spectra for wrought and EBM specimens precharged with hydrogen for four days. The spectra were measured at a heating rate of 15 °C/min. Reproduced from Navi [32] with permission from Elsevier.

temperatures compared with lower temperatures. This is different from steels, which are more sensitive to HE at relatively lower temperatures ranging from −200 °C to 200 °C. Hence, HE is an important factor to consider for safe design in a hydrogen environment since most superalloys are often used at high temperatures [294]. Thus, HE study of the Ni-based alloys, including fabricating by AM and conventional methods, is necessary [21–23,26,185,295,296].

Precipitation-hardenable IN 718 has high strength and is consequently hard to machine [290,297]. Hence, AM has received a lot of attention for fabricating IN 718 components in the last decade [26,298]. At present, AM of IN 718 alloy has been well established, and high-quality AM IN 718 parts can be produced with SLM [151,299] and EBM [154,300]. Hosseini and Popovich [301] reviewed the microstructural features, tensile strength, hardness, fatigue strength, and high-temperature creep behaviour of IN 718 alloy fabricatred with different AM techniques, different processing and post-processing conditions.

The HE susceptibility of IN 718 alloy produced with conventional processing has been studied by several researchers. Hirose et al. [302] reported that IN 718 alloy has a high sensitivity to HE, with hydrogen-induced cracking predominantly occurring at stress-concentrated regions on the surface of the alloy. They noted that reducing the local hardness of these surface regions can effectively prevent such cracking. It has been verified that the laser surface solutionizing technique is effective in reducing the HE susceptibility of IN 718. Tarzimoghadam et al. [184] studied the effects of γ and δ phases on the HE sensitivity of IN 718. They found that rapid and preferential desorption of diffusible hydrogen occurs in the γ-matrix, while δ/matrix interface decohesion serves as the primary damage mechanism in the presence of hydrogen. The role of the δ-phase in HE is attributed to hydrogen accumulation at the δ/matrix interfaces during loading, resulting in reduced bonding strength and concurrent decohesion upon straining. Furthermore, Li et al. [216] investigated the HE behaviour of IN 718 at room temperature, and reported that hydrogen charging deteriorates the mechanical

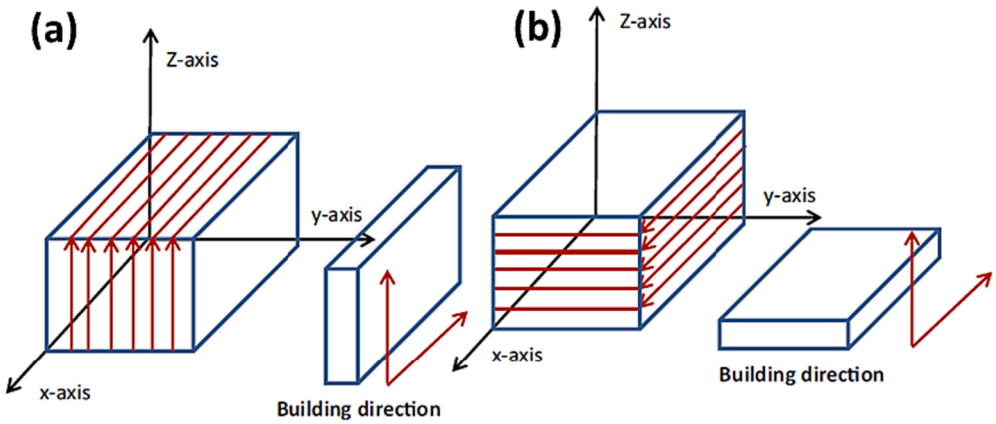

Fig. 19. Two different SLM-building directions: (a) SLM1 and (b) SLM2. Reproduced from Silverstein [30] with permission from Elsevier.

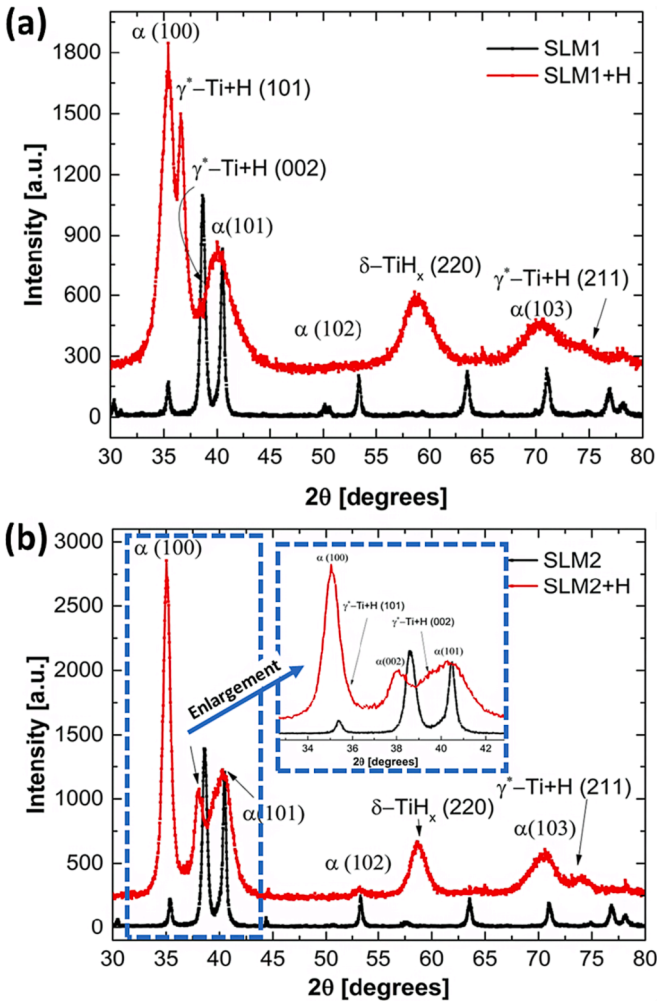

Fig. 20. X-ray diffractometer (XRD) pattern of Ti-6Al-4 V (a) SLM1 & (b) SLM2. Black: Uncharged & Red: gas phase hydrogen charged 10 h at 500 °C and 0.5 MPa. Reproduced from Silverstein [30] with permission from Elsevier. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

properties, as shown in Fig. 25 (a & b). They observed a partial Portevin-Le Chatelier (PLC) effect for hydrogen charging current density of 220 mA/cm^2 and 590 mA/cm^2. PLC is a type of plastic instability manifested as stress serrations during deformation. Additionally, they detected plastic deformation features characterised by dislocation cells in the hydrogen-induced brittle zone [303].

However, fewer work has been done on the HE susceptibility of IN 718 alloy processed by AM and no review paper has been published. Kolesov and Alekseeva [22] first reported that hydrogen accumulated around GBs in an SLM IN 718 sample after hydrogen charging, which compensated for the effect of high micro stresses. X-ray diffraction spectra taken from the SLM IN 718 samples and traditional manufactured IN 718 indicated that hydrogen charging in the SLM samples partially eliminated the residual stress/strain as the H-charged peaks in these samples were narrower than the initial as-built SLM samples without H-charging. In contrast, H-charging of the conventionally produced samples led to slightly peak widening. This can be considered as the result of hydrogen dissolution in the lattice, which decreased the micro-strain.

Li et al. [23] reported that hydrogen reduced the UTS, YS, and elongation of AM IN 718 samples, including both as-built (as-deposited) samples along both perpendicular to and parallel with the building directions, and heat-treated horizontal samples. Fig. 26 (c & d) show the results of tensile properties originated from their stress–strain curves. For uncharged specimens, YS and UTS of the as-built horizontal plane (HP), as-built vertical plane (VP) and heat-treated-HP samples were 737 MPa and 1078 MPa, 678 MPa and 975 MPa, 1260 MPa and 1475 MPa, respectively. High YS and UTS of the heat-treated-HP samples were attributed to the formation of the γ' and γ'' strengthening precipitates [304,305]. After hydrogen charging, both YS and UTS in the as-built VP specimens were not greatly affected but reduced by approximately 100 MPa in the as-built HP samples and heat-treated samples.

Additionally, for comparison of the hydrogen-induced degradation in mechanical properties, Li and his co-workers [23] used the index of HE (*HEI*) which was calculated using Eq. 3 [306] to calculate the relative fracture strain loss of the specimens:

$$HEI = \frac{\varepsilon_{f(uncharged)} - \varepsilon_{f(charged)}}{\varepsilon_{f(uncharged)}} \quad (3)$$

where $\varepsilon_{f(uncharged)}$ and $\varepsilon_{f(charged)}$ are the fracture strain of hydrogen uncharged and hydrogen charged specimens, respectively.

The HEI value in the as-built VP specimens was 23%, which was the lowest compared to the as-built HP specimens at 27% and heat-treated HP specimens at 37%. This indicated that the as-built VP specimens exhibited the highest resistance to HE. The synergistic effect of HELP and HEDE were used to explain the cracking of the AM IN 718 samples in this study. The presence of high-density dislocations in the hydrogen-induced brittle zone of both the as-built and heat-treated samples indicates that plastic deformation is a contributing factor to HE failure, aligning with the HELP mechanism. Concurrently, in the as-built samples, γ-matrix/Laves phase interface decohesion, and in the heat-treated-HP samples, intergranular cracking, both support the HEDE mechanism.

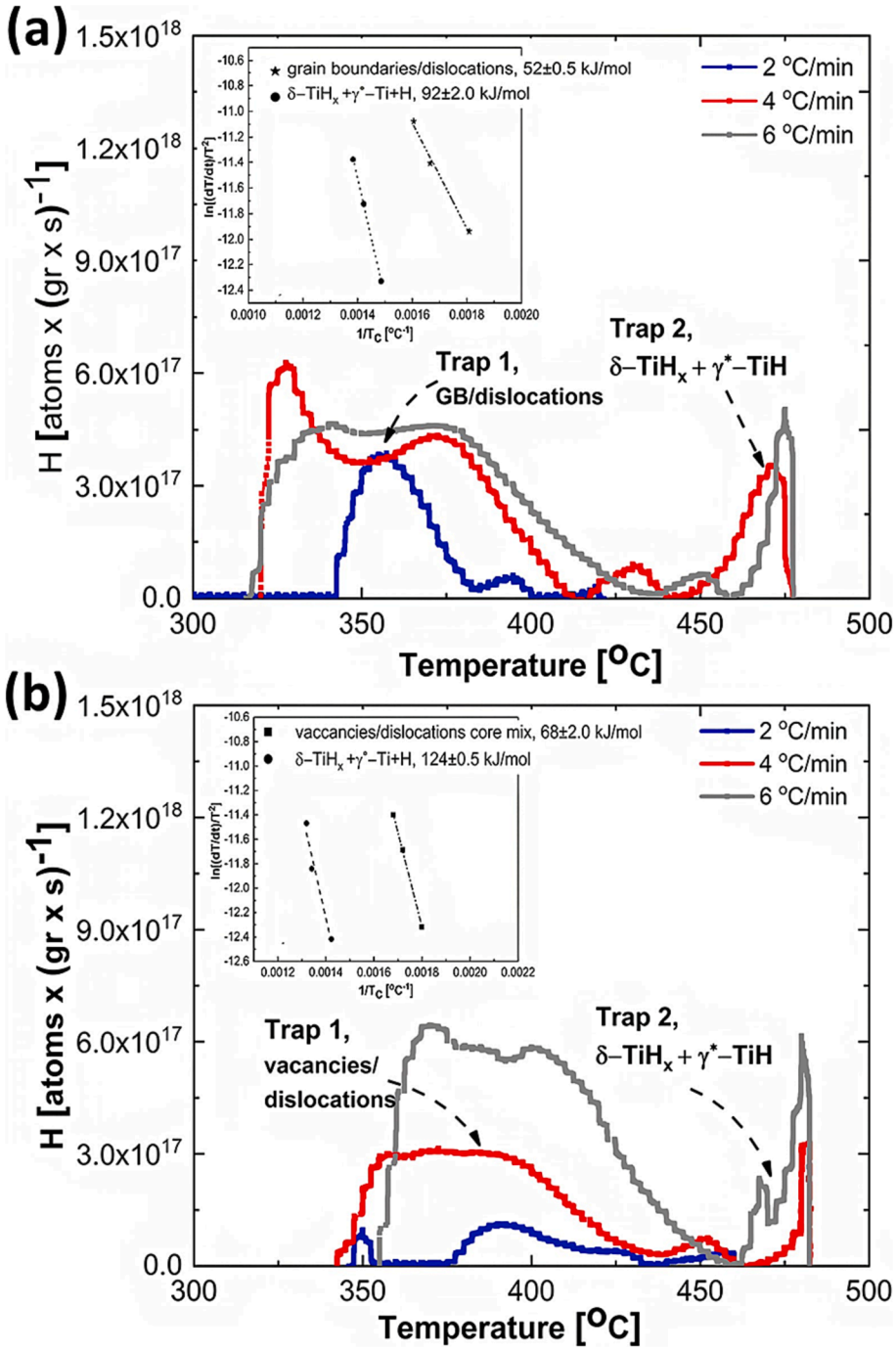

Fig. 21. Hydrogen thermal desorption curve and determination of the activation energies for two fitted hydrogen desorption peaks of the TDS spectra from hydrogenated Ti-6Al-4 V (a) SLM1 & (b) SLM2. Reproduced from Silverstein [30] with permission from Elsevier.

Fig. 27 shows that, for the as-built HP and VP samples, crack initiation sites were located at γ-matrix/Laves phase interfaces, but the crack propagation path in both as-built samples was different. The as-built HP samples had a zigzag crack propagation path, whereas the VP samples had a straight crack propagation path. In the as-built HP sample, the crack growth was in the form of decohesion along the interfaces. Because the intersection angle between the maximum shear stress direction and tensile direction was 45°, the cracks extended to the γ-matrix along a specific direction. In contrast, in the as-built VP sample, the long axis direction of Laves phases was almost vertical to the tensile direction, and this orientation was conducive to crack growth via void coalescence along long-chained γ-matrix/Laves phase interfaces, resulting in a straight cracking feature. Thus, the vertical samples had a stronger resistance to HE than the horizontal samples due to the suppression of hydrogen entrance by columnar dendrites. For the heat-treated HP samples, hydrogen-induced cracking initiated from γ-matrix/δ phase interfaces and propagated along GBs. As a result, the heat-treated horizontal samples had a higher susceptibility to HE due to their GB precipitates and high strength level. Hesketh et al. [24] reported similar results in the effect of printing direction to those of Li et al. [23].

Xu et al. [27] also investigated the HE behaviour of the as-built SLM and post-heat-treated IN 718 in terms of microstructural evolution and crack characterization using a low strain rate tensile (SSRT) tests under in situ electrochemical hydrogen charging. The solution treatment (ST)

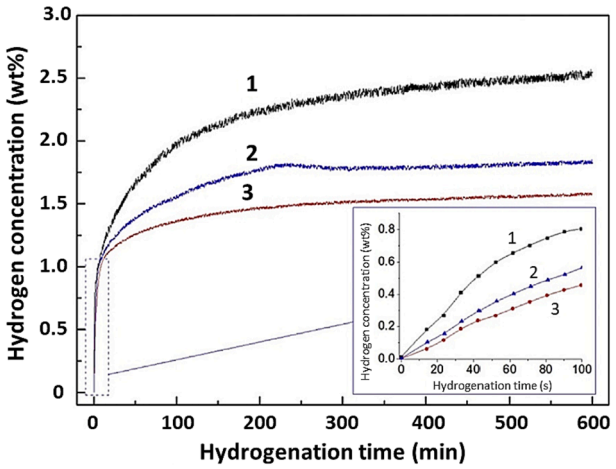

Fig. 22. The kinetic curves of hydrogen sorption of EBM Ti-6.5Al-3.5Mo-1.5Zr-0.3Si parts produced by EBM with different electron beam current: (1 – 3 mA, 2 – 6 mA & 3 – 4 mA). Reproduced from Stepanova [59] with permission from Elsevier.

and solution-aging treatment (SAT) were carried out as post-heat treatments. There was a substantial impact of post-heat treatment on the hydrogen content of the SLM IN 718. The hydrogen atoms preferred to diffuse in the ST samples compared to the as-built or SA samples due to the higher hydrogen desorption rate (as shown in Fig. 28 (a)), which revealed that the ST samples had the lowest HE susceptibility. Fig. 28 (b - d) show the SSRT test results obtained in air and under in-situ hydrogen charging. The results indicate that post-heat treatment significantly affected the tensile properties of both conditions. Compared with the as-built SLM samples, the ST process decreased the fracture strength from 1053 to 904 MPa in air and 858 to 679 MPa in hydrogen, while the SAT process increased the fracture strength to 1357 and 1154 MPa in air and hydrogen, respectively. Li et al. [23] ascribed the improvement in strength to the precipitation strengthening of γ'/γ'' phases [304,305].

To understand the hydrogen-induced degradation of the tensile properties, Xu and his co-workers [27] used I_{HE} susceptibility index which was calculated using Eq. 4 [307] to evaluate the effect of hydrogen on strength and plasticity.

$$I_{HE} = 1 - \frac{\sigma_{fH}(1+\delta_{fH})}{\sigma_{fa}(1+\delta_{fa})} \tag{5}$$

where σ_{fa} and σ_{fH} are the fracture strengths obtained in air and hydrogen, respectively, and δ_{fa} and δ_{fH} are the uniform plasticity strains obtained in air and hydrogen, respectively.

The SAT samples had the largest I_{HE} index of 0.87, followed by the as-built SLM and ST samples, which indicated that the SAT samples had the highest HE susceptibility. This was attributed to the high dislocation density and local strain along the cellular structure walls in the as-built SLM sample, which increased the local hydrogen concentration and resulted in cracking along the cell walls. The ST process led to the dissolution of the Laves phase and a reduction in the dislocation density, which slowed down the diffusion of hydrogen along the cell walls. Subsequent aging of the SAT samples caused precipitation of the γ', γ'' and δ phases. The formation of the δ phase along the GBs and the cell walls led to δ/γ interface cracking. Consequently, the SAT samples exhibited the highest strength as well as the highest susceptibility to HE among these samples [27].

However, Yoo et al. [25] proposed that if the temperature of heat treatment was sufficiently high, the hydrogen diffusivity of LPBF IN 718 was comparable to that of conventional IN 718. At the sufficiently high temperature of heat-treatment conditions, continuous γ'' precipitation along the sub-GB can reduce the density of vacancies and dislocations, weakening the role of grain/sub-GBs in hydrogen transport as well as in hydrogen trapping.

Additionally, Kevinsannya et al. [26] discovered that the fatigue limit and fatigue life of the AM IN 718 was not influenced by hydrogen, which was similar to the observation for forged IN 718, because hydrogen had little influence on the crack growth threshold. However, the mechanism for determining whether hydrogen deteriorates the crack growth threshold in IN 718 has not yet been clarified in their work.

Based on the current studies on HE of Ni-based superalloys fabricated by AM, two main HE mechanisms can be suggested, namely, HEDE and HELP [23,184–188,308]. There is a synergistic effect between these two mechanisms [23]. Many studies have also been conducted on the influence of printing direction [23,24] and post-heat treatment [23,25,27] on AM Ni-based alloys. However, the main knowledge gap on HE of AM Ni-based superalloys exists in the comparison of alloys fabricated by AM with those fabricated through conventional routes. Furthermore, the

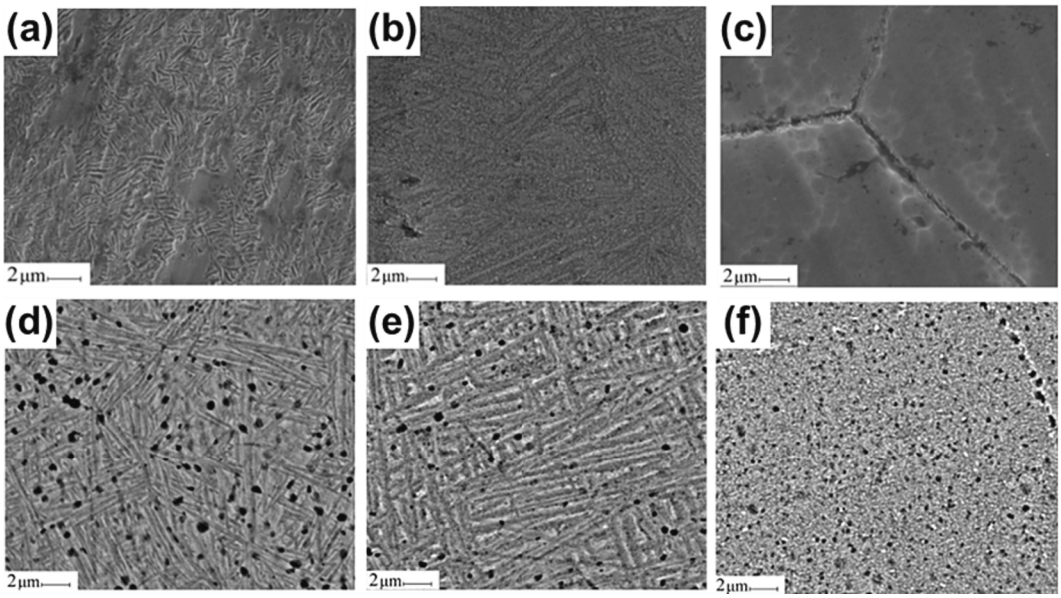

Fig. 23. SEM images of the surface of EBM Ti-6.5Al-3.5Mo-1.5Zr-0.3Si at different electron beam currents (a & d) 3 mA, (b & e) 4 mA, (c & f) 6 mA; (a-c) as-built (d-f) after hydrogenation. Reproduced from Stepanova [59] with permission from Elsevier.

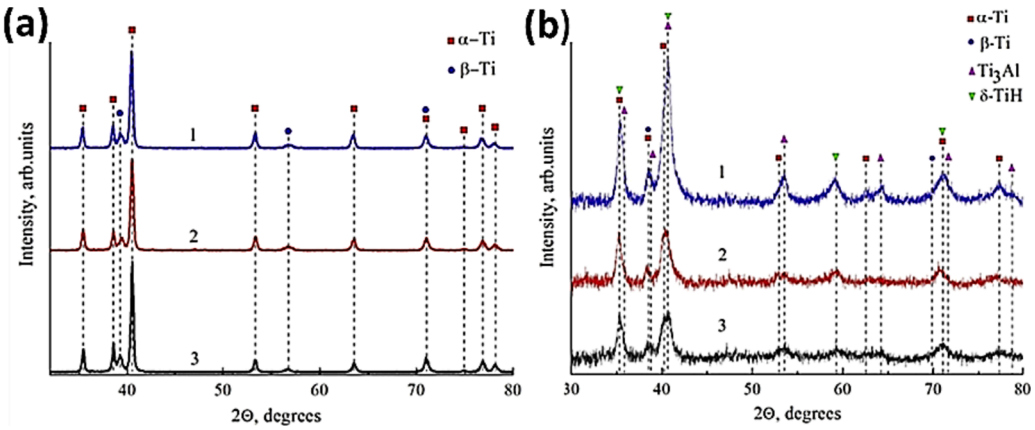

Fig. 24. Portions of diffraction patterns of the Ti-6.5Al-3.5Mo-1.5Zr-0.3Si samples produced by EBM a different electron beam current (1 – 6 mA, 2 – 4 mA & 3 – 3 mA) (a) before and (b) after hydrogenation. Reproduced from Stepanova [59] with permission from Elsevier.

Table 5
Summary of Ni-based superalloys processed through powder-based AM modalities [290].

Alloys	AM processes
Inconel 625	BJ, DED, EBM, SLM
Hastelloy X	EBM, SLM
Inconel 718	BJ, DED, EBM, SLM
Haynes 282	DED, EBM
Inconel 738	SLM, EBM
Mar-M247/CM274 * related to chemistry	SLM, EBM
Rene 142	EBM
Rene N5	EBM
CMSX-4	DED, EBM

current studies are limited to the IN 718 alloy, and the HE susceptibility of other AM Ni-based superalloys is still unknown.

4.3. Stainless steels

Stainless steel (SS) contains at least 12% chromium to ensure high corrosion resistance [309]. SS is often used for applications in environments that may contain hydrogen or moisture. Thus, HE of this type of steels is often a concern. There are several comprehensive books and reviews on the HE of SSs [310–312].

Most SSs have good AM processability, and therefore AM of SS is an active topic. Table 6 lists typical SSs that are commonly processable with AM. There are a few integrated reviews summarizing the research progress in this field [8,313–316]. In addition, Laleh et al. [317] compared the mechanisms of corrosion behaviour of AM SSs with their conventionally produced counterparts, including the deleterious effect of hydrogen. AM SS products generally have finer microstructure

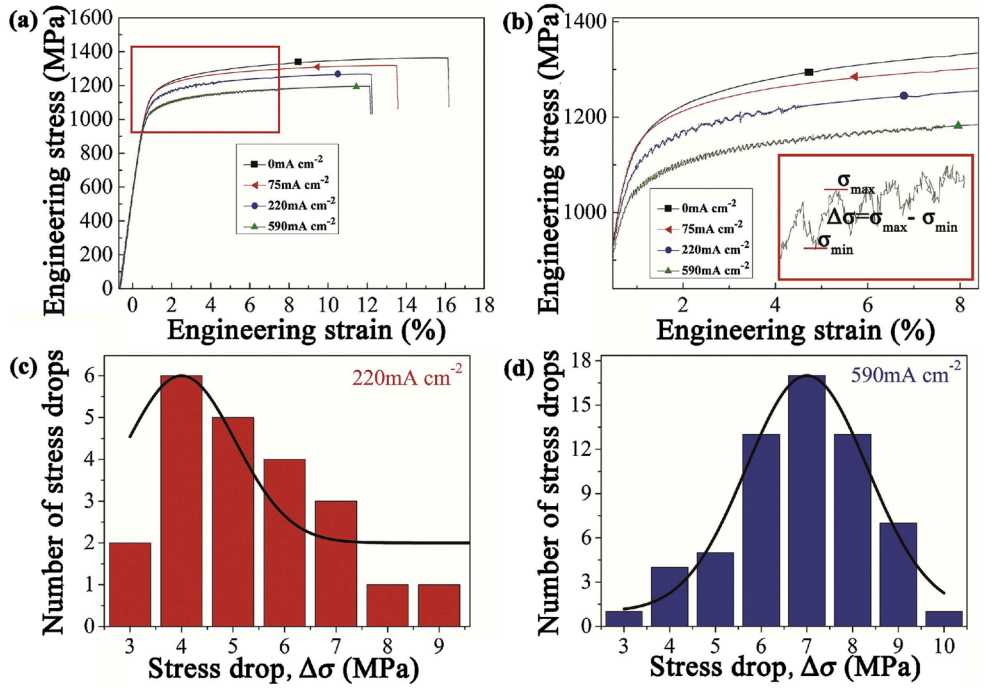

Fig. 25. (a) Engineering stress–strain curves of hydrogen free and hydrogen charged samples; (b) High magnification image of red rectangle region in (a). Reproduced from Li [216] with permission from Elsevier. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

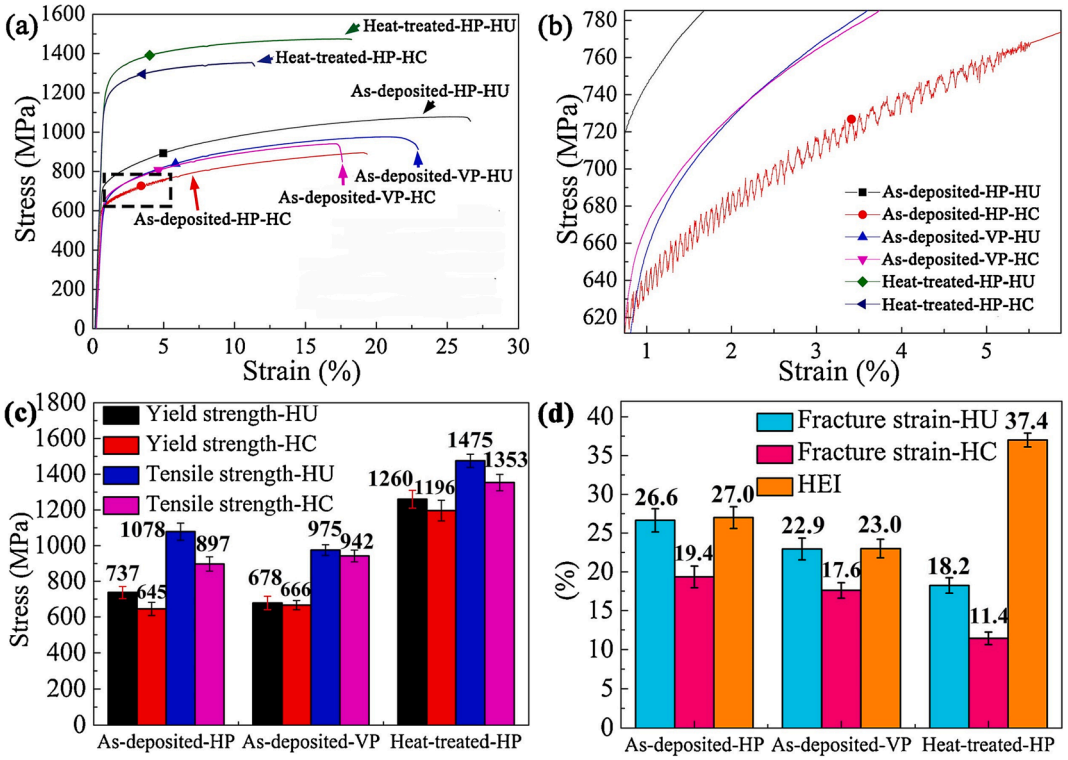

Fig. 26. (a) Stress–strain curves of IN 718 alloy with and without hydrogen; (b) high magnification image of signed region in (a); (c) YS and UTS of samples; (d) fracture strain and HEI of samples. HU: hydrogen uncharged; HC: hydrogen charged. Reproduced from Li [23] with permission from Elsevier.

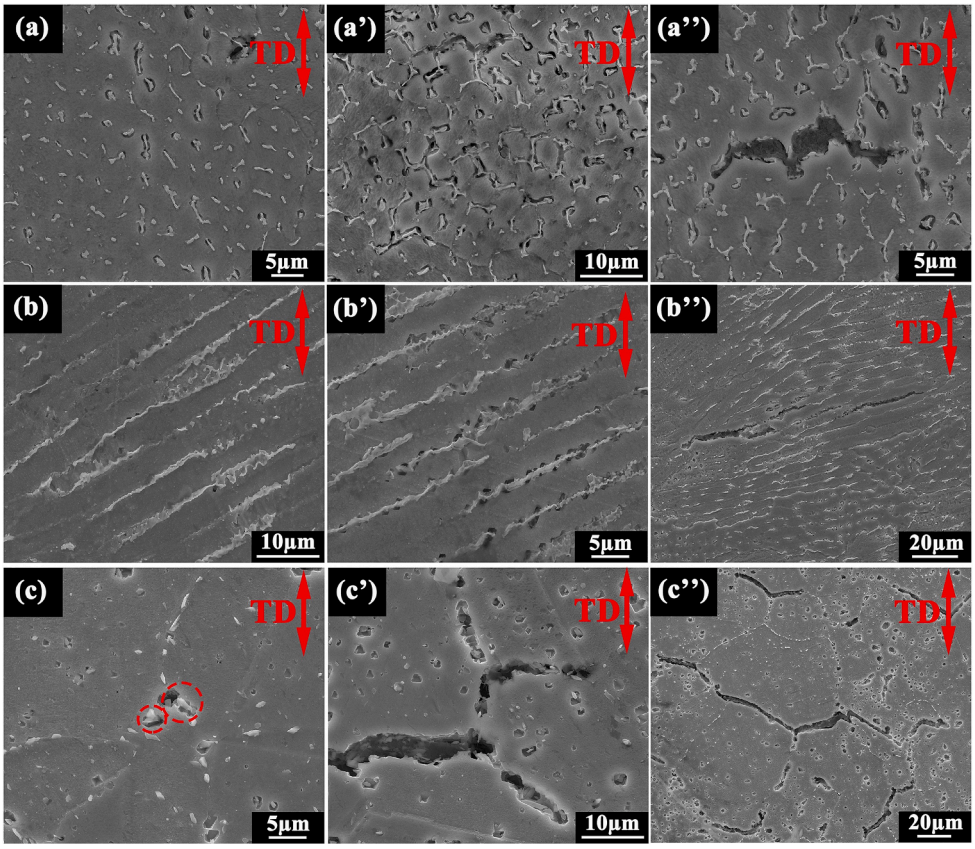

Fig. 27. Hydrogen-assisted cracking of fractured samples along lateral surfaces (a-a'') as-built HP sample; (b-b'') as-built VP sample; (c-c'') heat-treated-HP sample (a) crack initiation at the matrix/discrete Laves phase interfaces; (a') crack growth along matrix/Laves phase interfaces; (a'') cracks propagation through the γ-matrix; (b) crack initiation at matrix/long-chained Laves phase interfaces; (b') crack growth; (b'') cracks propagation through the coalescence of voids along matrix/Laves interfaces; (c) crack initiation at GBs; (c') crack growth along GBs; (c'') cracks propagation along GBs. TD: tensile direction. Reproduced from Li [23] with permission from Elsevier.

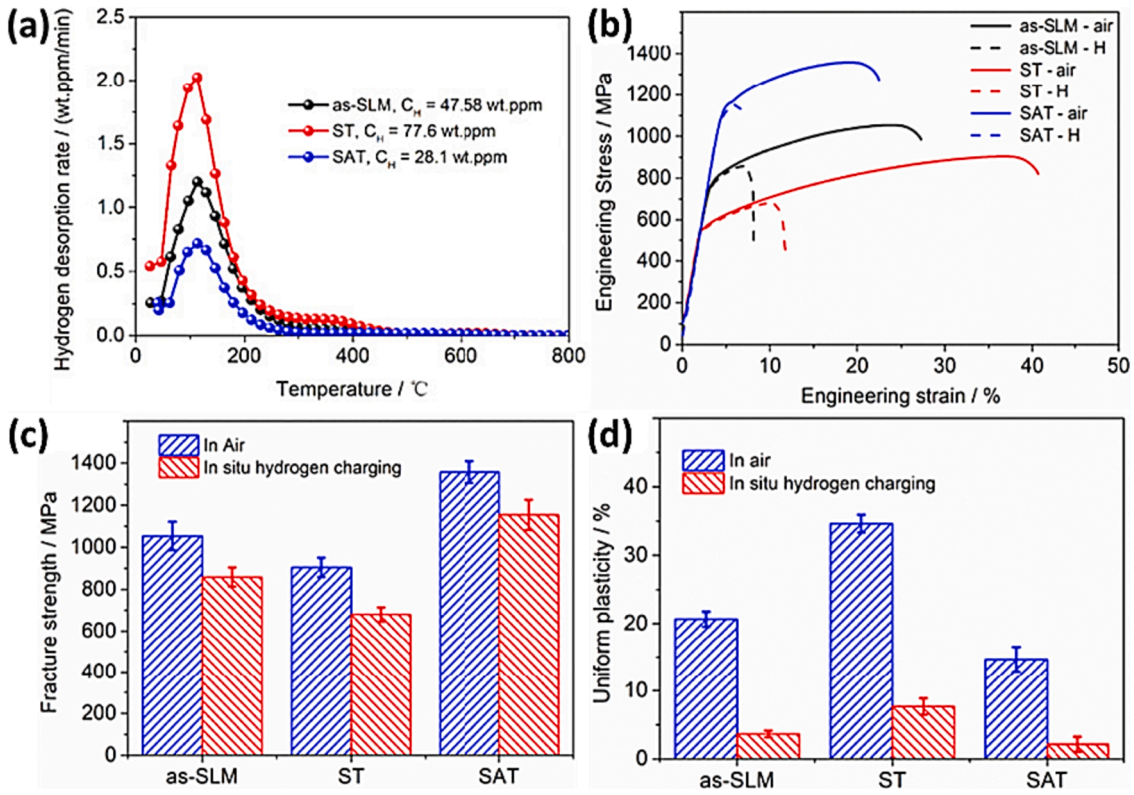

Fig. 28. (a) Hydrogen desorption curves – pre-charged at 5 mA/cm^2 for 24 h, (b) engineering stress–strain curves, (c) fracture strength, and (d) uniform plasticity for various processed IN 718. as-SLM: as-built SLM, ST: solution treatment, and SAT: solution aging treatment. [27].

Table 6
Summary of stainless steels processed using powder-based AM [8,316].

Alloys	Microstructure	Type	AM processes
304L	Austenite	Austenitic SS	DED, PBF
316L	Austenite	Austenitic SS	DED, PBF
17-4PH	Martensite	PH SS	PBF
AISI 420	Martensite	Martensitic SS	DED, PBF
AISI 431	Martensite	Martensitic SS	DED
AISI 441	Ferrite	Ferritic SS	PBF
2507	Duplex	Duplex SS	PBF
2205	Duplex	Duplex SS	PBF

because of the high cooling rate. But they are also highly anisotropic due to the formation of columnar grains, which differs from their wrought counterparts [95,318–321]. Thus, the HE susceptibility of AM SS is expected to be different from the wrought alloy [55]. Studies on HE of AM SS are limited. The existing studies have focused on austenitic SS and precipitation hardening (PH) SS.

4.3.1. *Austenitic stainless steels*

Austenitic stainless steel (ASS), particularly the AISI (American Iron and Steel Institute) 300 series, are the most common [322]. Hydrogen diffusivity in the FCC crystal structure of the ASS is lower than in the BCC lattice, which makes austenitic SS potentially more resistant to HE than ferritic stainless steels, particularly resistant to environment hydrogen embrittlement (EHE) [16]. Thus, ASS is commonly used to make parts exposed to gaseous hydrogen, such as gaseous hydrogen storage and gaseous hydrogen transportation applications [16,323–325]. Thermodynamically stable ASS, such as the AISI 316S, is less susceptible to HE even though severe HE has been reported in metastable ASSs, such as AISI 304 in which strain-induced α′ martensite forms during deformation [326–328]. The strain-induced α′ martensite provides a fast diffusion path for hydrogen, resulting in hydrogen enrichment at critical sites of the microstructure, such as the micro-mechanically stress concentrated regions in front of hetero-interfaces, and thus leading to H-assisted cracking [329,330]. For the AM ASS, the fast cooling may lead to solute segregation or the incompletion of the δ to γ transformation during AM. The small amount of δ ferrite retained in metastable ASSs may increase the HE susceptibility, providing crack initiation sites [16,331,332]. The current research on HE of AM ASS focuses on AISI 304L SS [13–16] and AISI 316L SS [4,17,54]. he reported results are discussed in the following sections.

4.3.1.1. AISI 304L. AISI 304, the most common ASS, has low cost, mechanical properties similar to construction steel, good corrosion resistance and high weldability [333]. The austenitic microstructure in 304L SS is subject to strain-induced transformation during service or forming due to the relatively low stacking fault energy [334]. This transformation is also dependent on microstructure, particularly texture [335] and grain size [336]. Highly dense AM 304L SS exhibits tensile and fatigue performance that are comparable to the wrought counterpart in both ambient and severe hydrogen environments. Generally, internal hydrogen causes tensile ductility loss in both PBF and DED-manufactured 304L SS. Furthermore, defects, and the micro-segregation in the AM parts can promote microcrack coalescence, leading to a more severe reduction in ductility [15].

In 2017, Baek et al. [13] first reported the dependency of HE resistance on AM processing of 304L SS. They compared the HE behaviour of 304L processed using direct metal tooling (DMT) with the rolled counterpart and found that the DMT processed 304L had higher HE resistance than the conventionally processed 304L. Fig. 29 shows that the elongation to fracture of the DMT 304L was similar in air (52%) and with hydrogen (50%). In contrast, the strain to failure of the rolled 304L was significantly reduced (68% reduced to 19%). In addition, the DMT specimens showed the same ductile fracture behaviour in both air (Fig. 30 (a)) and hydrogen (Fig. 30 (b)): ductile fracture with small

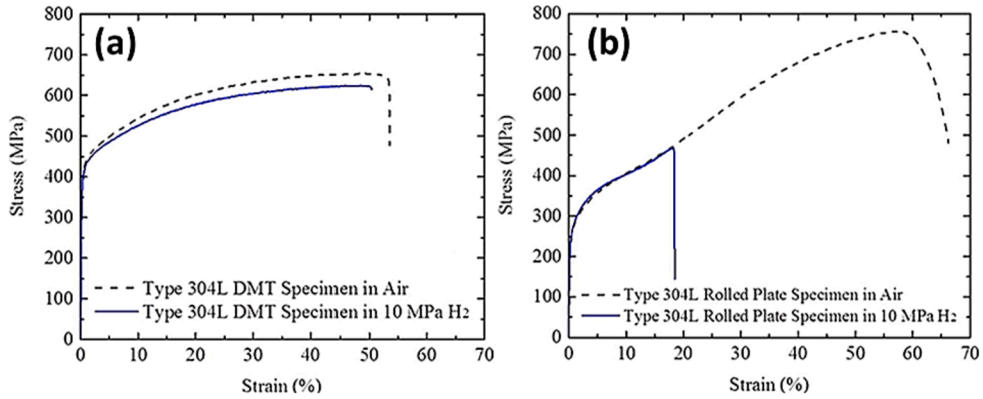

Fig. 29. Stress–strain curves of (a) type 304L STS DMT manufactured specimen, (b) type 304L STS rolled plate specimen. Reproduced from Baek [13] with permission from Elsevier.

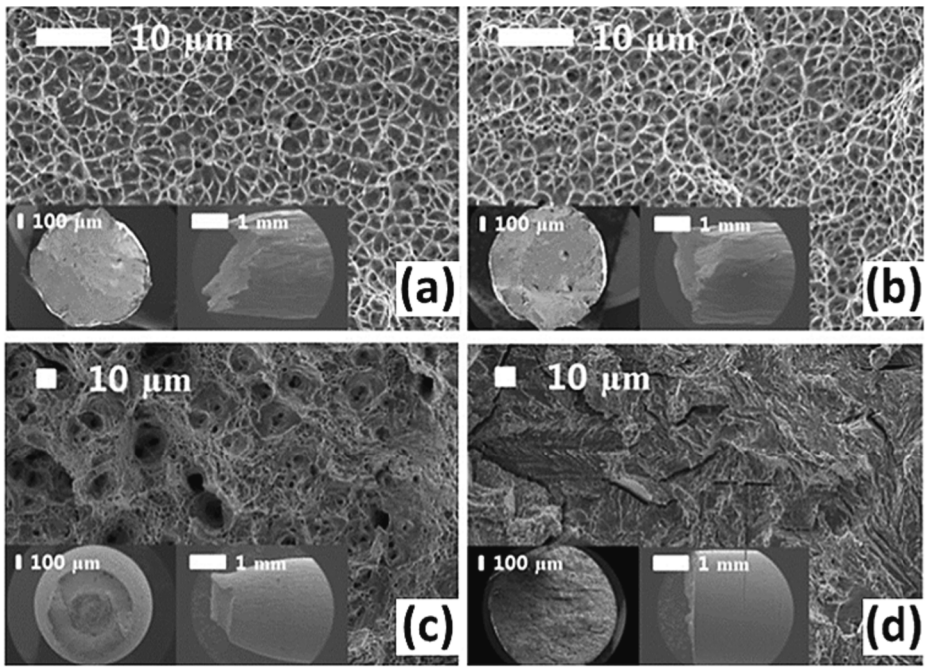

Fig. 30. Fractography of 304L SS specimens tested in different atmospheres (a) DMT manufactured specimen (air), (b) DMT manufactured specimen (10 MPa hydrogen), (c) rolled plate specimen (air), rolled plate specimen (10 MPa hydrogen). Reproduced from Baek [13] with permission from Elsevier.

uniform dimples and no evidence of cleavage or secondary cracks. The rolled 304L also exhibited ductile fracture in air, characterised by a cup-cone fracture mode (Fig. 30 (c)), but brittle cleavage fracture in hydrogen (Fig. 30 (d)). The HELP and HEDE mechanisms can be used to elucidate HE in this material. According to the HELP model, a locally elevated hydrogen concentration (such as the local tip) augments dislocation activity, resulting in localized stress concentration and the initiation of failure at planar defects [177,337]. Fig. 31 presents the EBSD images and phase maps of the fractured surfaces of both the DMT and rolled 304L. The strain-induced martensitic transformation did not occur in the DMT specimen in the gaseous hydrogen environment. In contrast, there was BCC strain-induced martensite in the rolled 304L. The finer microstructure of the DMT 304L might be responsible for the higher mechanical stability of the austenite. This study concluded that AM technology could be more suitable for creating 304L SS products for hydrogen usage and service [13].

Lee et al. [16] also investigated the microstructure and HE susceptibility of 304L SS manufactured using different processes. Fig. 32 shows the typical engineering stress- strain curves, the SEM images of the fracture surfaces (a-c), and the measured HE susceptibility of steel for each condition (d). The HE susceptibility of the 304L was closely related to the production route. For the casting and annealed (CA) 304L, H-charging resulted in a significant loss of ductility. For the LPBF and thermomechanical treatment (TMT) CA samples, hydrogen had a minor influence on ductility. EBSD analysis indicated that the volume fraction of strain-induced α' martensite was higher in the CA samples (~16%) than in the LPBF and CA-TMT samples (respectively ~ 3% and ~ 1%) at the same strain level of 60%. This suggests that the low resistance of the CA sample to HE could be attributed to the low local mechanical stability of the austenite. This instability resulted from chemical inhomogeneity in the austenite regions, notably around the retained δ zones in the initial state. The application of TMT after CA improved the chemical homogeneity of the microstructure. This effectively restored the stability of the austenite and consequently enhanced the HE resistance of the CA-TMT sample. In contrast, the LPBF sample exhibited a completely austenitic structure due to the high cooling rate, which also resulted in a finer and more homogenous microstructure with a distinctive solidification cellular structure. These features increased the

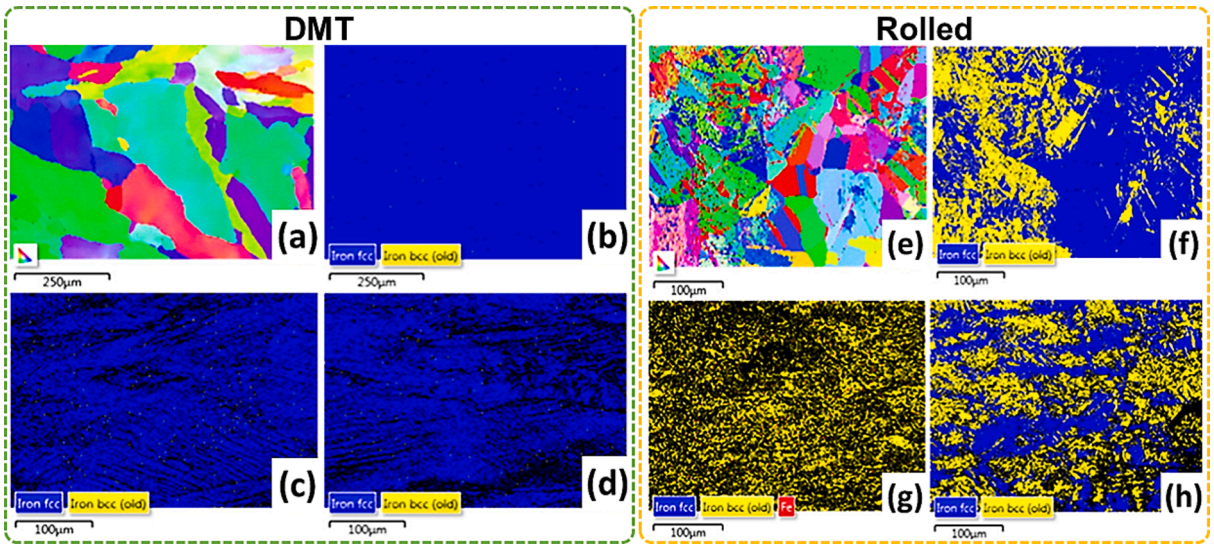

Fig. 31. EBSD images of fractured surface for DMT specimen (a –d) and rolled plate specimen (e –h); (a) DMT origin (inverse pole figure image), (b) DMT origin (phase map), (c) DMT specimen (air), (d) DMT specimen (hydrogen), (e) rolled (inverse pole figure image), (f) rolled plate origin (phase map), (g) rolled plate specimen (air), (h) rolled plate specimen (hydrogen). Reproduced from Baek [13] with permission from Elsevier.

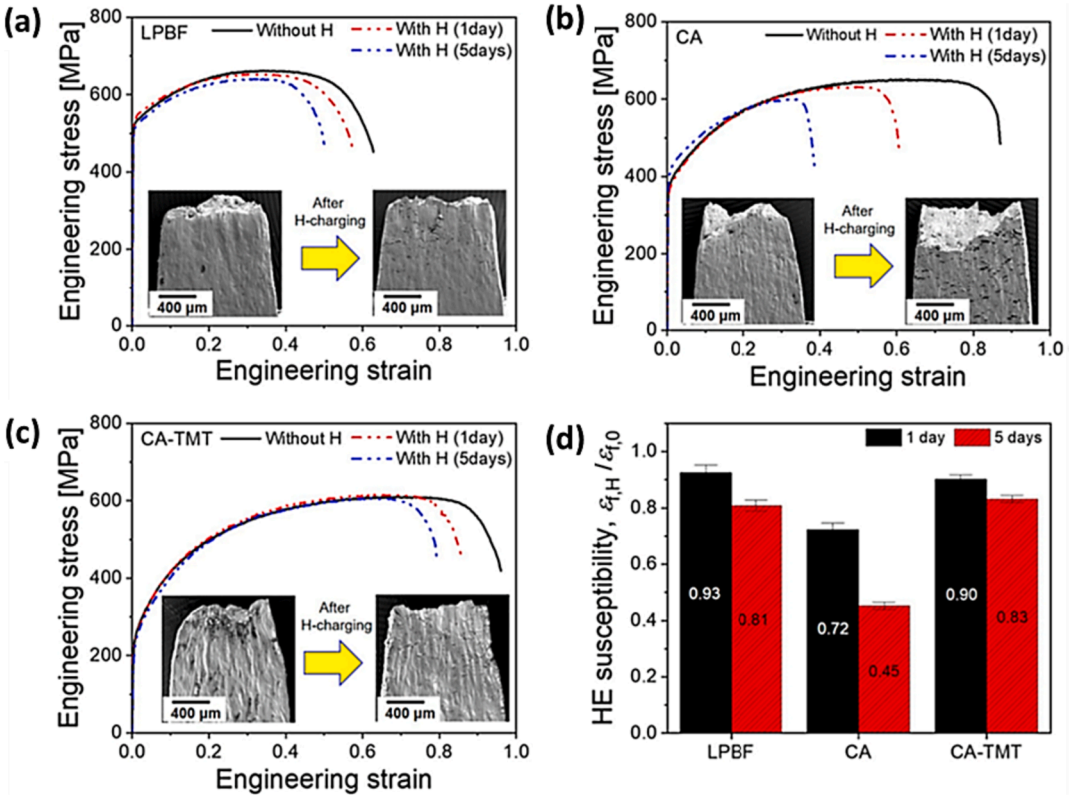

Fig. 32. Typical engineering stress-engineering strain curves of the (a) LPBF, (b) CA, and (c) CA-TMT samples with and without hydrogen. The SEM images show the surface of samples after tensile testing at the fracture site. (d) Quantification of HE susceptibility for each condition. Reproduced from Lee [16] with permission from Elsevier.

strength, stabilised the austenite, and enhanced the resistance to HE. Lee and co-workers [16] also found that the majority of the H-assisted cracks were initiated in the α′ martensite in all samples, further indicating the importance of strain-induced α′ martensitic transformation in the HE of 304L. Thus, the presence of δ and chemical inhomogeneity within the γ matrix were revealed to promote HE by causing the deformation-induced martensitic transformation and the concomitant hydrogen enrichment at the γ-α′ interface.

4.3.1.2. AISI 316L. 316L SS is commonly used in demanding applications like heat exchangers, jet engines, military applications, and biomedical services. But, 316L has limitations in some environments [338,339] due to its relatively low wear resistance and strength, particularly at elevated temperatures as SS components are susceptible

to oxidation, corrosion, carburization, sensitization, and HE [4,340]. AM can be used to fabricate 316L parts. The strength is typically higher because of the finer microstructure produced by AM processing [82]. The corrosion behaviour of the AM 316L SS has been well documented [4,17,54]. In contrast, there is less research on HE, although it is expected that the HE sensitivity of AM 316L SS would be different to 316L produced by conventional methods.

Miller et al. [17] investigated the HE of 316L fabricated by AM. Fig. 33 shows that the LPBF 316L had lower ductility (due to the high residual stress and higher porosity) and higher HE susceptibility than the thermomechanical (TM) 316L. The LPBF 316L showed lower strength and lower strain to fracture after exposure to the air for 2184 h, whereas, both strength and ductility were increased for the TM 316L. It was considered that the faster diffusion of hydrogen in the LPBF 316L was responsible for the HE. Nevertheless, it is hard to understand the increase in tensile properties of the TM 316L after exposure to air, and clarification work is needed (see Fig. 34).

Bertsch et al. [18] also studied the HE behaviour AM 316L produced using SLM and DED. Fig. 32 shows that hydrogen charging reduced the ductility of the as-built SLM and DED 316L. In contrast, post-processing heat treatment (HT) eliminated the hydrogen sensitivity of the SLM samples and weakened the HE sensitivity for the DED samples. The hydrogen sensitivity of the DED 316L seemed higher than for SLM 316L. It was considered that the HT reduced or even eliminated the AM defects such as porosity and residual stress, and therefore enhanced the HE-resistance. The smaller melt pools and faster cooling of the SLM process compared with the DED process may be responsible for the lower HE susceptivity of SLM 316L compared with the DED 316L.

Fig. 35 shows that the hydrogen charging did not change the microstructures of as-built SLM and DED 316L SS. However, the deformed microstructures varied with hydronation as shown in Fig. 36. The microstructure of the SLM 316L remained unchanging after deformation and hydrogen charging. In contrast, equiaxed dislocation cells were present during the later stages of the deformation of DED 316L and hydrogen charging promoted the development of texture within twin-free regions. This indicated that the DED 316L may be more sensitive to hydrogen. Hydrogen has the ability to be attracted to the stress fields surrounding dislocations due to the HELP mechanism, resulting in a decrease in the spacing between dislocations and an increase in their mobility [201]. Several studies have demonstrated that when materials are subjected to strain, hydrogen can expedite the formation of dislocation microstructures. This is accomplished by reducing the average size of dislocation cells in various materials such as Ni and Fe at a given level of strain [236,341,342].

The resulted published so far indicate that the HE susceptivity of AM 316L SS is closely related to the initial microstructure, such as the morphology and density of dislocations. While, the deformation twinning, dendritic segregation, and precipitates are considered to have little effect. Further detailed studies on HE in the as-built DED 316L is still needed to fully elucidate the mechanisms that are responsible for the embrittlement [18]. In addition, previous results suggested that SLM 316L SS can be a candidate for use in hydrogen environments because of its relatively high YS and resistance to HE.

4.3.2. *Precipitation-Hardening stainless steels*

Precipitation-hardening stainless steel (PHSS) has ultrahigh strength and good corrosion resistance [8] and is prevalently used in aerospace, oil and gas industries, chemical plants, and the marine industries. AM of PHSS has attracted significant interest from researchers [8,343], with greatest attention devoted to 17-4PH and 15-5PH [164,168,319,344–346]. Both steels are strengthened by precipitation of extensively dispersed Ni_3Al, Ni_3Ti, Ni_3Mo, B_2A Laves phase and other intermetallic precipitates in the martensite matrix [347,348]. 17-4PH SS contains 15 ~ 17.5 wt% Cr, 3 ~ 5 wt% Ni and 3 ~ 5 wt% Cu and is typically aged at 480 to 620 °C after solid solution heat treatment, which achieves a YS of 1370 MPa, a UTS of 1450 MPa and an elongation of 9% for the wrought alloy [168,347,349,350]. Research has indicated that the solid solution treated wrought 17-4PH steel has higher HE resistance than the peak-aged steel [351]. This indicates that the hydrogen sensitivity of this steel also depends on the microstructure.

Research has studied the susceptibility of AM 17-4PH to pitting, environmentally assisted cracking (EAC) [352,353], and stress corrosion cracking (SCC) [354]. Guennouni et al. [352] found that AM 17-4PH steel exhibited improved resistance to pit initiation due to the presence of finer and fewer NbC particles. However, they found that stable pits propagated faster for AM 17-4PH steel due to the larger amount of retained austenite. Stress was found to stabilize metastable pits, but to accelerate the propagation of stable pits, resulting in an increased susceptibility to EAC in AM 17-4PH. Shoemaker et al. [354] reported that the SCC performance of AM 17-4PH was consistently inferior to wrought steel, with crack growth rates up to 5 times higher. This degradation was attributed to an elevated oxygen content, porosity induced by argon processing, and fast cooling rate during AM processing.

Alnajjar et al. [55] comparative the HE of SLM and wrought 17-4PH. The SLM processed 17-4PH has a δ-ferritic microstructure with relatively fine grains (as shown in Fig. 37) due to the rapid cooling during

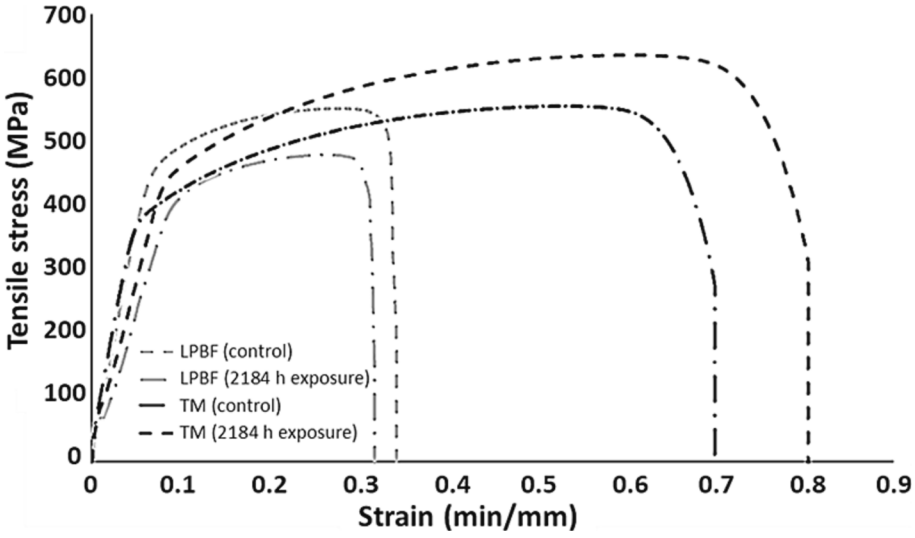

Fig. 33. Stress-Strain graphs for traditional manufacturing (TM) and LPBF samples before exposure (control) and after 2184 h of exposure. Reproduced from Miller [17] with permission from Elsevier.

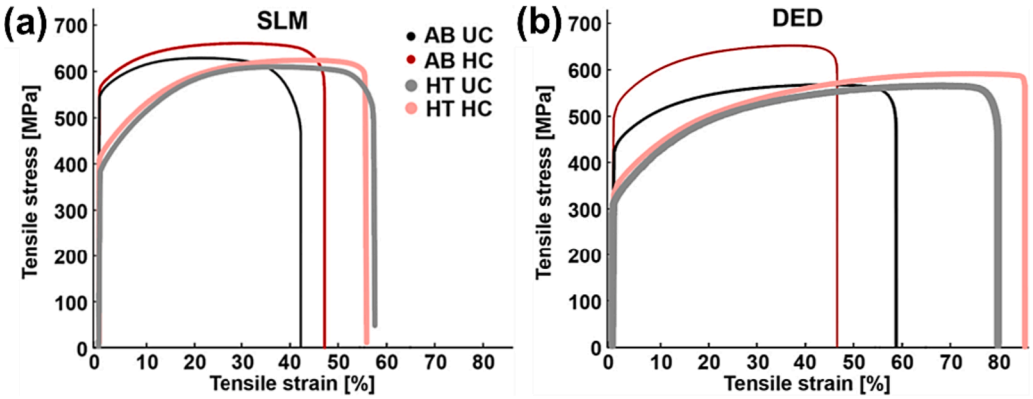

Fig. 34. Engineering stress–strain curves for specimens of (a) SLM and (b) DED 316L. Thin curves represent as-built 316L (black for uncharged, red for hydrogen-charged), and thicker curves represent heat-treated 316L (grey for uncharged, pink for hydrogen-charged), for both a and b. Reproduced from Bertsch [18] with permission from Elsevier. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

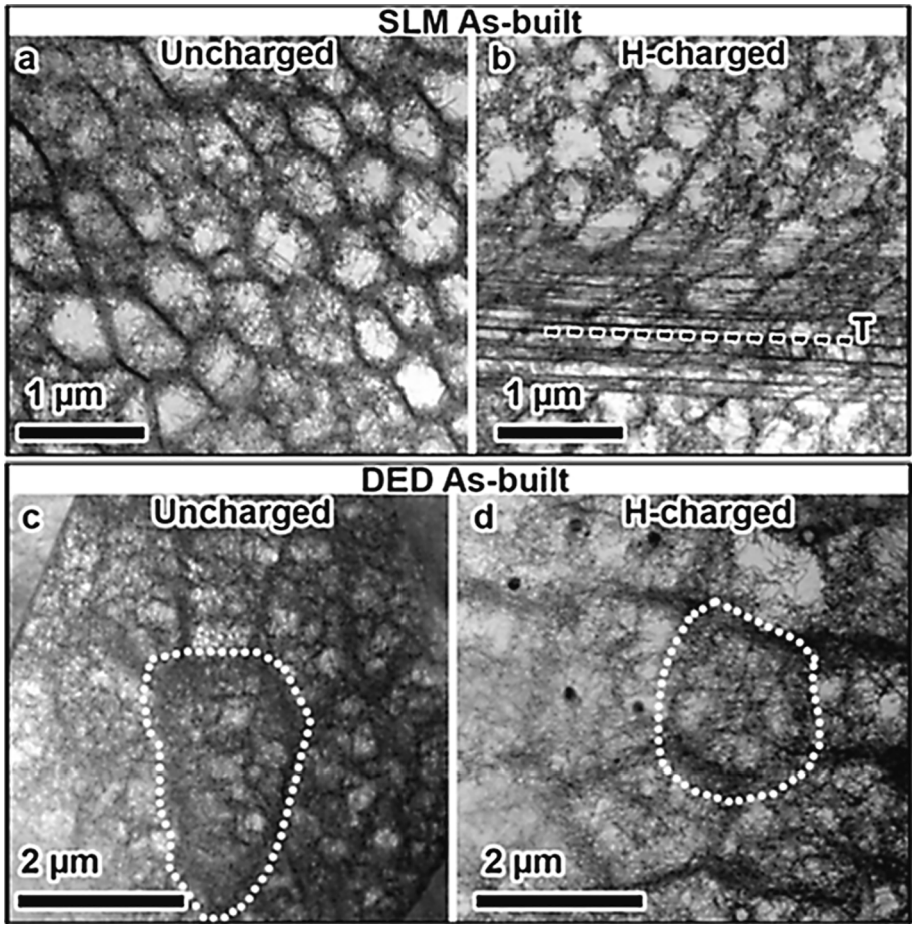

Fig. 35. Bright-field STEM micrographs of dislocation structures at 15% elongation in (a & b) as-built SLM, and (c & d) as-built DED 316L SS. Dashed line in (b) parallel to deformation twins, dotted lines in (c & d) indicate dislocation cell walls with dendritic segregation profiles. Reproduced from Bertsch [18] with permission from Elsevier.

SLM, which suppressed the formation of austenite and retained δ ferrite at room temperature. Aging both the wrought and SLM 17-4PH steels at 580 °C for 4 h caused precipitation of nanoscale Cu-containing precipitates in both steels. SSRT tests were used to study the HE behaviour in an acidic chloride solution at room temperature with cathodic charging. Fig. 38 presents the slow strain rate stress–strain curves of specimens in the following conditions: (i) uncharged, (ii) pre-charged and (iii) pre-charged plus continually charging during the tensile testing (pre + charged). These results allow identification of the difference in HE susceptibility. Both the wrought and the SLM steels were susceptible to HE evidenced by a significant reduction in elongation of the H-charged samples. Nevertheless, the SLM ferritic steel showed higher strength and better ductility for all testing conditions (uncharged, pre-charged, and pre-charged with continually charging during testing and different strain rate) than the martensitic wrought steel. This was attributed to the fine microstructure of the SLM alloy.

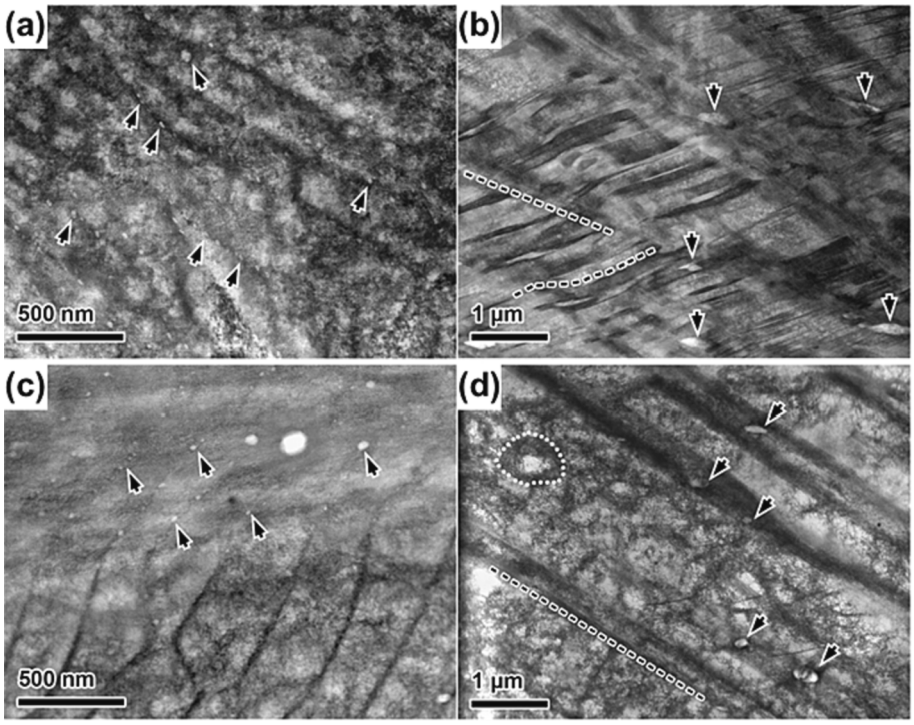

Fig. 36. Bright-field STEM micrographs of dislocation structures up to 250 μm away from the fracture surfaces in (a & b) uncharged & (c & d) hydrogenated as-built SLM (left) and DED (right) 316L. Arrows indicate voids formed along dislocation walls. Reproduced from Bertsch [18] with permission from Elsevier.

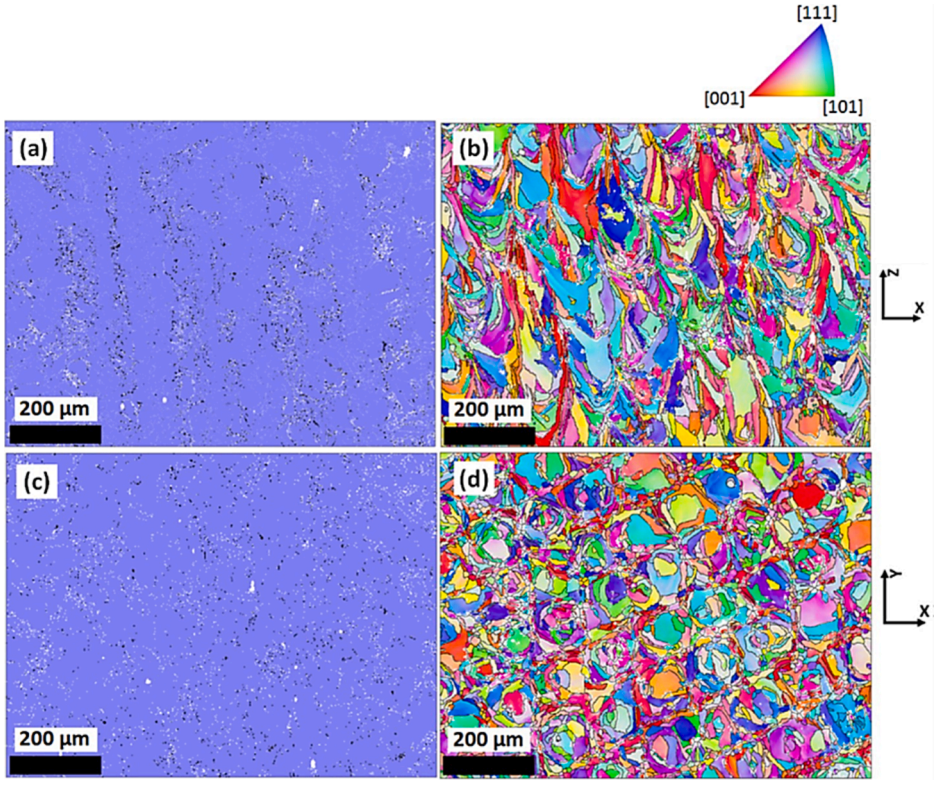

Fig. 37. EBSD data obtained on the SLM 17-4PH steel aged 4 h at 580 °C. (a) phase map of the X-Z plane. The BCC and FCC phases are in blue and black respectively. The non-indexed areas are in white. (b) orientation map of the X-Z plane (IPF Z, where Z is the building direction), (c) phase map of the X–Y plane. (d) orientation map of the X–Y plane. Reproduced from Alnajjar [55] with permission from Elsevier. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.4. Ultrahigh strength steels

Ultrahigh-strength steels are predominantly used in aerospace, defence and marine industries to fabricate engineering components subjected to high load and with high geometrical complexity. Hence, additive manufacturing of ultrahigh strength steels provides advantages for integrated part production. Work on AM has focused on maraging steels, 4340 steel, and tool steel. High strength steels are associated with high susceptibility to hydrogen because of their high strength. Therefore, investigation and understanding of the HE behaviour of these AM

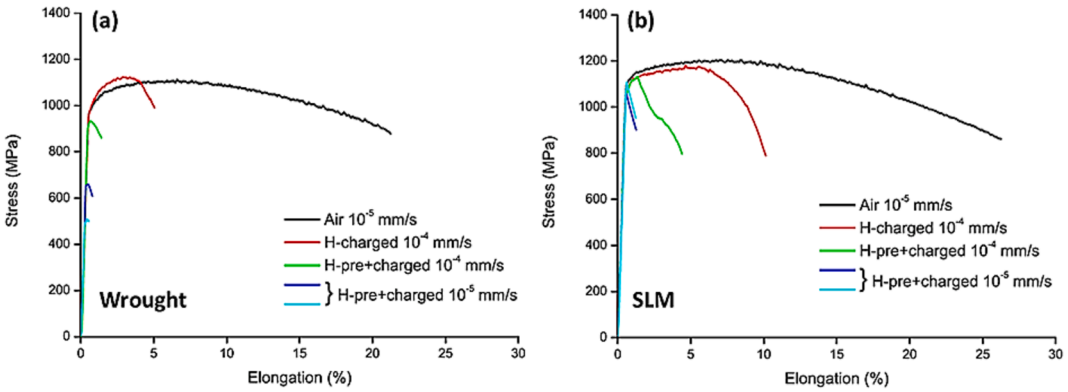

Fig. 38. Stress–strain curves of (a) wrought and (b) SLM smooth 17-4PH specimens. Reproduced from Alnajjar [55] with permission from Elsevier.

steels are essential before they can be practically used. HE of wrought ultrahigh strength steels has been comprehensively studied. However, research on HE of AM steels is limited. The following sections review the currently published results on maraging steels, 4340 steel, and tool steel.

4.4.1. Maraging steels

Maraging steels contain almost no carbon but a high content of other alloying elements, including nickel, cobalt, and molybdenum. The ultrahigh strength is achieved through precipitation of intermetallic compounds, such as Ni_3Ti, Ni_3Mo Fe_2Mo and Fe_7Mo_6, that form during aging after the steels are quenched to martensite [355]. Maraging steels combine ultrahigh strength and high toughness, which is critically important in the aerospace sector. The most typical maraging steel is the 18Ni, which can achieve tensile strength over 2000 MPa after proper heat treatment.

Several studies have shown that SLM processing can produce fully dense maraging steel, which showed mechanical properties comparable to those produced utilizing traditional manufacturing [356–359]. However, maraging steels produced either using traditional routes [54,57,356,360] or using AM [225] are very susceptible to HE, manifesting as a reduction in mechanical properties [54,57,356,360]. In addition, HE of maraging steel is increased by the presence of pores produced during SLM or casting because these defects can accumulate hydrogen and promote crack propagation and brittle fracture [225].

The HE behaviours of grade 300 maraging steel processed by SLM were investigated by Li et al. [254] and Kwon et al. [57]. Li et al. [254] examined the microstructures and mechanical properties of the grade 300 maraging steel, processed by SLM, before and after hydrogen charging. The hydrogen charged specimen suffered severe HE as the strength and plasticity decreased significantly. In situ high energy X-ray diffraction (XRD) and EBSD were used to study the mechanism of the fracture mode transition from ductile fracture to brittle fracture after hydrogen charging. The results are shown in Fig. 39. The SLM 300 maraging steel has a martensitic matrix similar to that of a conventional grade 300 maraging steel, with irregular coarse plate martensite [361]. Hydrogen charging did not cause any phase transformation. Compared to the uncharged sample, the hydrogen charged steel has significantly lower tensile strength and elongation at break and no apparent plastic deformation at failure. They found that the ductile to brittle fracture mode transition in the SLM maraging steel was induced by hydrogen charging. There were non-uniform dimples, a typical feature of ductile fracture, on the fracture surface of the uncharged specimen, whereas the hydrogen charged specimen showed a brittle fracture mode with a mixture of dimples and cleavage planes, as presented in Fig. 40. They

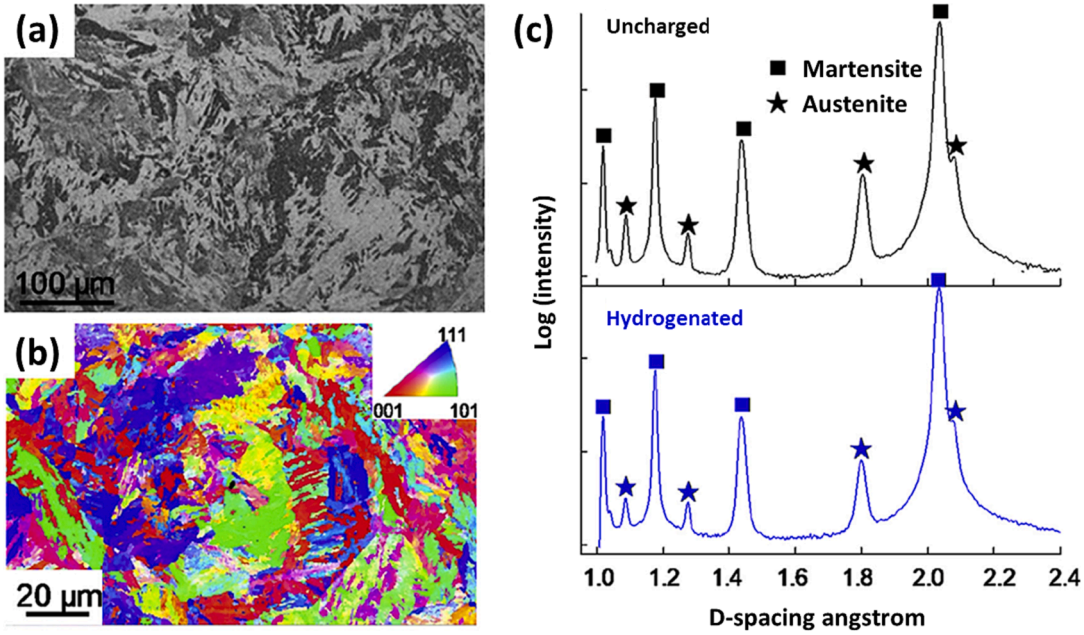

Fig. 39. (a) OM and (b) IPF micrographs of the studied AM grade 300 maraging steel, and (c) high energy XRD profiles of the hydrogen-free and hydrogen-charged specimens before deformation. Reproduced from Li [254] with permission from Elsevier.

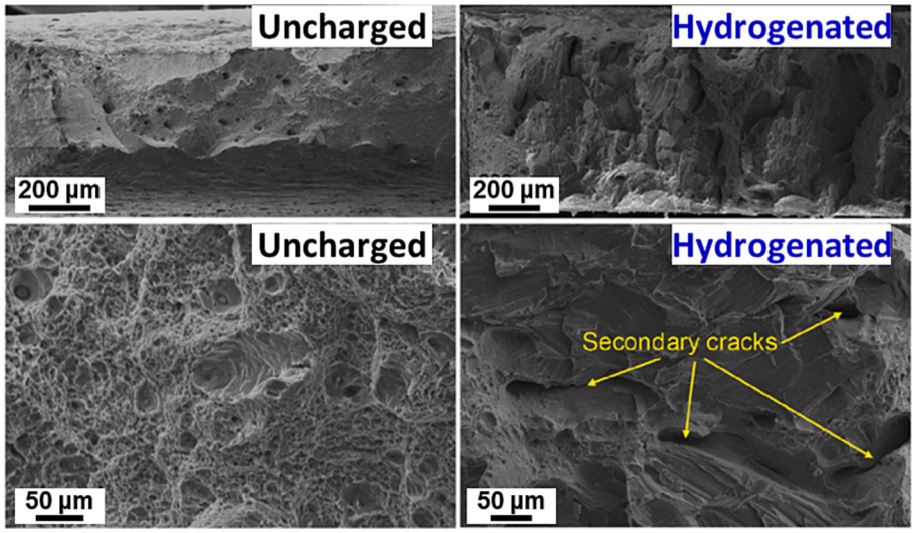

Fig. 40. The macroscopic and microscopic morphologies of the fracture surfaces of the uncharged and hydrogenated samples showing the obvious HE. Reproduced from Li [254] with permission from Elsevier.

proposed that the ductile to brittle fracture mode transition in SLM maraging steel in the presence of hydrogen was due to the locally premature martensitic transformation (as in Fig. 41) and cracking along the {1 1 1} retained austenite plane that was perpendicular to the loading direction. The microcracks released the localised stress fields in retained austenite and thus inhibited the further phase transformation from retained austenite to the newly formed martensite. These microcracks coalesced and propagated rapidly under the applied stress, leading to the brittle fracture of the hydrogen charged specimen.

In addition, Kwon et al. [57] also studied how post-heat treatment affected the HE of the SLM 18Ni 300. They found the reduction in the amount of retained austenite that was replaced with martensite at the borders of the solidification cells. Compared to the hydrogen charged as-built counterparts, the hydrogen charged heat-treated samples had a smaller number of trapped hydrogen atoms and showed improved ductility in the tensile tests. The reduction of the austenite content and the dislocation density after heat treatment led to the decrease in hydrogen susceptibility of the steels. Furthermore, they reported that isothermal aging improved the strength of the uncharged steel, but resulted in more hydrogen trapped and thus reduced HE resistance. Thus, it was concluded that artificial aging had a negative effect on the HE resistance of steel, whereas heat treatment had a positive effect. The same tendency was observed in maraging steels manufactured using traditional methods [57,356,357].

However, there is still no systematic comparison and analysis between maraging steels fabricated by AM and conventional methods. Thus, further investigation is necessary.

4.4.2. *AISI 4340 – High strength low alloy steel*

AISI 4340 steel is a typical medium carbon low alloy ultra-high strength steel with YS over 1860 MPa [94]. The shape complexity of engineering parts made of this steel drives the research of AM of 4340 steel. Currently, the LAM techniques used for 4340 steel include laser welding and laser cladding for repair. The current major research effort is undertaken to evaluate the microstructure evolution during the repair of some high-value AISI 4340 steel components by the DMD process [155,362–364]. However, there are only a few studies that investigated the mechanical properties and microstructure of 4340 steel processing by the PBF methods [155,362,363,365].

AISI 4340 steel is prone to HE, which is a common cause of failure [197,366,367]. Generally, higher strength steels are more susceptible to HE [367]. During the steel manufacturing process, hydrogen is easily absorbed by liquid Fe, which results in IHE $C+H_2O \leftrightarrow CO+H_2$ and $Fe+H_2O \leftrightarrow FeO+H_2$ [197]. Internal hydrogen has a substantial influence on high-strength steels. Because of the ease of microcrack growth, IHE by the mechanism of high-pressure bubble formation might be

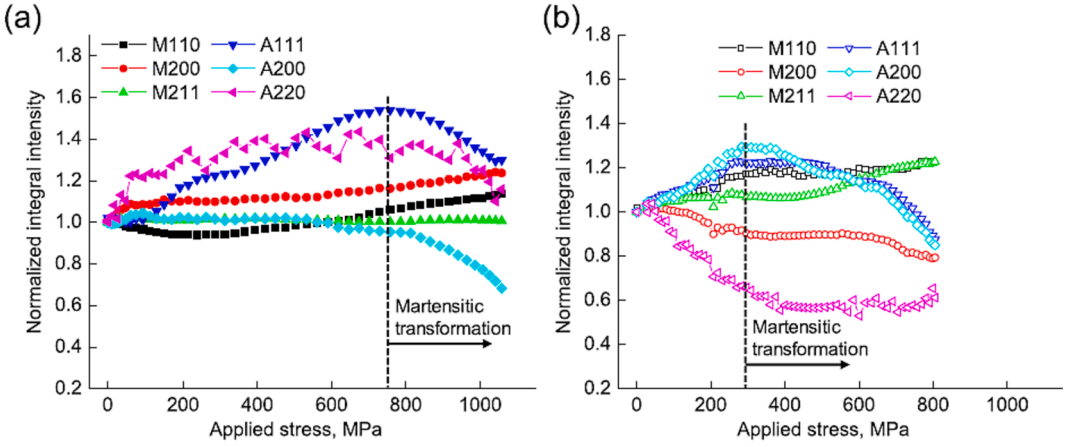

Fig. 41. Lattice strain evolution of the martensite and retained austenite in the (a) uncharged and (b) hydrogenated specimens with the applied stress along the loading direction. Reproduced from Li [254] with permission from Elsevier.

extremely severe [366]. From the diffusion perspective, the major H-trap sites in low alloy carbon steels are most likely associated with dislocations, within the interfaces of the martensite nano-structure, and with the interface between the matrix and particles including carbides, nitrides and carbonitrides. In the AISI 4340 steel, dislocations are suggested as the major H-trap sites [368,369].

Liu and co-workers [369] reported two-stage permeation and trapping of hydrogen in as-received commercial 4340 steel attributed to the presence of two types of hydrogen trapping sites, which were identified as ferrite GBs and dislocations (trap 1) and ferrite/cementite interfaces (trap 2). Fig. 42 shows that the apparent diffusivity of hydrogen in trap 1 was higher than that in trap 2, while the saturation concentration of hydrogen for trap 1 was lower than that for trap 2.

Fig. 43 shows that the representative microstructure from the research of Liu and co-workers [369] of as-received commercial 4340 steel (in the hot rolled and slow cooled condition) consisted of ferrite and pearlitic (lamellar cementite), while the microstructure of the PBF 4340 steel was predominantly martensitic with fine microstructure because of the rapid cooling of the melt pools [155]. It should also be mentioned that 4340 is typically used in the quenched and tempered condition, for which the microstructure is tempered martensite. As mentioned before, HE is sensitive to microstructure, and microstructural alteration may lead to improvement in resistance to HE [34]. Thus, it could be expected that the interaction of hydrogen with microstructural defects of 4340 manufactured by AM is different to that produced using conventional methods. Although the HE of AISI 4340 fabricated by conventional methods has been considerably investigated [197,366,367,369–371], no literature has been dedicated to the study of HE of AM AISI 4340. In order to advance the research of AM 4340 steel and realise its industrial applications, it is necessary and essential to investigate the HE performance (see Fig. 44)

4.4.3. AISI H13 – Tool steel

Tool steel refers to a variety of carbon and alloy steels that are particularly well-suited to make tools, such as cutting tools, dies, and moulds (TDM). Complicated TDM manufacturing can take months or even years and entails high costs. AM offers a new tool-making route, which can shorten the production cycle by around 40%. In addition, AM methods can also directly fabricate geometrically complex shapes, such as internal cooling channels, enhancing the design function [8].

AISI H13 steel is one of the most common tool steels, and is characterised with high strength, high toughness, high hardenability and good resistance to thermal softening. AISI H13 has been extensively studied after production by both the PBF and DED processes [8,93,158,372,373]. The typical microstructure is characterised by a martensite matrix, a small amount of retained austenite, and composition-dependent carbides [157,160,373–375]. Such a microstructure also occurs in laser-welded or laser-cladding parts [376,377]. The microstructure of AM H13 steel is characterised by ultrafine cellular or dendritic sub-grain structures due to the fast cooling ($10^3 \sim 10^6$ K/s). These microstructures are not common in cast H13 [119]. In PBF H13 steel, the intrinsic heat treatment can decompose martensite into fine ferrite and Fe_3C precipitates, which is similar to traditional tempering [160]. DED H13 steel specimens had strengths similar to their wrought counterparts [93,157], but, the elongation (approx. 5% ~ 6%) was lower than traditionally manufactured parts (approx. 9%) [378]. The ductility for the PBF H13 steel (<4%) was even lower than that for the DED-built components. The high brittleness of AM H13 is related to the high content of carbon martensite, and high residual stress, up to 1420 MPa, because of the fast-cooling of the AM process [379–381]. The YS and UTS of H13 vary from about 830 to 1564 MPa and from 900 to 2064 MPa, respectively, indicating the high fluctuation of mechanical properties in the AM H13 components [8].

HE is not a major cause of failure for tool steels in service. Thus, there is a lack of research on the HE properties of tool steels [382]. However, as previously stated, high-strength steels are highly susceptible to HE [244]. This high hydrogen sensitivity is often linked to the presence of martensite in the microstructure [227]. For example, AISI H13 is a high-strength martensitic steel, and thus is likely to be sensitive to HE. The AM can impart better properties to H13 steel, including high strength, good anisotropy performance and high design freedom, which expands the application range of this steel. To date, highly dense H13 steel parts have been successfully produced using AM by optimizing the processing parameters [117]. However, there is no literature dedicated to the study of the HE of AM H13 steel, even in commercial H13 steel processed by conventional methods. For the development of industrial applications of AISI H13 steel, it is necessary to systematically and comprehensively investigate the HE performance.

In summary, the studies on the HE performance of AM steels has mostly focussed on austenitic stainless steels, including AISI 304L and AISI 316L. Precipitation-hardening stainless steel such as 17-4PH and grade 300 maraging steel have also received some attention. The effect of post-heat treatment [16–18,55,57] and different AM processes [18] on the HE properties of AM steels have been investigated. Most studies showed that the strength and plasticity of steel specimens decrease after charging with hydrogen [13,55,254]. But, the majority of steels fabricated by SLM have better resistance to HE than steels produced by conventional fabrication methods [18,55]. Further research is necessary due to the insufficient number of studies that systematically compare and analyse the HE performance of steels fabricated using conventional

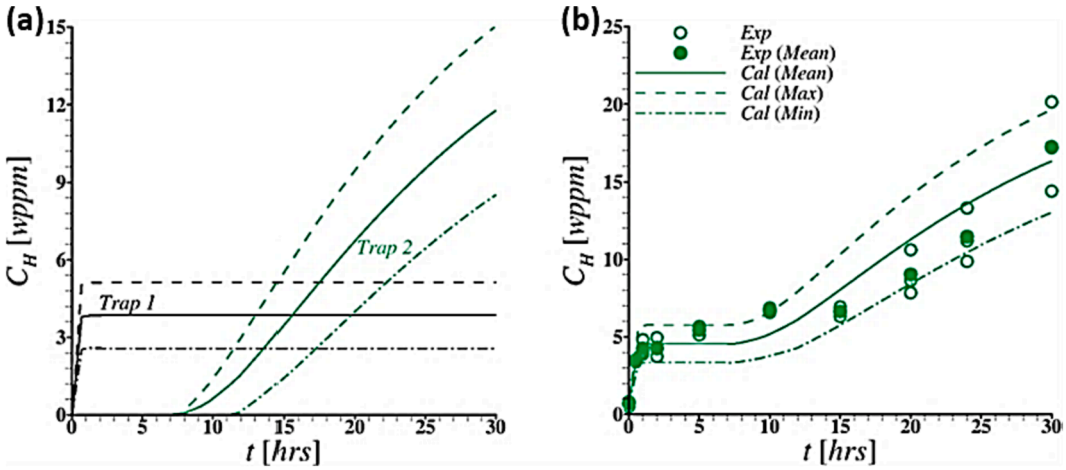

Fig. 42. (a) Calculated variation of hydrogen concentration, C_H, with electrochemical charging time, t, in the two traps present in 4340 steel, (b) comparison of calculated (cal) and experimentally (exp) measured variation of total hydrogen concentration, C_H, with electrochemical charging time, t. [369].

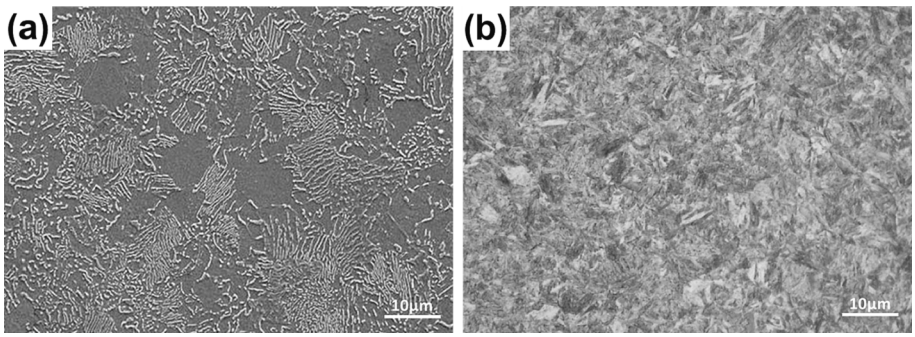

Fig. 43. Micrographs of (a) as-received commercially AISI 4340 steel (hot rolled and slow cooled) [369] and (b) as-built PBF AISI 4340 steel. [363].

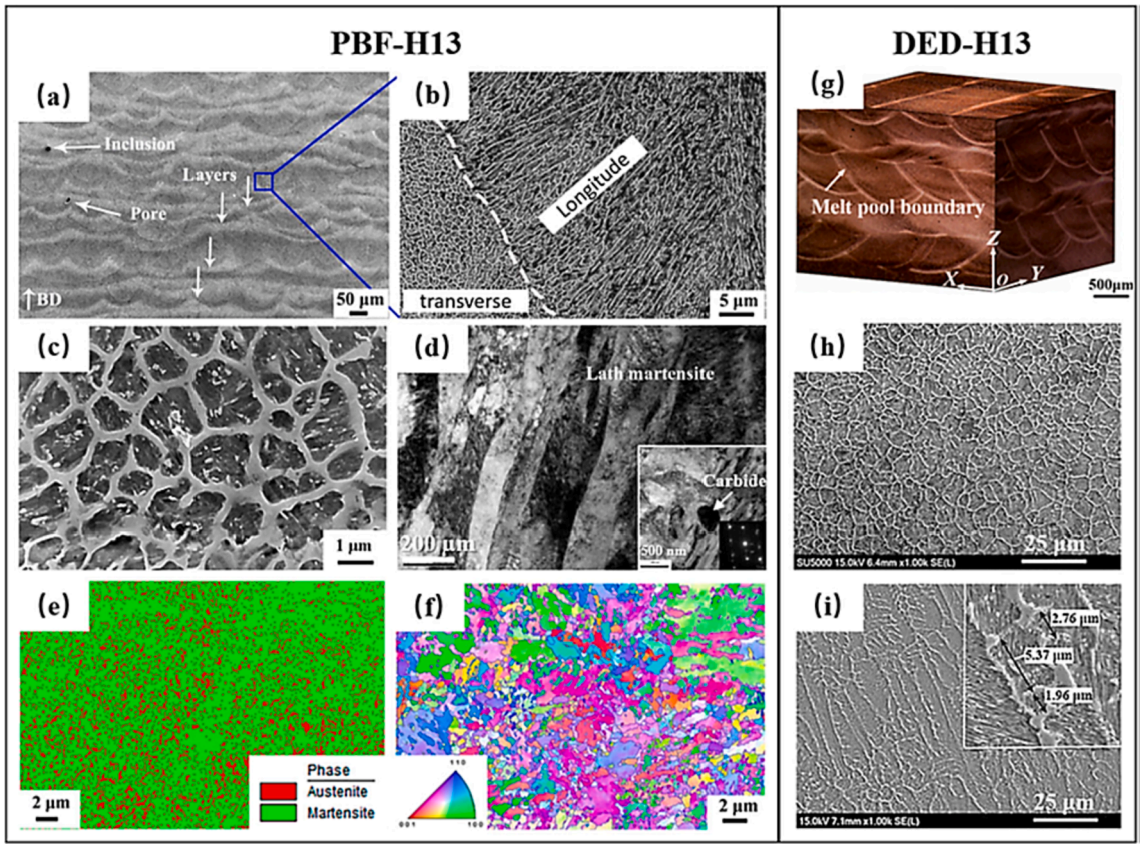

Fig. 44. (a, b & c) SEM images on the cross-section of an LPBF-built H13 steel; (d) TEM image of another LPBF-built H13 steel showing the lath martensite and $M_{23}C_6$ carbide; EBSD phase map (e) and orientation colour map (f) of an LPBF fabricated H13 steel; (g) Low magnification image of the microstructures in an LDED built H13 sample; (h & i) SEM image of microstructure of the as-built LDED H13 sample: (h) the top, (i) the middle. Reproduced from Yin [8] with permission from Taylor & Francis, and Tan is the author of the paper.

methods and AM methods. Moreover, there is a lack of studies on using AM methods to manufacture hydrogen-sensitive high-strength AISI 4340 steel, and H13 steel.

4.5. *High-Entropy alloys (HEAs)*

High-entropy alloys (HEAs) are a relatively new alloy class with five or more principal elements [383]. The concentration of each element in the HEA is 5 ~ 35 atomic% [384]. This innovative alloy design method has resulted in several HEA systems with good properties, including good corrosion resistance [385], good fracture toughness at cryogenic temperatures [386], and high strength at elevated temperatures [387]. Thus, HEAs may find much potential in harsh environment applications. Several detailed review papers on HEAs have been published [199,388–392].

AM of HEA has attracted considerable attention [393]. Compared with conventional techniques for making HEAs, such as the plasma arc melting, AM of HEA allows the creation of non-equilibrium microstructures and may produce comparable or better mechanical properties [394,395]. The current mainstream AM processing methods for HEAs are DLD (also known as LMD, DMD, LENS and laser cladding), SLM (also known as LBM, DMLS, LMF) and EBM, as listed in Table 7 [80,393]. A few HEAs [199], including CoCrFeNiMn [396–398], CrMnFeCoNi [396,399], CrFeCoNi [400], AlCoCrFeNi [401,402], $AlCoCrFeNiTi_{0.5}$ [403], compositionally graded HEAs [404,405], and laminated HEAs [406], have been successfully produced with AM using either pre-alloyed HEA powders or mixed metallic powders. Moghaddam et al. [80] published a review of HEAs processed using AM, including the

Table 7
Summary of HEAs processed through powder-based AM methods [80,393].

Alloys	AM processes
CoCrFeNiMn (Cantor)-based	LMD, SLM
$Al_xCoCrFeNi$-based	LMD, DMD, SLM
MoNbTaW	LMD
FeCoCrAlCu	LMD
TiZrNbWMo	Laser Cladding
CrMoTaWZr	Laser Cladding
AlTiVMoNb	Laser Cladding
MoNbTaW	SLM
$Ni_6Cr_4WFe_9Ti$	SLM
CoCrFeNi	SLM
$Al_xCoCrFeNi$	LENS, SLM, EBM
CoCrFeNiTi-based	EBM
$Al_{0.5}CrMoNbTa_{0.5}$	EBM

feedstock requirements, common processing methods (LMD, SLM and EBM), microstructures features along with their properties and underlying mechanisms, and a conclusion of the current state and prospects of AM HEAs. This review summarises the responses of HEAs to hydrogen.

4.5.1. CoCrFeNiMn

The equiatomic CoCrFeNiMn HEA alloy has a single-phase FCC crystal structure, which has a good combination of strength-ductility at room and cryogenic temperatures. CoCrFeNiMn HEA was reported as having a HE susceptibility lower than that of the 304 and 316 ASSs [407–411]. Moreover, the HE resistance of CoCrFeNiMn HEA can be increased by increasing the nano-twin density [412].

However, the strength of this alloy, when processed using traditional methods, is relatively low, with an YS of approximately 300 MPa and UTS of approximately 530 MPa. This limitation has drawn the attention of many researchers aiming to enhance the mechanical properties [413]. AM offers advantages in improving material properties due to the rapid cooling, which facilitates the formation of finer microstructure [258,414]. The rapid cooling also causes a significant accumulation of dislocation groups, leading to the formation of a cellular structure with a high density of dislocations at cell walls [415–417]. In addition, traditional manufacturing methods such as casting, melting, and powder metallurgy cannot achieve the production of precise parts with complex structures. Typically, products manufactured through traditional methods require additional machining before they can be put into service. This not only leads to material wastage but also increases costs, thereby restricting the development and application of HEAs [258]. Currently, the AM processes have been successfully applied to prepare the CoCrFeMnNi [396–398].

Due to the superior HE resistance of conventionally produced CoCrFeNiMn HEAs, several studies have investigated the HE performance of AM CoCrFeNiMn HEAs. Fu et al. [257] investigated the HE behaviour of the as-built SLM and post-annealed equiatomic CoCrFeMnNi HEA samples using SSRT (initial strain rate of $1 \times 10^{-5}s^{-1}$) tests under in situ electrochemical hydrogen charging. The testing results in air (as shown in Fig. 45) showed that annealing at 900 ℃ for 1 h increased the uniform plasticity from 20% of the as-built SLM sample to 3%, but slightly reduced the UTS from 751 to 707 MPa. However, the UTS of the annealed sample was still significantly higher than that of the traditional manufactured CoCrFeMnNi. Hydrogen charging remarkably reduced the plasticity, indicating the high hydrogen susceptivity of this alloy. The stress–strain curves with in situ hydrogen charging (Fig. 45 (a)) showed only 3.3% uniform plasticity of as-built SLM samples. In contrast, annealed samples exhibited about 10% uniform plasticity. This implied that annealing improved the HE resistance. The annealed SLM CoCrFeNiMn had delayed initiation and propagation of cracks after hydrogen charging, which effectively improved the HE resistance of the SLM samples. Investigations on both as-built SLM and annealed prototype CoCrFeMnNi HEA samples suggested that annealing preserved the cellular structures while effectively reduced the dislocation densities. Fig. 46 shows that a high dislocation density was distributed along the fusion boundary in as-built SLM CoCrFeNiMn, while annealed SLM

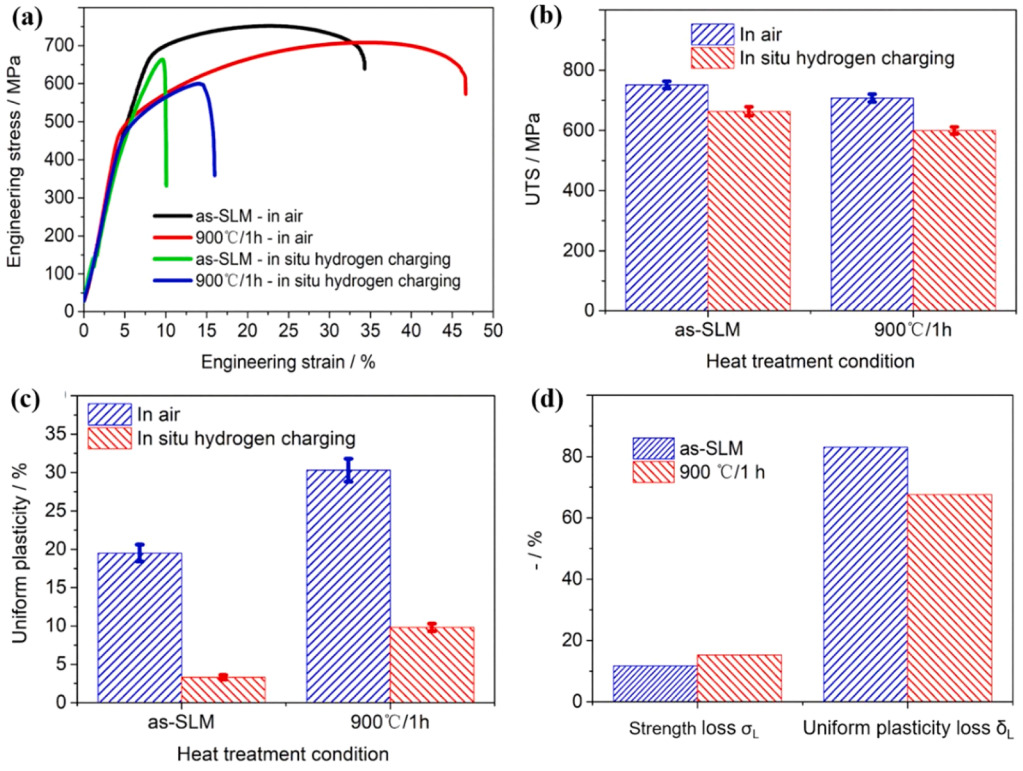

Fig. 45. Results from the SSRT tests in air and under in situ hydrogen charging of as-built SLM and 900 ℃ for 1 h annealed samples: (a) engineering stress–strain curves; (b) UTS values; (c) uniform plasticity values; (d) strength loss (σ_L) and uniform plasticity loss (δ_L). Reproduced from Fu [257] with permission from Elsevier.

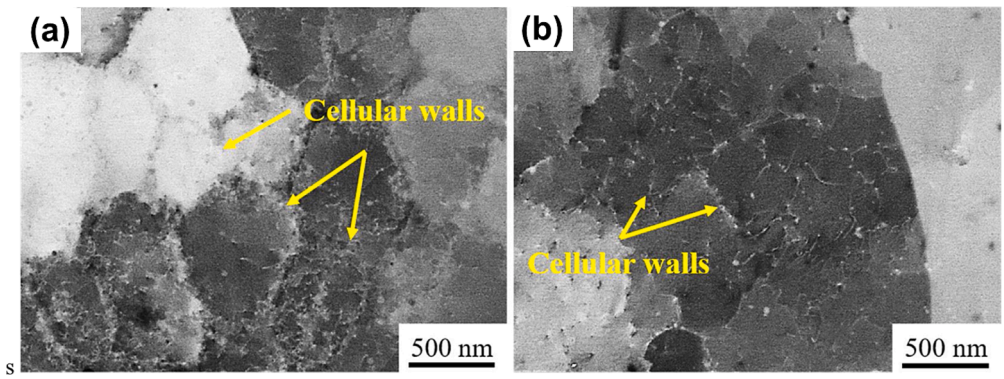

Fig. 46. High magnification electron channelling contrast imaging (ECCI) on X-Y plane of the uncharged (a) as-built SLM and (b) annealing at 900 ℃ for 1 h samples. Reproduced from Fu [257] with permission from Elsevier.

CoCrFeNiMn was associated with a relatively low-density of dislocations. Thus, annealing slightly reduced the strength, annealing significantly increased the ductility and HE resistance as indicated by the SSRT tests with in situ electrochemical hydrogen charging.

Fu et al. [257] also reported that crack initiation and propagation were delayed by the HELP mechanism for the same alloy because of the formation of nano-twins and dislocation cells in the modified structures. During testing with hydrogen charging, a region with nano-sized deformation twins and dislocation cells formed beneath the surface of the annealed sample. This region was also enriched with hydrogen. Hydrogen can enhance the motion of dislocations, increase their mobility, reduce the stacking fault energy and facilitate the formation of nano-sized twins and dislocation cells. This deformation area delayed the initiation of HE cracks by reducing the hydrogen concentration at the GBs. However, as deformation and hydrogen concentration increased near the sample surface, HE cracks initiated along specific crystal planes and propagated. Along the crack path, GBs captured more hydrogen, leading to the formation of slip bands and dislocation cells that slowed the crack propagation. Thus, it is expected that the HE resistance of SLM CoCrFeMnNi can be improved by post-processing.

Lin et al. [223] also confirmed that a heat treatment of CoCrFeMnNi at 900 °C reduced dislocation density at the cell walls, which consequently reduced the metal's ability to trap hydrogen. They attributed the decrease in trapping capability in the heat-treated SLM sample to both decrease in dislocation density and removal of Mn segregation. They discovered that a change from HE in the as-built SLM sample to hydrogen-induced plasticity in the heat-treated SLM sample was achieved by modifying the cell structure through heat treatment. There were higher amounts of hydrogen in the hydrogen affected region of the as-built sample compared to that of the heat-treated sample aggravated the HEDE mechanism, which consequently enhanced the formation of intergranular cracking with relatively low strain localisation and degraded HE resistance. The HELP mechanism is also activated in the as-built HEA sample. Hydrogen amounts were initially higher in the as-built SLM sample, and the hydrogen concentration at the GB could further increase when hydrogen atoms were transported to the GB intercepted by deformation twins. Therefore, the GBs intercepted by deformation twins with high concentrations of hydrogen were more susceptible to the HEDE mechanism and high stress concentrations, making them the preferential initiation sites and propagation paths for intergranular cracking. This combined interplay of the HELP and HEDE mechanisms is referred to as HELP-mediated HEDE model [44,212]. Therefore, the pronounced twinability in the as-built SLM CoCrFeMnNi sample results in intensified intergranular cracking.

Furthermore, Cheng et al. [258] studied the effect of different processing parameters on the HE performance of CoCrFeNiMn HEAs fabricated by LPBF using SSRT tests with and without hydrogen. They characterised the processing parameters using volumetric energy density (VED) evaluated using the following empirical equation (Eq.5) [418].

$$VED = \frac{P}{vht} \tag{5}$$

where P is the laser power, v is the scanning speed, t is the layer thickness, and h is the hatch spacing.

It was found that the improper LPBF parameters led to increased numbers of microcracks and holes, resulting in a slight decrease in the HE resistance of the material. Fig. 47 shows that the sample processed using the medium VED had the highest UTS and elongation before and after hydrogen charging, which were 663 MPa, 24% and 646 MPa, 24%, respectively. In contrast, the low VED sample had significant processing defects, had the lowest UTS of 633 MPa, the highest elongation loss ratio of 17% and strength loss ratio of 3% after charging hydrogen. Hydrogen charging did not produce a significant effect on the YS, indicating that hydrogen did not cause solid solution strengthening, which is consistent with some previous related studies [257,258,412]. Thus, they concluded that proper process parameters can produce better mechanical properties, and even after hydrogen charging, there can be uniform plastic deformation. In addition, the samples with three different processes have low strength loss and elongation loss ratio, reflecting significant hydrogen embrittlement resistance. In addition, the elongation of the high VED sample was increased after charging hydrogen, which indicates that hydrogen increased ductility for LPBF CoCrFeNiMn [258].

In addition, Cheng et al. [258] found that hydrogen charging increased the strain hardening rate of the LPBF CoCrFeNiMn HEA. This increase was associated with a notable increase in the proportion of deformation twins in the presence of hydrogen. This strengthening behaviour became evident after the emergence of the strain hardening platform, enabling the strain hardening rate to reach and surpass that of the sample without hydrogen charging. This phenomenon played a vital role in strengthening the material, contributing significantly to the strengthening mechanism. In addition, the extra work hardening resulting from the presence of deformation twins is commonly associated with the "dynamic Hall-Petch" effect [419]. The formation and propagation of nanotwins introduced new interfaces that led to grain fragmentation, thereby reducing the mean free path of dislocations. In summary, the strengthening effect caused by a significant quantity of deformation twins increased the HE resistance of LPBF CoCrFeNiMn HEA, to maintain comprehensive mechanical properties in a hydrogen environment. However, this positive effect may be diminished under unfavourable process parameters.

4.5.2. *AlCoCrFeNi2.1*

The $AlCoCrFeNi_{2.1}$ eutectic HEA has gained significant attention due to its distinctive in-situ composite structure, which effectively addresses the limitations of inadequate strength in FCC single-phase HEA and

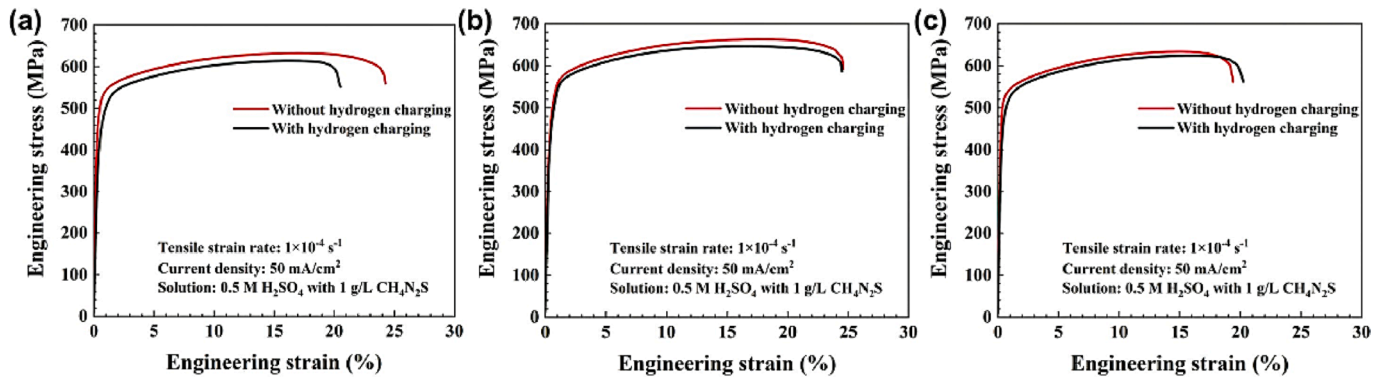

Fig. 47. Mechanical properties of LPBF samples under different processes with and without hydrogen processed using VED values of (a) 71 J/mm^3, (b) 81 J/mm^3 and (c) 90 J/mm^3. Reproduced from Cheng [258] with permission from Elsevier.

insufficient ductility in BCC single-phase HEA [420]. Several researchers have successfully produced AlCoCrFeNi$_{2.1}$ HEA using AM methods, including LMD [421], SLM [420,422,423], and PAAM (powder-bed arc additive manufacturing) [424].

Wan et al. [199] used in-situ small-scale tensile testing with and without cathodic hydrogen charging to investigate the HE behaviour of the AlCoCrFeNi$_{2.1}$ eutectic HEA, which was fabricated with LENS using pre-alloyed powder. As-cast samples were also tested as a reference. The hydrogen charging led to a drop in the strain to failure of the LENS HEA from 10% to 7% with an embrittlement factor of 33%. In comparison, the as-cast sample exhibited a reduction in ductility from 19% to 16%, with an embrittlement factor of about 20%. Both samples exhibited a decrease in the UTS (AM sample: from 1214 MPa to 1126 MPa and as-cast sample: from 1007 MPa to 959 MPa) but a slight reduction in the YS (AM sample: 820 MPa & as-cast sample: 480 MPa). The fractured H-charged specimens showed surface fissures that were not present in the hydrogen-free specimens, indicating brittle behaviour. TDS analysis indicated a larger concentration of hydrogen (approx. 12 wppm) in the AM specimen than in the as-cast specimen (approx. 9 wppm). Thus, they concluded that the finer microstructure and increased density of boundaries in the AM sample contributed to the larger hydrogen uptake.

In addition, Wan et al. [199] used finite element analysis (FEA) to simulate the hydrogen diffusion in both AM and as-cast samples and suggested an inhomogeneous hydrogen diffusion that was strongly dependent on the local microstructure. Fig. 48 indicates that the interfacial cracking of the H-charged specimens can be attributed to a locally increased hydrogen concentration at the phase boundaries between the FCC/L12 and BCC/B2 phases. Fig. 49 shows representative cases of the phase morphology dependence of the deformation and crack initiation. The cracking susceptibility was also influenced by the relative orientation of the phase boundary and the loading direction. When the phase boundary was nearly perpendicular to the loading direction, fracture was more easily initiated. In this case, cracks highly likely initiated in the BCC/B2 phase. In the phase boundaries of the examined eutectic HEA, there was a Kudjumov-Sachs (K-S) type orientation relationship [425] between FCC/L1$_2$ and the BCC/B2 phases, which may allow for slip transfer between the two phases. They concluded that the regular high-angle GBs were more vulnerable to cracking than low-angle GBs and phase boundaries, according to the EBSD study of hydrogen-induced surface cracks in the AM samples. Both the HELP and HEDE mechanisms are typically activated in this HEA. Within the FCC/L12 phase, hydrogen restricts the deformation to specific slip systems, intensifying local deformation and leading to localized necking and failure. This process results in cracking along the slip traces. Whereas hydrogen also diminishes the bonding energy at boundaries, making them more susceptible to opening as cracks approach due to the stress field created by

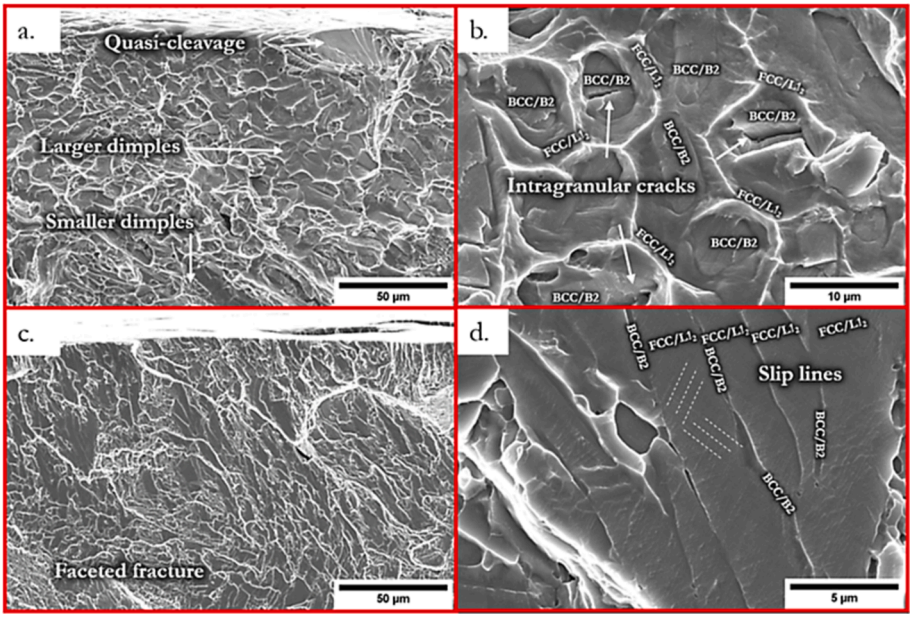

Fig. 48. Detailed fractography of the tested specimens (H-charged): (a & b) as-cast; (c & d) AM. [199].

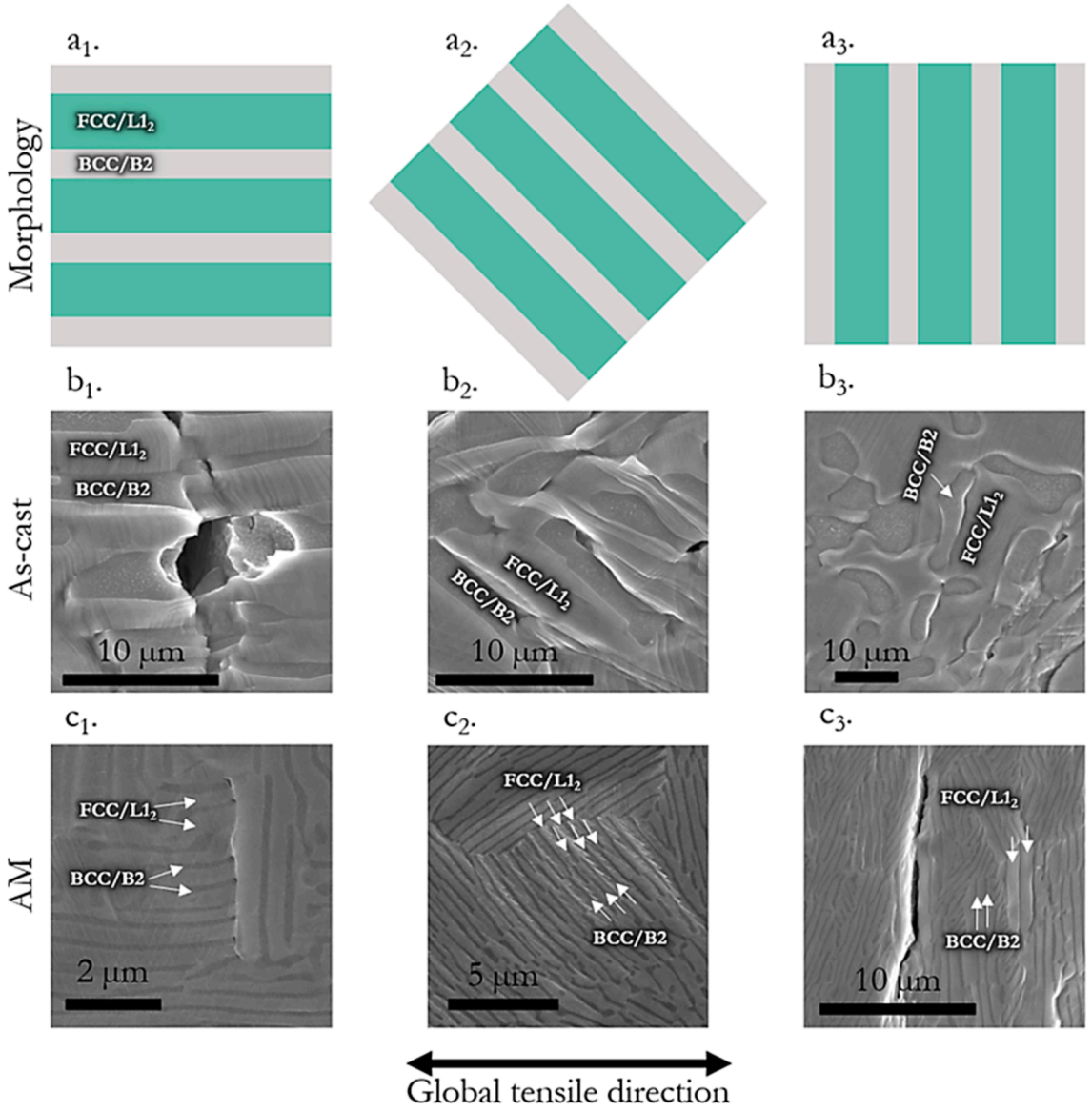

Fig. 49. Representative cases showing the phase morphology dependence of the deformation and crack initiation. (a_1-a_3) representative phase morphology; (b_1-b_3) exemplary SEM images of the as-cast specimen; (c_1-c_3) exemplary SEM images of the AM specimen. The global tensile direction is horizontal. [199].

these cracks.

Overall, HEAs can be high-strength materials. High-strength materials are critical when serving in a hydrogen-containing environment, as they might experience higher mechanical stress and thus a higher susceptibility to HE owing to their high strength [199,408,409,426]. Currently, although several attempts to produce high-strength HEAs by AM have been successful, the study of HE behaviour of the AM HEAs is still limited. The knowledge gaps of the current research on the effect of the HE performance of HEAs include but are not limited to different feeding mechanisms, the type of heat sources, and the direction of printing. Therefore, for the development of HEAs' practical applications, it is necessary to systematically and comprehensively investigate the HE properties of this material.

4.6. Summary

4.6.1. The dominant HE mechanism of AM metals

Table 8 lists several alloys that have been identified the prevailing HE mechanisms and have been fabricated using AM and traditional methods. The predominant HE mechanism for AM metals is HELP. But, some alloys also cite the synergistic action of HELP and HEDE. For Ti alloys, the mechanism involves HFC.

Currently, there is no obvious discernible distinction in HE

Table 8
The dominant HE mechanisms of metals by additive manufacturing (AM) and traditional (TR) methods.

Materials	AM	Ref.	TR	Ref.
Ti-6Al-4 V	HFC	[29–37]	HFC	[32]
IN 718	HELP + HEDE	[23,28]	HELP + HEDE	[216,217]
AISI 304L	HELP + HEDE	[13,16]	HELP + HEDE	[16,427]
AISI 316L	HELP	[17–20]	HELP	[217,428]
17-4PH	HELP	[55]	HELP	[55,429]
18Ni 300	HELP	[57,254–256]	HELP	[356,361]
CoCrFeNiMn	HELP[1]	[257,258][1]	HELP	[407–411]
	HELP-mediated HEDE[2]	[223][2]		
$AlCoCrFeNi_{2.1}$	HELP + HEDE	[199]	HELP + HEDE	[199]

mechanisms between materials manufactured through AM and those produced using traditional methods. Nevertheless, the susceptibility of metals to HE remains disparate across various manufacturing methods, largely attributable to the distinct microstructural characteristics inherent in materials produced through AM.

4.6.2. Influences of microstructural features of AM metals on HE

As previously mentioned, AM processes are characterised with rapid cooling and directional solidification within small melt pools. Consequently, the microstructures and properties of metals produced through AM significantly differ from those manufactured using conventional methods [8,51]. It is widely acknowledged that HE is influenced by a number of factors, including sample density and microstructural characteristics such as dislocation density, grain boundaries, and internal cracks and voids [40]. The major differences in microstructural features between AM metals and conventionally produced metals, and their influence on HE are summarised as follows:

4.6.2.1. Metallurgical defects. AM processes have been known to introduce metallurgical defects, including porosities, cracks, and inclusions [51,99]. These defects can serve as sites for hydrogen absorption and subsequently promote embrittlement [40]. In addition, defects and micro-segregation in AM components can promote microcrack coalescence, resulting in a more severe reduction in ductility [15,258]. However, the majority of the current research primarily focuses on evaluating the HE performance of high-density samples, with limited consideration to the influence of material defects on HE. Given that defects are a prevalent microstructural feature in AM metals, it is imperative to place greater emphasis on investigating their roles in HE.

4.6.2.2. The finer microstructure. Finer microstructures associated with AM metals also influence HE susceptibility by increasing the boundary density [199] and dislocation density [55]. Previous studies have demonstrated that finer microstructure in AM samples often results in increased hydrogen uptake [199]. However, in some cases, AM samples with finer microstructure may exhibit a lower HE susceptibility, such as austenite in AM 304L SS [13,16]. The high cooling rate during the AM process generally results in a finer and more homogeneous microstructure with a distinctive solidification cellular structure. This type of microstructure can suppress dislocation slip and deformation twinning during deformation, resulting in the reduction in the nucleation sites of α' martensite. Thus, compared with traditionally fabricated ASSs, the strain-induced α' martensitic transformation in AM ASSs is suppressed, and the austenitic stability is enhanced [13,16,430]. As martensite is more susceptible to hydrogen in comparison to other ferrous microstructures, and materials undergoing martensitic transformation are notably sensitive to HE [225–227]. Therefore, the utilisation of AM to suppress strain-induced martensitic transformation presents a feasible approach for enhancing the HE resistance of austenitic steels, deserving further research attention. In addition, extensive research conducted on typical martensitic steels has consistently demonstrated that materials produced through AM exhibit substantially finer microstructure, such as smaller martensitic laths and blocks, when compared to materials manufactured using traditional methods [106,117,431]. It can be anticipated that this discrepancy in microstructure will also result in discernible differences in their resistance to HE. Nevertheless, the relevant research in this topic remains relatively limited, underscoring the need for further extensive investigations.

Moreover, materials produced with different AM methods also differ in their microstructures. For instance, EBM Ti-6Al-4 V has a lower amount of discontinuous β-phase, absorbing less hydrogen than the wrought alloy, while the SLM Ti-6Al-4 V demonstrated a higher trend of hydrogen absorption and hydride precipitation than the EBM counterparts because of the α'-martensite microstructure [31,34]. Thus, subsequent research cannot be limited to comparing the HE performance of a single AM method with traditional methods.

4.6.2.3. Anisotropic properties. In addition to the finer microstructure, most AM metals also exhibit pronounced anisotropy due to the directional solidification feature that results in the formation of columnar grains, distinguishing them from their wrought counterparts [95,318–321]. This inherent anisotropy can introduce variations in susceptibility to HE along different orientations, necessitating a comprehensive study of the build orientations effects on the hydrogen sensitivity of the AM metals. Currently, several studies have indicated that vertical (i.e., obtained parallel to the printing direction) AM IN718 samples exhibit superior HE resistance compared to their horizontal (i.e., obtained normal to the printing direction) counterparts. This improved resistance can be attributed to the suppression of hydrogen entrance by columnar dendrites [23,24]. However, tests on AM Ti-6Al-4 V showed contrasting results, with vertical samples displaying lower HE resistance compared to samples oriented horizontally. This disparity can be ascribed to the horizontal direction may have a high-stress intensity environment that reduced the probability of the stress-induced hydride mechanism [30]. As such, the current body of research cannot definitively determine whether vertically constructed samples outperform horizontally constructed ones in terms of HE resistance. Nonetheless, it is evident that a strong correlation exists between HE performance and the anisotropic characteristics of the samples, emphasising the need for further investigation in future studies.

In summary, the microstructural characteristics of AM metals make their HE susceptibility differing to the counterpart metals produced using conventional methods. A comprehensive understanding of AM microstructural features that affect HE is essential for advancing the knowledge and mitigation strategies related to HE in AM metals. Further research is needed to explore these intricate relationships and develop effective solutions to enhance the resistance of AM metals to HE.

5. Conclusions and outlook

The present state of research on the HE of metals fabricated using AM is presented in this review with a focus on the mechanisms and primary influencing factors, including Ti alloys, Ni-based superalloys, steels, and high-entropy alloys. This paper also reviews some common materials that are sensitive to hydrogen but lack research on processing by AM, including AISI 4340, and some promising AM metals which lack literature dedicated to the study of HE properties, including $AlCoCrFeNi_{2.1}$ eutectic HEA and H13.

Currently, research on HE of AM metals has primarily identified the predominant HE mechanism to be the hydrogen-enhanced localised plasticity (HELP) mechanism, though others also identified the synergistic action of HELP and HEDE. For titanium alloys, the predominant mechanism involved hydride formation and cleavage (HFC). At present, there is no discernible distinction in HE mechanisms between materials manufactured through AM and those produced through traditional methods. The most recent studies have found better HE resistance for AM metals, for example, DMT and SLM 304L SS [13,16] and SLM 17-4PH [55]. In contrast, some of the researchers have found the opposite, such as EBM Ti-6Al-4 V was more prone to hydrogen-induced cracking at interphase boundaries [32], and DED 316L SS had enhanced HE susceptibility [18]. In addition, the manufacturing methods, including the direction of printing (horizontal or vertical) [23,24,30], feeding mechanism (DED or PBF) [15,18], the type of heat source (laser beam or electron beam) [34] and post-heat treatment conditions [16,23,25,27,57,257], can also significantly affect the HE behaviour of the AM alloys.

However, investigations on the HE susceptibility of numerous materials processed by AM remain scarce, with few studies dedicated to understanding the HE characteristics of AM metals [13,32,54,59,286]. This review highlights the following research gaps that merit future

exploration:

(i) There is a lack of literature dedicated to the study of HE behaviours of some typical metals that can be fabricated by AM such as H13 and $AlCoCrFeNi_{2.1}$.
(ii) Due to the variations in hydrogen-charging conditions, chemical composition, or the testing environment, there is a lack of an adopted standard for assessing HE in AM metals at present, which is causing difficulty in comparing results, even resulting in conflicting conclusions.
(iii) The influence of microstructure on HE of AM metals is not yet well established, as evidenced by conflicting conclusions, which needs clarification to fully understand the principles responsible for the embrittlement.
(iv) Comparative studies on the initial microstructure differences between AM and traditionally processed materials are lacking, including the form and spacing of dislocation structures which are related to the variances in hydrogen response.

This review has provided in-depth insights into the current state of knowledge regarding HE in metals produced through AM, based on previously published results on this topic. While the investigation of HE in AM metals is experiencing substantial growth, a notable gap in comprehensive reviews in this area persists. This paper successfully fills this gap. The following observations were drawn from this review, and can be considered for future study in this domain:

(i) There exists a disparity in HE susceptibility between AM metals and conventionally manufactured metals primarily stems from the microstructural features formed during the AM fabrication process
(ii) For the same metal, employing different AM techniques leads to differing HE susceptibility
(iii) Conflicting results from different studies appear due to the differences in experimental conditions
(iv) AM has the potential to create HE-resistant metals

In conclusion, based on the overview of current research progress in the hydrogen embrittlement of additively manufactured metals, it is foreseeable that additive manufacturing will emerge as a prominent processing method in the future for improving material resistance to hydrogen embrittlement.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This work was financially supported by the ARC Discovery Project for funding support (No. DP210103162). The authors are grateful to the ARC Discovery Project (No. DP210103162) for funding support. Jeffrey Venezuela acknowledges the support of the Advance Queensland Industry Research Fellowship.

References

[1] N. Bailey, 1 - Factors influencing weldability, in: N. Bailey (Ed.), Weldability of Ferritic Steels, Woodhead Publishing, 1994, 1–44.
[2] 11 - Steels for aircraft structures, in: A.P. Mouritz (Ed.), Introduction to Aerospace Materials, Woodhead Publishing, 2012, pp. 232-250.
[3] S.G.K. Manikandan, D. Sivakumar, M. Kamaraj, 7 - Hydrogen embrittlement sensitivity of Laves phase, in: S.G.K. Manikandan, D. Sivakumar, M. Kamaraj (Eds.), Welding the Inconel 718 Superalloy, Elsevier, 2019, pp. 231-263.
[4] S. Mirzababaei, S. Pasebani, A review on binder jet additive manufacturing of 316L stainless steel, J. Manuf. Mater. Proc. 3 (3) (2019) 36.
[5] B. Sun, D. Wang, X. Lu, D. Wan, D. Ponge, X. Zhang, Current challenges and opportunities toward understanding hydrogen embrittlement mechanisms in advanced high-strength steels: a review, Acta Metall. Sinica (English Letters) 34 (6) (2021) 741–754.
[6] W.H. Johnson II, On some remarkable changes produced in iron and steel by the action of hydrogen and acids, Proc. R. Soc. Lond. 23 (156–163) (1875) 168–179.
[7] I. Gibson, D.W. Rosen, B. Stucker, M. Khorasani, D. Rosen, B. Stucker, M. Khorasani, Additive manufacturing technologies, Springer, 2021.
[8] Y. Yin, Q. Tan, M. Bermingham, N. Mo, J. Zhang, M.-X. Zhang, Laser additive manufacturing of steels, Int. Mater. Rev. (2021) 1–87.
[9] D. Herzog, V. Seyda, E. Wycisk, C. Emmelmann, Additive manufacturing of metals, Acta Mater. 117 (2016) 371–392.
[10] J.O. Milewski, Additive manufacturing of metals, Appl. Mech. Mater. (2017).
[11] T. DebRoy, H.L. Wei, J.S. Zuback, T. Mukherjee, J.W. Elmer, J.O. Milewski, A. M. Beese, A. Wilson-Heid, A. De, W. Zhang, Additive manufacturing of metallic components – Process, structure and properties, Prog. Mater Sci. 92 (2018) 112–224.
[12] M.M. Francois, A. Sun, W.E. King, N.J. Henson, D. Tourret, C.A. Bronkhorst, N. N. Carlson, C.K. Newman, T. Haut, J. Bakosi, Modeling of additive manufacturing processes for metals: Challenges and opportunities, Curr. Opin. Solid State Mater. Sci. 21 (4) (2017) 198–206.
[13] S.W. Baek, E.J. Song, J.H. Kim, M. Jung, U.B. Baek, S.H. Nahm, Hydrogen embrittlement of 3-D printing manufactured austenitic stainless steel part for hydrogen service, Scr. Mater. 130 (2017) 87–90.
[14] A. McWilliams, M. Morgan, P. Korniko, Hydrogen effects on fracture toughness of additively manufactured type 304L stainless steel, ASME Pressure Vessels and Piping Conference, Amer Soc Mechanical Engineers, San Antonio, TX, 2019.
[15] T.R. Smith, C. San Marchi, J.D. Suga, D.K. Balch, Effects of extreme hydrogen environments on the fracture and fatigue behavior of additively manufactured stainless steels, ASME Pressure Vessels and Piping Conference, Amer Soc Mechanical Engineers, San Antonio, TX, 2019.
[16] D.-H. Lee, B. Sun, S. Lee, D. Ponge, E.A. Jägle, D. Raabe, Comparative study of hydrogen embrittlement resistance between additively and conventionally manufactured 304L austenitic stainless steels, Mater. Sci. Eng. A 803 (2021).
[17] J.T. Miller, H.J. Martin, E. Cudjoe, Comparison of the effects of a sulfuric acid environment on traditionally manufactured and additive manufactured stainless steel 316L alloy, Addit. Manuf. 23 (2018) 272–286.
[18] K.M. Bertsch, A. Nagao, B. Rankouhi, B. Kuehl, D.J. Thoma, Hydrogen embrittlement of additively manufactured austenitic stainless steel 316L, Corros. Sci. 192 (2021).
[19] P. Metalnikov, G. Ben-Hamu, D. Eliezer, Hydrogen Trapping in Laser Powder Bed Fusion 316L Stainless Steel, Metals 12 (10) (2022).
[20] J. Liu, H. Yang, L. Meng, D. Liu, T. Xu, D. Xu, X. Shao, C. Shao, S. Li, P. Zhang, Z. Zhang, Significance of mMelt Pool structure on the hydrogen embrittlement behavior of a selective laser-melted 316L austenitic stainless steel, Materials 16 (4) (2023) 1741.
[21] N. Ehrlin, M. Fisk, C. Bjerken, Flow stress model for hydrogen degraded Inconel 718, Mech. Mater. 119 (2018) 56–64.
[22] S. Kolesov, E. Alekseeva, Evaluation of the Stress-Strain State in Alloy 718 After Hydrogen Charging, 9th International Symposium on Superalloy 718 and Derivatives - Energy, Aerospace, and Industrial Applications, Springer International Publishing Ag, Champion, PA, 2018, pp. 553–563.
[23] X.F. Li, Q.Z. Li, T. Wang, J. Zhang, Hydrogen-assisted failure of laser melting additive manufactured IN718 superalloy, Corros. Sci. 160 (2019) 14.
[24] J. Hesketh, N. McClelland, Y. Zhang, C. Green, A. Turnbull, Influence of additive manufacturing by laser powder bed fusion on the susceptibility of Alloy 718 to hydrogen embrittlement, Corros. Eng. Sci. Technol. 56 (6) (2021) 565–574.
[25] J. Yoo, S. Kim, M.C. Jo, H. Park, J.E. Jung, J. Do, D.W. Yun, I.S. Kim, B.-G. Choi, Investigation of hydrogen embrittlement properties of Ni-based alloy 718 fabricated via laser powder bed fusion, Int. J. Hydrogen Energy 47 (43) (2022) 18892–18910.
[26] S. Kevinsannya, O. Okazaki, Y. Takakuwa, Y. Ogawa, H. Funakoshi, S. Kawashima, H. Matsuoka, Matsunaga, Defect tolerance and hydrogen susceptibility of the fatigue limit of an additively manufactured Ni-based superalloy 718, Int. J. Fatigue 139 (2020) 10.
[27] J. Xu, Z. Hao, Z. Fu, X. He, H. Wang, G. Xu, Hydrogen embrittlement behavior of selective laser-melted Inconel 718 alloy, J. Mater. Res. Technol. 23 (2023) 359–369.
[28] N.K. Mohandas, A. Giorgini, M. Vanazzi, T. Riemslag, S.P. Scott, V. Popovich, Hydrogen embrittlement of Inconel 718 manufactured by laser powder bed fusion using sustainable feedstock: effect of heat treatment and microstructural anisotropy, Metals 13 (2) (2023) 418.
[29] N. Pushilina, A. Panin, M. Syrtanov, E. Kashkarov, V. Kudiiarov, O. Perevalova, R. Laptev, A. Lider, A. Koptyug, Hydrogen-induced phase transformation and microstructure evolution for Ti-6Al-4V parts produced by electron beam melting, Metals 8 (5) (2018) 16.
[30] R. Silverstein, D. Eliezer, Hydrogen trapping in 3D-printed (additive manufactured) Ti-6Al-4V, Mater. Charact. 144 (2018) 297–304.
[31] P. Metalnikov, D. Eliezer, G. Ben-Hamu, E. Tal-Gutelmacher, Y. Gelbstein, C. Munteanu, Hydrogen embrittlement of electron beam melted Ti-6Al-4V, J. Mater. Res. Technol. 9 (6) (2020) 16126–16134.

[32] N.U. Navi, J. Tenenbaum, E. Sabatani, G. Kimmel, R. Ben David, B.A. Rosen, Z. Barkay, V. Ezersky, E. Tiferet, Y.I. Ganor, N. Eliaz, Hydrogen effects on electrochemically charged additive manufactured by electron beam melting (EBM) and wrought Ti-6Al-4V alloys, Int. J. Hydrogen Energy 45 (46) (2020) 25523–25540.
[33] E. Brosh, N.U. Navi, B.A. Rosen, N. Eliaz, Microvoids in electrochemically hydrogenated titanium-based alloys, Int. J. Hydrogen Energy 46 (53) (2021) 27234–27242.
[34] P. Metalnikov, D. Eliezer, G. Ben-Hamu, Hydrogen trapping in additive manufactured Ti-6Al-4V alloy, Mater. Sci. Eng. A-Struct. Mater. Prop. Microstruct. Process. 811 (2021) 10.
[35] Q.M. Shi, Y. Sun, S.F. Yang, J. Van Dessel, H.T. Lubbers, S.P. Zhong, Y.F. Gu, M. Bila, T. Dormaar, J. Schoenaers, C. Politis, Failure analysis of an in-vivo fractured patient-specific Ti6Al4V mandible reconstruction plate fabricated by selective laser melting, Eng. Fail. Anal. 124 (2021) 16.
[36] D. Kong, D. Zhao, G. Zhu, X. Ni, L. Zhang, W. Wu, C. Man, Y. Zhou, C. Dong, B. Sun, Heat treatment effects on the hydrogen embrittlement of Ti6Al4V fabricated by laser beam powder bed fusion, Addit. Manuf. 50 (2022).
[37] N. Lulu-Bitton, E. Sabatani, B.A. Rosen, N. Kostirya, G. Agronov, E. Tiferet, N. Eliaz, N.U. Navi, Mechanical behavior of electrochemically hydrogenated electron beam melting (EBM) and wrought Ti–6Al–4V using small punch test, Int. J. Hydrogen Energy 47 (9) (2022) 6388–6403.
[38] S.K. Dwivedi, M. Vishwakarma, Hydrogen embrittlement in different materials: A review, Int. J. Hydrogen Energy 43 (46) (2018) 21603–21616.
[39] J. Venezuela, Q. Liu, M. Zhang, Q. Zhou, A. Atrens, A review of hydrogen embrittlement of martensitic advanced high-strength steels, Corros. Rev. 34 (3) (2016) 153–186.
[40] S. Lynch, Hydrogen embrittlement phenomena and mechanisms, Corros. Rev. 30 (3–4) (2012).
[41] L.E. Murr, S.M. Gaytan, D.A. Ramirez, E. Martinez, J. Hernandez, K.N. Amato, P. W. Shindo, F.R. Medina, R.B. Wicker, Metal fabrication by additive manufacturing using laser and electron beam melting technologies, J. Mater. Sci. Technol. 28 (1) (2012) 1–14.
[42] Q. Liu, Q. Zhou, J. Venezuela, M. Zhang, J. Wang, A. Atrens, A review of the influence of hydrogen on the mechanical properties of DP, TRIP, and TWIP advanced high-strength steels for auto construction, Corros. Rev. 34 (3) (2016) 127–152.
[43] P. Bajaj, A. Hariharan, A. Kini, P. Kürnsteiner, D. Raabe, E.A. Jägle, Steels in additive manufacturing: a review of their microstructure and properties, Mater. Sci. Eng. A 772 (2020), 138633.
[44] M.B. Djukic, G.M. Bakic, V. Sijacki Zeravcic, A. Sedmak, B. Rajicic, The synergistic action and interplay of hydrogen embrittlement mechanisms in steels and iron: Localized plasticity and decohesion, Eng. Fracture Mech. 216 (2019).
[45] Q. Liu, A. Atrens, A critical review of the influence of hydrogen on the mechanical properties of medium-strength steels, 31(3-6) (2013) 85-103.
[46] S. Ramamurthy, A. Atrens, Stress corrosion cracking of high-strength steels, Corros. Rev. 31 (1) (2013) 1–31.
[47] N. Winzer, A. Atrens, W. Dietzel, G. Song, K.U. Kainer, Evaluation of the delayed hydride cracking mechanism for transgranular stress corrosion cracking of magnesium alloys, Mater. Sci. Eng. A 466 (1) (2007) 18–31.
[48] J.J. Lewandowski, M. Seifi, Metal additive manufacturing: a review of mechanical properties, Annu. Rev. Mat. Res. 46 (1) (2016) 151–186.
[49] Z. Zeng, M. Salehi, A. Kopp, S. Xu, M. Esmaily, N. Birbilis, Recent progress and perspectives in additive manufacturing of magnesium alloys, J. Magnesium Alloys 10 (6) (2022) 1511–1541.
[50] A.C.F.o.A.M. Technologies, A.C.F.o.A.M.T.S.F.o. Terminology, Standard terminology for additive manufacturing technologies, Astm International, 2012.
[51] S. Gorsse, C. Hutchinson, M. Goune, R. Banerjee, Additive manufacturing of metals: a brief review of the characteristic microstructures and properties of steels, Ti-6Al-4V and high-entropy alloys, Sci. Technol. Adv. Mater. 18 (1) (2017) 584–610.
[52] I. Buj-Corral, A. Tejo-Otero, F. Fenollosa-Artés, Development of AM technologies for metals in the sector of medical implants, Metals 10 (5) (2020).
[53] F. Neugebauer, N. Keller, V. Ploshikhin, F. Feuerhahn, H. Köhler, Multi scale FEM simulation for distortion calculation in additive manufacturing of hardening stainless steel, International Workshop on Thermal Forming and Welding Distortion, Bremen, Germany, 2014.
[54] S.L. Li, M.M. Liu, Y. Ren, Y.D. Wang, Hydrogen embrittlement behaviors of additive manufactured maraging steel investigated by in situ high-energy X-ray diffraction, Mater. Sci. Eng. A-Struct. Mater. Prop. Microstruct. Process. 766 (2019), 138341.
[55] M. Alnajjar, F. Christien, C. Bosch, K. Wolski, A comparative study of microstructure and hydrogen embrittlement of selective laser melted and wrought 17–4 PH stainless steel, Mater. Sci. Eng. A-Struct. Mater. Prop. Microstruct. Process. 785 (2020) 12.
[56] Z.J. Sun, Y.Y. Xu, F.D. Chen, L.D. Shen, X.B. Tang, L.W. Sun, M.Y. Fan, P. Huang, Effects of ion irradiation on microstructure of 316L stainless steel strengthened by disperse nano TiC through selective laser melting, Mater Charact 180 (2021) 10.
[57] Y.J. Kwon, R. Casati, M. Coduri, M. Vedani, C.S. Lee, Hydrogen embrittlement behavior of 18ni 300 maraging steel produced by selective laser melting, Materials (Basel) 12 (15) (2019) 12.
[58] C. Ledford, C. Rock, P. Carriere, P. Frigola, D. Gamzina, T. Horn, Characteristics and processing of hydrogen-treated copper powders for eb-pbf additive manufacturing, Appl. Sci.-Basel 9 (19) (2019) 22.
[59] E. Stepanova, N. Pushilina, M. Syrtanov, R. Laptev, E. Kashkarov, Hydrogen effect on Ti-6.5Al-3.5Mo-1.5Zr-0.3Si parts produced by electron beam melting, Int. J. Hydrogen Energy 44 (55) (2019) 29380–29388.
[60] M.Y. Panchenko, G.G. Maier, E.V. Melnikov, V.A. Moskvina, K.A. Reunova, S. V. Astafurov, V.E. Rubtsov, E.A. Kolubaev, E.G. Astafurova, The Peculiarities of Hydrogen Embrittlement of Nb-alloyed Stainless Steel Fabricated by Electron-beam Additive Manufacturing, Amer Inst Physics, Tomsk, RUSSIA, 2020.
[61] P.A. Hooper, Melt pool temperature and cooling rates in laser powder bed fusion, Addit. Manuf. 22 (2018) 548–559.
[62] M. Uddagiri, O. Shchyglo, I. Steinbach, B. Wahlmann, C. Koerner, phase-field study of the history-effect of remelted microstructures on nucleation during additive manufacturing of Ni-based superalloys, Metall. Mater. Trans. A 54 (5) (2023) 1825–1842.
[63] E. Yasa, Chapter 3 - Selective laser melting: principles and surface quality, in: J. Pou, A. Riveiro, J.P. Davim (Eds.), Addit Manuf, Elsevier2021, pp. 77-120.
[64] D. Ding, Z. Pan, D. Cuiuri, H. Li, Wire-feed additive manufacturing of metal components: technologies, developments and future interests, Int. J. Adv. Manuf. Technol. 81 (1–4) (2015) 465–481.
[65] K. Taminger, R.A. Hafley, Electron beam freeform fabrication: a rapid metal deposition process, 3rd annual automotive composites conference, 2003.
[66] S.L. Sing, C.F. Tey, J.H.K. Tan, S. Huang, W.Y. Yeong, 3D printing of metals in rapid prototyping of biomaterials: Techniques in additive manufacturing, Rapid Prototyping of Biomaterials (2020) 17–40.
[67] A. Singh, S. Kapil, M. Das, A comprehensive review of the methods and mechanisms for powder feedstock handling in directed energy deposition, Addit. Manuf. 35 (2020).
[68] X. Zhao, S. Li, M. Zhang, Y. Liu, T.B. Sercombe, S. Wang, Y. Hao, R. Yang, L. E. Murr, Comparison of the microstructures and mechanical properties of Ti–6Al–4V fabricated by selective laser melting and electron beam melting, Mater. Des. 95 (2016) 21–31.
[69] S. Lambrakos, K. Cooper, An algorithm for inverse modeling of layer-by-layer deposition processes, J. Mater. Eng. Perform. 18 (3) (2009) 221–230.
[70] S. Lambrakos, K. Cooper, A general algorithm for inverse modeling of layer-by-layer liquid-metal deposition, J. Mater. Eng. Perform. 19 (3) (2010) 314–324.
[71] I. Yadroitsev, A. Gusarov, I. Yadroitsava, I. Smurov, Single track formation in selective laser melting of metal powders, J Mater Process Tech 210 (12) (2010) 1624–1631.
[72] J.W. Xie, P. Fox, W. O'Neill, C. Sutcliffe, Effect of direct laser re-melting processing parameters and scanning strategies on the densification of tool steels, J Mater Process Tech 170 (3) (2005) 516–523.
[73] J. Choi, Y. Chang, Characteristics of laser aided direct metal/material deposition process for tool steel, Int J Mach Tool Manu 45 (4–5) (2005) 597–607.
[74] K. Zhang, W. Liu, Influences of processing parameters on microstructure and properties of laser direct deposited stainless steel components, 2009 Second International Conference on Intelligent Computation Technology and Automation, 2009, pp. 856-859.
[75] Z.L. Lu, D.C. Li, Z.Q. Tong, Q.P. Lu, M.M. Traore, A.F. Zhang, B.H. Lu, Investigation into the direct laser forming process of steam turbine blade, Opt Laser Eng 49 (9–10) (2011) 1101–1110.
[76] K. M., K. M., T. T., Influence of technological parameters on additive manufacturing steel parts in selective laser sintering, Archives Mater. Sci. Eng. 67 (2) (2014) 84-92.
[77] V.P. Matilainen, H. Piili, A. Salminen, O. Nyrhila, Preliminary investigation of keyhole phenomena during single layer fabrication in laser additive manufacturing of stainless steel, Physcs Proc 78 (2015) 377–387.
[78] Z.Q. Wang, T.A. Palmer, A.M. Beese, Effect of processing parameters on microstructure and tensile properties of austenitic stainless steel 304L made by directed energy deposition additive manufacturing, Acta Mater. 110 (2016) 226–235.
[79] N. Shamsaei, A. Yadollahi, L. Bian, S.M. Thompson, An overview of direct laser deposition for additive manufacturing; part ii: mechanical behavior, process parameter optimization and control, Addit. Manuf. 8 (2015) 12–35.
[80] A.O. Moghaddam, N.A. Shaburova, M.N. Samodurova, A. Abdollahzadeh, E. A. Trofimov, Additive manufacturing of high entropy alloys: A practical review, J. Mater. Sci. Technol. 77 (2021) 131–162.
[81] S. Sanchez, P. Smith, Z. Xu, G. Gaspard, C.J. Hyde, W.W. Wits, I.A. Ashcroft, H. Chen, A.T. Clare, Powder bed fusion of nickel-based superalloys: A review, Int J Mach Tool Manu 165 (2021), 103729.
[82] N. Haghdadi, M. Laleh, M. Moyle, S. Primig, Additive manufacturing of steels: a review of achievements and challenges, J. Mater. Sci. 56 (1) (2020) 64–107.
[83] S. Liu, Y.C. Shin, Additive manufacturing of Ti6Al4V alloy: A review, Mater. Des. 164 (2019), 107552.
[84] Y. Li, X. Liang, Y. Yu, D. Wang, F. Lin, Review on additive manufacturing of single-crystal nickel-based superalloysa, Chin. J. Mech. Eng.: Additive Manuf. Front. 1 (1) (2022), 100019.
[85] S.M. Yusuf, N. Gao, Influence of energy density on metallurgy and properties in metal additive manufacturing, Mater. Sci. Technol. 33 (11) (2017) 1269–1289.
[86] G. Pi, A. Zhang, G. Zhu, D. Li, B. Lu, Research on the forming process of three-dimensional metal parts fabricated by laser direct metal forming, Int. J. Adv. Manuf. Technol. 57 (9–12) (2011) 841–847.
[87] M. Griffith, D. Keicher, C.L. Atwood, Free form fabrication of metallic components using laser engineered net shaping (LENS), Sandia National Lab. (SNL-NM), Albuquerque, NM (United States), 1996.
[88] J. Smugeresky, D. Keicher, J. Romero, M. Griffith, L. Harwell, Laser engineered net shaping (LENS) process: optimization of surface finish and microstructural

properties, Sandia National Lab.(SNL-NM), Albuquerque, NM (United States), 1997.
[89] M. Ma, Z. Wang, D. Wang, X. Zeng, Control of shape and performance for direct laser fabrication of precision large-scale metal parts with 316L stainless steel, Opt. Laser Technol. 45 (2013) 209–216.
[90] F. Wang, J. Mei, X. Wu, Direct laser fabrication of Ti6Al4V/TiB, J Mater Process Tech 195 (1–3) (2008) 321–326.
[91] J. Milewski, G. Lewis, D. Thoma, G. Keel, R. Nemec, R. Reinert, Directed light fabrication of a solid metal hemisphere using 5-axis powder deposition, J Mater Process Tech 75 (1–3) (1998) 165–172.
[92] R. Mah, Directed light fabrication, Adv. Mater. Process. 151 (3) (1997) 31–34.
[93] J. Mazumder, J. Choi, K. Nagarathnam, J. Koch, D. Hetzner, The direct metal deposition of H13 tool steel for 3-D components, JOM 49 (5) (1997) 55–60.
[94] S. Bhattacharya, G.P. Dinda, A.K. Dasgupta, J. Mazumder, Microstructural evolution of AISI 4340 steel during Direct Metal Deposition process, Mater. Sci. Eng. A-Struct. Mater. Prop. Microstruct. Process. 528 (6) (2011) 2309–2318.
[95] T. Niendorf, S. Leuders, A. Riemer, H.A. Richard, T. Tröster, D. Schwarze, Highly anisotropic steel processed by selective laser melting, Metall. Mater. Trans. B 44 (4) (2013) 794–796.
[96] Y. Zhang, W. Jarosinski, Y.-G. Jung, J. Zhang, Additive manufacturing processes and equipment, Addit. Manuf. (2018) 39–51.
[97] E. Tascioglu, Y. Karabulut, Y. Kaynak, Influence of heat treatment temperature on the microstructural, mechanical, and wear behavior of 316L stainless steel fabricated by laser powder bed additive manufacturing, Int. J. Adv. Manuf. Technol. 107 (5) (2020) 1947–1956.
[98] M. Seifi, A. Salem, J. Beuth, O. Harrysson, J.J. Lewandowski, Overview of materials qualification needs for metal additive manufacturing, JOM 68 (3) (2016) 747–764.
[99] J. Delgado, J. Ciurana, L. Serenó, Comparison of forming manufacturing processes and selective laser melting technology based on the mechanical properties of products: in this work, the superior property of the selective laser melting technology is presented by comparing four real parts manufactured using forming processes and selective laser melting technology and analysed for tension, compression and flexural, Virtual Phys. Prototyping 6 (3) (2011) 167–178.
[100] L.E. dos Santos Paes, M. Pereira, F.A. Xavier, W.L. Weingaertner, L.O. Vilarinho, Lack of fusion mitigation in directed energy deposition with laser (DED-L) additive manufacturing through laser remelting, J. Manuf. Process. 73 (2022) 67–77.
[101] J. Zhou, X. Han, H. Li, S. Liu, J. Yi, Investigation of layer-by-layer laser remelting to improve surface quality, microstructure, and mechanical properties of laser powder bed fused AlSi10Mg alloy, Mater. Des. 210 (2021), 110092.
[102] J.H. Martin, B.D. Yahata, J.M. Hundley, J.A. Mayer, T.A. Schaedler, T.M. Pollock, 3D printing of high-strength aluminium alloys, Nature 549 (7672) (2017) 365–369.
[103] Y. Kok, X.P. Tan, P. Wang, M.L.S. Nai, N.H. Loh, E. Liu, S.B. Tor, Anisotropy and heterogeneity of microstructure and mechanical properties in metal additive manufacturing: A critical review, Mater. Des. 139 (2018) 565–586.
[104] Q. Tan, J. Zhang, N. Mo, Z. Fan, Y. Yin, M. Bermingham, Y. Liu, H. Huang, M.-X. Zhang, A novel method to 3D-print fine-grained AlSi10Mg alloy with isotropic properties via inoculation with LaB6 nanoparticles, Addit. Manuf. 32 (2020), 101034.
[105] A. Takaichi, T. Suyalatu, N. Nakamoto, N. Joko, Y. Nomura, S. Tsutsumi, H. Migita, S. Doi, A. Kurosu, N. Chiba, Y. Wakabayashi, T. Igarashi, Hanawa, Microstructures and mechanical properties of Co-29Cr-6Mo alloy fabricated by selective laser melting process for dental applications, J. Mech. Behav. Biomed. Mater. 21 (2013) 67–76.
[106] Q. Tan, H. Chang, Y. Yin, F. Wang, D. Huang, G. Liang, T. Wu, M. Yan, X. Cheng, M.-X. Zhang, Simultaneous enhancements of strength and ductility of a selective laser melted H13 steel through inoculation treatment, Scr. Mater. 219 (2022), 114874.
[107] S.A. Mantri, T. Alam, D. Choudhuri, C.J. Yannetta, C.V. Mikler, P.C. Collins, R. Banerjee, The effect of boron on the grain size and texture in additively manufactured β-Ti alloys, J. Mater. Sci. 52 (20) (2017) 12455–12466.
[108] M.J. Bermingham, D.H. StJohn, J. Krynen, S. Tedman-Jones, M.S. Dargusch, Promoting the columnar to equiaxed transition and grain refinement of titanium alloys during additive manufacturing, Acta Mater. 168 (2019) 261–274.
[109] C.H. Ng, M.J. Bermingham, L. Yuan, M.S. Dargusch, Towards β-fleck defect free additively manufactured titanium alloys by promoting the columnar to equiaxed transition and grain refinement, Acta Mater. 224 (2022), 117511.
[110] S.Y. Zhou, Y. Su, H. Wang, J. Enz, T. Ebel, M. Yan, Selective laser melting additive manufacturing of 7xxx series Al-Zn-Mg-Cu alloy: Cracking elimination by co-incorporation of Si and TiB2, Addit. Manuf. 36 (2020), 101458.
[111] X.P. Li, G. Ji, Z. Chen, A. Addad, Y. Wu, H.W. Wang, J. Vleugels, J. Van Humbeeck, J.P. Kruth, Selective laser melting of nano-TiB2 decorated AlSi10Mg alloy with high fracture strength and ductility, Acta Mater. 129 (2017) 183–193.
[112] Q. Tan, J. Zhang, Q. Sun, Z. Fan, G. Li, Y. Yin, Y. Liu, M.-X. Zhang, Inoculation treatment of an additively manufactured,, aluminium alloy with titanium nanoparticles, Acta Mater. 196 (2020) (2024) 1–16.
[113] B. Li, B. Qian, Y. Xu, Z. Liu, J. Zhang, F. Xuan, Additive manufacturing of ultrafine-grained austenitic stainless steel matrix composite via vanadium carbide reinforcement addition and selective laser melting: Formation mechanism and strengthening effect, Mater. Sci. Eng. A 745 (2019) 495–508.
[114] W. Zhai, Z. Zhu, W. Zhou, S.M.L. Nai, J. Wei, Selective laser melting of dispersed TiC particles strengthened 316L stainless steel, Compos. B Eng. 199 (2020), 108291.
[115] Y. Wang, Z. Liu, Y. Zhou, X. Yang, J. Tang, X. Liu, J. Li, G. Le, Microstructure and mechanical properties of TiN particles strengthened 316L steel prepared by laser melting deposition process, Mater. Sci. Eng. A 814 (2021), 141220.
[116] A. Durga, N.H. Pettersson, S.B.A. Malladi, Z. Chen, S. Guo, L. Nyborg, G. Lindwall, Grain refinement in additively manufactured ferritic stainless steel by in situ inoculation using pre-alloyed powder, Scr. Mater. 194 (2021), 113690.
[117] Q. Tan, Y. Yin, F. Wang, H. Chang, S. Liu, G. Liang, T. Wu, M.-X. Zhang, Rationalization of brittleness and anisotropic mechanical properties of H13 steel fabricated by selective laser melting, Scr. Mater. 214 (2022), 114645.
[118] C.-Y. Chou, N.H. Pettersson, A. Durga, F. Zhang, C. Oikonomou, A. Borgenstam, J. Odqvist, G. Lindwall, Influence of solidification structure on austenite to martensite transformation in additively manufactured hot-work tool steels, Acta Mater. 215 (2021), 117044.
[119] M. Narvan, K.S. Al-Rubaie, M. Elbestawi, Process-structure-property relationships of AISI H13 tool steel processed with selective laser melting, Materials 12 (14) (2019) 2284.
[120] J. Yan, H. Song, Y. Dong, W.-M. Quach, M. Yan, High strength (~2000 MPa) or highly ductile (~11%) additively manufactured H13 by tempering at different conditions, Mater. Sci. Eng. A 773 (2020), 138845.
[121] J. Krell, A. Röttger, K. Geenen, W. Theisen, General investigations on processing tool steel X40CrMoV5-1 with selective laser melting, J Mater Process Tech 255 (2018) 679–688.
[122] V. Popovich, E. Borisov, V. Heurtebise, T. Riemslag, A. Popovich, V.S. Sufiiarov, Creep and thermomechanical fatigue of functionally graded Inconel 718 produced by additive manufacturing, TMS 2018 147th Annual Meeting & Exhibition Supplemental Proceedings, Springer, 2018, pp. 85-97.
[123] N.E. Uzan, B. Ratzker, P. Landau, S. Kalabukhov, N. Frage, Compressive creep of AlSi10Mg parts produced by selective laser melting additive manufacturing technology, Addit. Manuf. 29 (2019), 100788.
[124] P. Aliprandi, F. Giudice, E. Guglielmino, A. Sili, Tensile and creep properties improvement of Ti-6Al-4V alloy specimens produced by electron beam powder bed fusion additive manufacturing, Metals 9 (11) (2019) 1207.
[125] Y.-L. Kuo, S. Horikawa, K. Kakehi, Effects of build direction and heat treatment on creep properties of Ni-base superalloy built up by additive manufacturing, Scr. Mater. 129 (2017) 74–78.
[126] A. Röttger, K. Geenen, M. Windmann, F. Binner, W. Theisen, Comparison of microstructure and mechanical properties of 316L austenitic steel processed by selective laser melting with hot-isostatic pressed and cast material, Mater. Sci. Eng. A 678 (2016) 365–376.
[127] Y.M. Wang, T. Voisin, J.T. McKeown, J. Ye, N.P. Calta, Z. Li, Z. Zeng, Y. Zhang, W. Chen, T.T. Roehling, R.T. Ott, M.K. Santala, P.J. Depond, M.J. Matthews, A. V. Hamza, T. Zhu, Additively manufactured hierarchical stainless steels with high strength and ductility, Nat. Mater. 17 (2017) 63.
[128] A. Mertens, S. Reginster, Q. Contrepois, T. Dormal, O. Lemaire, J. Lecomte-Beckers, Microstructures and mechanical properties of stainless steel AISI 316L processed by selective laser melting, Mater. Sci. Forum 783–786 (2014) 898–903.
[129] ASM handbook, ASM International. Handbook Committee., Materials Park, Ohio, 1990.
[130] W. Zhai, W. Zhou, Z. Zhu, S.M.L. Nai, Selective laser melting of 304L and 316L stainless steels: a comparative study of microstructures and mechanical properties, Steel Res. Int. 93 (7) (2022) 2100664.
[131] P. Guo, B. Zou, C.Z. Huang, H.B. Gao, Study on microstructure, mechanical properties and machinability of efficiently additive manufactured AISI 316L stainless steel by high-power direct laser deposition, J Mater Process Tech 240 (2017) 12–22.
[132] M. Jaskari, S. Ghosh, I. Miettunen, P. Karjalainen, A. Järvenpää, Tensile properties and deformation of AISI 316L additively manufactured with various energy densities, Materials (Basel) 14 (19) (2021).
[133] K. Zhang, S.J. Wang, W.J. Liu, X.F. Shang, Characterization of stainless steel parts by laser metal deposition shaping, Mater. Des. 55 (2014) 104–119.
[134] M. Ghayoor, K. Lee, Y. He, C.-H. Chang, B.K. Paul, S. Pasebani, Selective laser melting of 304L stainless steel: Role of volumetric energy density on the microstructure, texture and mechanical properties, Addit. Manuf. 32 (2020), 101011.
[135] J. Yu, M. Rombouts, G. Maes, Cracking behavior and mechanical properties of austenitic stainless steel parts produced by laser metal deposition, Mater. Des. 45 (2013) 228–235.
[136] P. Jamshidi, M. Aristizabal, W. Kong, V. Villapun, S.C. Cox, L.M. Grover, M. M. Attallah, Selective laser melting of Ti-6Al-4V: the impact of post-processing on the tensile, fatigue and biological properties for medical implant applications, Materials (Basel) 13 (12) (2020).
[137] C. Qiu, G.A. Ravi, C. Dance, A. Ranson, S. Dilworth, M.M. Attallah, Fabrication of large Ti–6Al–4V structures by direct laser deposition, J. Alloy. Compd. 629 (2015) 351–361.
[138] Y. Zhai, H. Galarraga, D.A. Lados, Microstructure, static properties, and fatigue crack growth mechanisms in Ti-6Al-4V fabricated by additive manufacturing: LENS and EBM, Eng. Fail. Anal. 69 (2016) 3–14.
[139] T. Vilaro, C. Colin, J.D. Bartout, As-fabricated and heat-treated microstructures of the Ti-6Al-4V alloy processed by selective laser melting, Metall. Mater. Trans. A 42 (10) (2011) 3190–3199.
[140] H. Rafi, N. Karthik, H. Gong, T.L. Starr, B.E. Stucker, Microstructures and mechanical properties of Ti6Al4V parts fabricated by selective laser melting and electron beam melting, J. Mater. Eng. Perform. 22 (12) (2013) 3872–3883.
[141] M. Shunmugavel, A. Polishetty, G. Littlefair, Microstructure and mechanical properties of wrought and additive manufactured Ti-6Al-4 V cylindrical bars, Procedia Technol. 20 (2015) 231–236.

[142] L. Facchini, E. Magalini, P. Robotti, A. Molinari, S. Höges, K. Wissenbach, Ductility of a Ti-6Al-4V alloy produced by selective laser melting of prealloyed powders, Rapid Prototyp. J. (2010).
[143] B. Wysocki, P. Maj, R. Sitek, J. Buhagiar, K. Kurzydłowski, W. Święszkowski, Laser and electron beam additive manufacturing methods of fabricating titanium bone implants, Appl. Sci. 7 (7) (2017).
[144] H. Galarraga, D.A. Lados, R.R. Dehoff, M.M. Kirka, P. Nandwana, Effects of the microstructure and porosity on properties of Ti-6Al-4V ELI alloy fabricated by electron beam melting (EBM), Addit. Manuf. 10 (2016) 47–57.
[145] N. Hrabe, T. Quinn, Effects of processing on microstructure and mechanical properties of a titanium alloy (Ti–6Al–4V) fabricated using electron beam melting (EBM), Part 2: Energy input, orientation, and location, Mater. Sci. Eng. A 573 (2013) 271–277.
[146] J. Yu, M. Rombouts, G. Maes, F. Motmans, Material properties of Ti6Al4V parts produced by laser metal deposition, Phys. Procedia 39 (2012) 416–424.
[147] B.E. Carroll, T.A. Palmer, A.M. Beese, Anisotropic tensile behavior of Ti–6Al–4V components fabricated with directed energy deposition additive manufacturing, Acta Mater. 87 (2015) 309–320.
[148] T. Trosch, J. Strößner, R. Völkl, U. Glatzel, Microstructure and mechanical properties of selective laser melted Inconel 718 compared to forging and casting, Mater. Lett. 164 (2016) 428–431.
[149] D. Zhang, W. Niu, X. Cao, Z. Liu, Effect of standard heat treatment on the microstructure and mechanical properties of selective laser melting manufactured Inconel 718 superalloy, Mater. Sci. Eng. A 644 (2015) 32–40.
[150] Z. Wang, K. Guan, M. Gao, X. Li, X. Chen, X. Zeng, The microstructure and mechanical properties of deposited-IN718 by selective laser melting, J. Alloy. Compd. 513 (2012) 518–523.
[151] J. Strößner, M. Terock, U. Glatzel, Mechanical and microstructural investigation of nickel-based superalloy IN718 manufactured by selective laser melting (SLM), Adv. Eng. Mater. 17 (8) (2015) 1099–1105.
[152] K.A. Unocic, L.M. Kolbus, R.R. Dehoff, S.N. Dryepondt, B.A. Pint, High-temperature performance of UNS N07718 processed by additive manufacturing, NACE corrosion 2014 (2014).
[153] A. Strondl, M. Palm, J. Gnauk, G. Frommeyer, Microstructure and mechanical properties of nickel based superalloy IN718 produced by rapid prototyping with electron beam melting (EBM), Mater. Sci. Technol. 27 (5) (2011) 876–883.
[154] M.M. Kirka, F. Medina, R. Dehoff, A. Okello, Mechanical behavior of post-processed Inconel 718 manufactured through the electron beam melting process, Mater. Sci. Eng. A 680 (2017) 338–346.
[155] E. Jelis, M.R. Hespos, N.M. Ravindra, Process evaluation of AISI 4340 steel manufactured by laser powder bed fusion, J. Mater. Eng. Perform. 27 (1) (2017) 63–71.
[156] R. Mertens, B. Vrancken, N. Holmstock, Y. Kinds, J.P. Kruth, J. Van Humbeeck, Influence of powder bed preheating on microstructure and mechanical properties of H13 tool steel SLM parts, Phys. Procedia 83 (2016) 882–890.
[157] L. Xue, J. Chen, S.H. Wang, Freeform laser consolidated H13 and CPM 9V tool steels, Metallogr., Microstruct., Anal. 2 (2) (2013) 67–78.
[158] M. Mazur, P. Brincat, M. Leary, M. Brandt, Numerical and experimental evaluation of a conformally cooled H13 steel injection mould manufactured with selective laser melting, Int. J. Adv. Manuf. Technol. 93 (1) (2017) 881–900.
[159] R. Dörfert, J. Zhang, B. Clausen, H. Freiße, J. Schumacher, F. Vollertsen, Comparison of the fatigue strength between additively and conventionally fabricated tool steel 1.2344, Addit. Manuf. 27 (2019) 217–223.
[160] J. Lee, J. Choe, J. Park, J.-H. Yu, S. Kim, I.D. Jung, H. Sung, Microstructural effects on the tensile and fracture behavior of selective laser melted H13 tool steel under varying conditions, Mater Charact 155 (2019), 109817.
[161] Z. Hu, H. Zhu, H. Zhang, X. Zeng, Experimental investigation on selective laser melting of 17–4PH stainless steel, Opt. Laser Technol. 87 (2017) 17–25.
[162] P. Auguste, A. Mauduit, L. Fouquet, S. Pillot, Study on 17–4PH stainless steel produced by selective laser melting, Univ Politeh Buchar 80 (4) (2018) 197–210.
[163] L.E. Murr, E. Martinez, J. Hernandez, S. Collins, K.N. Amato, S.M. Gaytan, P. W. Shindo, Microstructures and properties of 17–4PH stainless steel fabricated by selective laser melting, J. Mater. Res. Technol. 1 (3) (2012) 167–177.
[164] A. Yadollahi, M. Mahmoudi, A. Elwany, H. Doude, L. Bian, J.C. Newman, Effects of crack orientation and heat treatment on fatigue-crack-growth behavior of AM 17–4PH stainless steel, Eng. Fract. Mech. 226 (2020), 106874.
[165] R. Shrestha, P.D. Nezhadfar, M. Masoomi, J. Simsiriwong, N. Phan, N. Shamsaei, Effects of design parameters on thermal history and mechanical behavior of additively manufactured 17-4PH stainless steel, Proceedings of the 29th Annual International Solid Freeform Symposium-An Additive Manufacturing Conference, 2018, pp. 1277-1289.
[166] J.H. Wu, C.K. Lin, Influence of high temperature exposure on the mechanical behavior and microstructure of 17–4PH stainless steel, J. Mater. Sci. 38 (5) (2003) 965–971.
[167] F. Ning, W. Cong, Microstructures and mechanical properties of Fe-Cr stainless steel parts fabricated by ultrasonic vibration-assisted laser engineered net shaping process, Mater. Lett. 179 (2016) 61–64.
[168] A. Yadollahi, N. Shamsaei, S.M. Thompson, A. Elwany, L. Bian, Effects of building orientation and heat treatment on fatigue behavior of selective laser melted 17–4PH stainless steel, Int. J. Fatigue 94 (2017) 218–235.
[169] K. Kempen, E. Yasa, L. Thijs, J.P. Kruth, J. Van Humbeeck, Microstructure and mechanical properties of Selective Laser Melted 18Ni-300 steel, Lasers in Manufacturing 2011: Proceedings of the Sixth International Wlt Conference on Lasers in Manufacturing, Vol 12, Pt A 12 (2011) 255-263.
[170] J. Song, Q. Tang, Q.X. Feng, S. Ma, R. Setchi, Y. Liu, Q.Q. Han, X.J. Fan, M. X. Zhang, Effect of heat treatment on microstructure and mechanical behaviours of 18Ni-300 maraging steel manufactured by selective laser melting, Opt. Laser Technol. 120 (2019), 105725.
[171] R. Casati, J. Lemke, A. Tuissi, M. Vedani, Aging behaviour and mechanical performance of 18-ni 300 steel processed by selective laser melting, Metals 6 (9) (2016).
[172] L. Kučerová, I. Zetková, A. Jandová, M. Bystrianský, Microstructural characterisation and in-situ straining of additive-manufactured X3NiCoMoTi 18-9-5 maraging steel, Mater. Sci. Eng. A 750 (2019) 70–80.
[173] C.L. Tan, K.S. Zhou, W.Y. Ma, P.P. Zhang, M. Liu, T.C. Kuang, Microstructural evolution, nanoprecipitation behavior and mechanical properties of selective laser melted high-performance grade 300 maraging steel, Mater. Des. 134 (2017) 23–34.
[174] J. Suryawanshi, K.G. Prashanth, U. Ramamurty, Tensile, fracture, and fatigue crack growth properties of a 3D printed maraging steel through selective laser melting, J. Alloy. Compd. 725 (2017) 355–364.
[175] M.J. Holzweissig, A. Taube, F. Brenne, M. Schaper, T. Niendorf, Microstructural Characterization and Mechanical Performance of Hot Work Tool Steel Processed by Selective Laser Melting, Metall. Mater. Trans. B 46 (2) (2015) 545–549.
[176] W.E. Frazier, Metal additive manufacturing: a review, J. Mater. Eng. Perform. 23 (6) (2014) 1917–1928.
[177] T. Michler, J. Naumann, Microstructural aspects upon hydrogen environment embrittlement of various bcc steels, Int. J. Hydrogen Energy 35 (2) (2010) 821–832.
[178] G.P. Tiwari, A. Bose, J.K. Chakravartty, S.L. Wadekar, M.K. Totlani, R.N. Arya, R. K. Fotedar, A study of internal hydrogen embrittlement of steels, Mater. Sci. Eng. A 286 (2) (2000) 269–281.
[179] D. Matejczyk, C. Bampton, M. Jacinto, C. Swingler, Issues in alloy development for space propulsion - the example of hydrogen-compatible thermo-span, 1st Space Exploration Conference: Continuing the Voyage of Discovery, American Institute of Aeronautics and Astronautics, 2005.
[180] L. Raymond, Hydrogen embrittlement testing, American Society for Testing & Materials, Place of publication not identified, 1974.
[181] I. Bernstein, R. Garber, G. Pressouyre, Effect of hydrogen on behaviour of materials, TMS-AIME, New York, 1976) p 37 (1976).
[182] D. Westphal, F. Worzala, Hydrogen in metals, IM Bernstein and AW Thompson, eds (1974) 78-89.
[183] I.M. Bernstein, A.W. Thompson, Hydrogen in metals : proceedings of an international conference on the effects of hydrogen on materials properties and selection and structural design, 23-27 September, 1973, Seven Springs Conference Center, Champion, Pa, American Society for Metals, Metals Park, Ohio, 1974.
[184] Z. Tarzimoghadam, M. Rohwerder, S.V. Merzlikin, A. Bashir, L. Yedra, S. Eswara, D. Ponge, D. Raabe, Multi-scale and spatially resolved hydrogen mapping in a Ni–Nb model alloy reveals the role of the δ phase in hydrogen embrittlement of alloy 718, Acta Mater. 109 (2016) 69–81.
[185] G.C. Obasi, Z. Zhang, D. Sampath, R. Morana, R. Akid, M. Preuss, Effect of microstructure and alloy chemistry on hydrogen embrittlement of precipitation-hardened Ni-based alloys, Metall. Mater. Trans. A 49 (4) (2018) 1167–1181.
[186] V. Demetriou, J.D. Robson, M. Preuss, R. Morana, Effect of hydrogen on the mechanical properties of alloy 945X (UNS N09945) and influence of microstructural features, Mater. Sci. Eng. A 684 (2017) 423–434.
[187] Z. Zhang, G. Obasi, R. Morana, M. Preuss, Hydrogen assisted crack initiation and propagation in a nickel-based superalloy, Acta Mater. 113 (2016) 272–283.
[188] Z. Tarzimoghadam, D. Ponge, J. Klöwer, D. Raabe, Hydrogen-assisted failure in Ni-based superalloy 718 studied under in situ hydrogen charging: The role of localized deformation in crack propagation, Acta Mater. 128 (2017) 365–374.
[189] G. Leyson, B. Grabowski, J. Neugebauer, Multiscale description of dislocation induced nano-hydrides, Acta Mater. 89 (2015) 50–59.
[190] Part II: Corrosion Topics, in: F. Khoshnaw, R. Gubner (Eds.), Corrosion Atlas Case Studies, Elsevier2022, pp. xliii-lxviii.
[191] M. Wasim, M.B. Djukic, T.D. Ngo, Influence of hydrogen-enhanced plasticity and decohesion mechanisms of hydrogen embrittlement on the fracture resistance of steel, Eng. Fail. Anal. 123 (2021).
[192] T. Wang, H. Zhang, W. Liang, Hydrogen embrittlement fracture mechanism of 430 ferritic stainless steel: The significant role of carbides and dislocations, Mater. Sci. Eng. A 829 (2022).
[193] L.B. Pfeil, The effect of occluded hydrogen on the tensile strength of iron, Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character 112(760) (1926) 182-195.
[194] R. Oriani, A mechanistic theory of hydrogen embrittlement of steels, Ber. Bunsen. Phys. Chem 76 (8) (1972) 848–857.
[195] A.R. Troiano, The role of hydrogen and other interstitials in the mechanical behavior of metals, Springer, 2016.
[196] S.P. Lynch, Hydrogen embrittlement (HE) phenomena and mechanisms, Stress Corrosion Cracking (2011) 90–130.
[197] C. Ícaro Zanetti de, F. Célia Marina de Alvarenga, F. Itamar, C. Margarita Ballester, S. Leonardo Taborda, Hydrogen embrittlement in AISI 4340 steel (AMS 6415S) quenched and tempered, 23rd ABCM International Congress of Mechanical Engineering, 2015.
[198] R. Wang, Effects of hydrogen on the fracture toughness of a X70 pipeline steel, Corros. Sci. 51 (12) (2009) 2803–2810.
[199] D. Wan, S. Guan, D. Wang, X. Lu, J. Ma, Hydrogen embrittlement of additively manufactured AlCoCrFeNi2.1 eutectic high-entropy alloy, Corros. Sci. 195 (2022).
[200] C.D. Beachem, A new model for hydrogen-assisted cracking (hydrogen "embrittlement"), Metall. Mater. Trans. B 3 (2) (1972) 441–455.

[201] H.K. Birnbaum, P. Sofronis, Hydrogen-enhanced localized plasticity — a mechanism for hydrogen-related fracture, Mater. Sci. Eng. A 176 (1–2) (1994) 191–202.
[202] J.P. Hirth, L. Kubin, Dislocations in solids, Elsevier, 2009.
[203] P. Sofronis, H.K. Birnbaum, Mechanics of the hydrogen dislocation impurity interactions — I. increasing shear modulus, J. Mech. Phys. Solids 43 (1) (1995) 49–90.
[204] X.Z. Ran, D. Liu, H.B. Tang, H.M. Wang, R.J. Santucci, J.R. Scully, Hydrogen diffusion and trapping in laser additively manufactured ultra-high strength AerMet100 steel, Corrosion 77 (10) (2021) 1043–1059.
[205] S. Cao, C.V.S. Lim, B. Hinton, X.H. Wu, Effects of microtexture and Ti3Al (alpha (2)) precipitates on stress-corrosion cracking properties of a Ti-8Al-1Mo-1V alloy, Corros. Sci. 116 (2017) 22–33.
[206] S. Cao, S.M. Zhu, C.V.S. Lim, X.G. Zhou, X.B. Chen, B.R.W. Hinton, R.R. Boyer, J. C. Williams, X.H. Wu, The mechanism of aqueous stress-corrosion cracking of alpha plus beta titanium alloys, Corros. Sci. 125 (2017) 29–39.
[207] J. von Pezold, L. Lymperakis, J. Neugebeauer, Hydrogen-enhanced local plasticity at dilute bulk H concentrations: The role of H-H interactions and the formation of local hydrides, Acta Mater. 59 (8) (2011) 2969–2980.
[208] D. Westlake, Generalized model for hydrogen embrittlement, Argonne National Lab., Ill, 1969.
[209] D.S. Shih, I.M. Robertson, H.K. Birnbaum, Hydrogen embrittlement of α titanium: in situ tem studies, Acta Metall. 36 (1) (1988) 111–124.
[210] V.A. Livanov, B.A. Kolachev, A.A. Buhanova, Hydrogen embrittlement of titanium and its alloys, in: R.I. Jaffee, N.E. Promisel (Eds.), The Science, Technology and Application of Titanium, Pergamon, 1970, pp. 561-576.
[211] D. Eliezer, T.H. Böllinghaus, 18 - Hydrogen effects in titanium alloys, in: R. P. Gangloff, B.P. Somerday (Eds.), Gaseous Hydrogen Embrittlement of Materials in Energy Technologies, Woodhead Publishing, 2012, pp. 668–706.
[212] A. Nagao, M. Dadfarnia, B.P. Somerday, P. Sofronis, R.O. Ritchie, Hydrogen-enhanced-plasticity mediated decohesion for hydrogen-induced intergranular and "quasi-cleavage" fracture of lath martensitic steels, J. Mech. Phys. Solids 112 (2018) 403–430.
[213] B.N. Popov, J.-W. Lee, M.B. Djukic, Chapter 7 - Hydrogen permeation and hydrogen-induced cracking, in: M. Kutz (Ed.), Handbook of Environmental Degradation of Materials, (Third Edition),, William Andrew Publishing, 2018, pp. 133–162.
[214] M.B. Djukic, G.M. Bakic, V.S. Zeravcic, A. Sedmak, B. Rajicic, Hydrogen embrittlement of industrial components: prediction, prevention, and models, Corrosion 72(7) (2016) 943-961.
[215] L. Wan, W.T. Geng, A. Ishii, J.-P. Du, Q. Mei, N. Ishikawa, H. Kimizuka, S. Ogata, Hydrogen embrittlement controlled by reaction of dislocation with grain boundary in alpha-iron, Int. J. Plast 112 (2019) 206–219.
[216] X. Li, J. Zhang, E. Akiyama, Q. Fu, Q. Li, Hydrogen embrittlement behavior of Inconel 718 alloy at room temperature, J. Mater. Sci. Technol. 35 (4) (2019) 499–502.
[217] F. Galliano, E. Andrieu, C. Blanc, J.-M. Cloue, D. Connetable, G. Odemer, Effect of trapping and temperature on the hydrogen embrittlement susceptibility of alloy 718, Mater. Sci. Eng. A 611 (2014) 370–382.
[218] M.F. Baltacioglu, B. Cetin, B. Bal, The effect of strain rate on the hdrogen embrittlement susceptibility of aluminum 7075, J. Eng. Mater. Technol. 145 (2) (2022).
[219] N. Winzer, A. Atrens, W. Dietzel, G. Song, K.U. Kainer, Fractography of stress corrosion cracking of Mg-Al alloys, Metall. Mater. Trans. A 39 (5) (2008) 1157–1173.
[220] M. López Freixes, X. Zhou, H. Zhao, H. Godin, L. Peguet, T. Warner, B. Gault, Revisiting stress-corrosion cracking and hydrogen embrittlement in 7xxx-Al alloys at the near-atomic-scale, Nat. Commun. 13 (1) (2022) 4290.
[221] A. Mohammadi, P. Edalati, M. Arita, J.W. Bae, H.S. Kim, K. Edalati, Microstructure and defect effects on strength and hydrogen embrittlement of high-entropy alloy CrMnFeCoNi processed by high-pressure torsion, Mater. Sci. Eng. A 844 (2022), 143179.
[222] H. Luo, Z. Pan, X. Wang, H. Cheng, A.A. Nazarov, X. Li, Hydrogen-assisted failure in partially recrystallized carbon alloyed equiatomic CoCrFeMnNi high-entropy alloy, Corros. Sci. 203 (2022), 110357.
[223] Y.-T. Lin, X. An, Z. Zhu, M.L.S. Nai, C.-W. Tsai, H.-W. Yen, Effect of cell wall on hydrogen response in CoCrFeMnNi high-entropy alloy additively manufactured by selective laser melting, J. Alloy. Compd. 925 (2022), 166735.
[224] I. Jeong, K.M. Ryu, D.G. Lee, Y. Jung, K. Lee, J.S. Lee, D.-W. Suh, Austenite morphology and resistance to hydrogen embrittlement in medium Mn transformation-induced plasticity steel, Scr. Mater. 169 (2019) 52–56.
[225] J. Venezuela, Q. Zhou, Q. Liu, H. Li, M. Zhang, M.S. Dargusch, A. Atrens, The influence of microstructure on the hydrogen embrittlement susceptibility of martensitic advanced high strength steels, Mater. Today Commun. 17 (2018) 1–14.
[226] G. Krauss, A. Marder, The morphology of martensite in iron alloys, Metall. Trans. 2 (9) (1971) 2343–2357.
[227] T. Zhang, W. Zhao, Q. Deng, W. Jiang, Y. Wang, Y. Wang, W. Jiang, Effect of microstructure inhomogeneity on hydrogen embrittlement susceptibility of X80 welding HAZ under pressurized gaseous hydrogen, Int. J. Hydrogen Energy 42 (39) (2017) 25102–25113.
[228] A.H.M. Krom, A. Bakker, Hydrogen trapping models in steel, Metall. Mater. Trans. B 31 (6) (2000) 1475–1482.
[229] W.-J. Hui, Z.-H. Wang, Z.-B. Xu, Y.-J. Zhang, X.-L. Zhao, Hydrogen embrittlement of a microalloyed bainitic forging steel, J. Iron Steel Res. Int. 26 (9) (2019) 1011–1021.
[230] M.A. Mohtadi-Bonab, J.A. Szpunar, S.S. Razavi-Tousi, A comparative study of hydrogen induced cracking behavior in API 5L X60 and X70 pipeline steels, Eng. Fail. Anal. 33 (2013) 163–175.
[231] A. Nagao, C.D. Smith, M. Dadfarnia, P. Sofronis, I.M. Robertson, The role of hydrogen in hydrogen embrittlement fracture of lath martensitic steel, Acta Mater. 60 (13–14) (2012) 5182–5189.
[232] A. Shibata, H. Takahashi, N. Tsuji, Microstructural and crystallographic features of hydrogen-related crack propagation in low carbon martensitic steel, ISIJ Int. 52 (2) (2012) 208–212.
[233] A. Shibata, T. Murata, H. Takahashi, T. Matsuoka, N. Tsuji, Characterization of hydrogen-related fracture behavior in as-quenched low-carbon martensitic steel and tempered medium-carbon martensitic steel, Metall. Mater. Trans. A 46 (12) (2015) 5685–5696.
[234] X.Y. Liu, J. Kameda, J.W. Anderegg, S. Takaki, K. Abiko, C.J. McMahon, Hydrogen-induced cracking in a very-high-purity high-strength steel, Mater. Sci. Eng. A 492 (1) (2008) 218–220.
[235] M.L. Martin, B.P. Somerday, R.O. Ritchie, P. Sofronis, I.M. Robertson, Hydrogen-induced intergranular failure in nickel revisited, Acta Mater. 60 (6) (2012) 2739–2745.
[236] S. Wang, M.L. Martin, P. Sofronis, S. Ohnuki, N. Hashimoto, I.M. Robertson, Hydrogen-induced intergranular failure of iron, Acta Mater. 69 (2014) 275–282.
[237] Y. Bai, Y. Momotani, M.C. Chen, A. Shibata, N. Tsuji, Effect of grain refinement on hydrogen embrittlement behaviors of high-Mn TWIP steel, Mater. Sci. Eng. A 651 (2016) 935–944.
[238] A. Macadre, N. Nakada, T. Tsuchiyama, S. Takaki, Critical grain size to limit the hydrogen-induced ductility drop in a metastable austenitic steel, Int. J. Hydrogen Energy 40 (33) (2015) 10697–10703.
[239] K. Takasawa, R. Ikeda, N. Ishikawa, R. Ishigaki, Effects of grain size and dislocation density on the susceptibility to high-pressure hydrogen environment embrittlement of high-strength low-alloy steels, Int. J. Hydrogen Energy 37 (3) (2012) 2669–2675.
[240] A. Begić Hadžipašić, J. Malina, M. Malina, The influence of microstructure on hydrogen diffusion and embrittlement of multiphase fine-grained steels with increased plasticity and strength, Chem. Biochem. Eng. Q. 25 (2) (2011) 159–169.
[241] M. Loidl, O. Kolk, S. Veith, T. Göbel, Characterization of hydrogen embrittlement in automotive advanced high strength steels, Mater. Werkst. 42 (12) (2011) 1105–1110.
[242] V. Olden, C. Thaulow, R. Johnsen, Modelling of hydrogen diffusion and hydrogen induced cracking in supermartensitic and duplex stainless steels, Mater. Des. 29 (10) (2008) 1934–1948.
[243] Z. Wang, J. Liu, F. Huang, Y.-J. Bi, S.-Q. Zhang, Hydrogen diffusion and its effect on hydrogen embrittlement in DP steels with different martensite content, Front. Mater. 7 (2020).
[244] J. Venezuela, Q. Zhou, Q. Liu, M. Zhang, A. Atrens, Hydrogen trapping in some automotive martensitic advanced high-strength steels, Adv. Eng. Mater. 20 (1) (2018).
[245] T. Mukherjee, W. Zhang, T. DebRoy, An improved prediction of residual stresses and distortion in additive manufacturing, Comput. Mater. Sci 126 (2017) 360–372.
[246] R.A. Oriani, The diffusion and trapping of hydrogen in steel, Acta Metall. 18 (1) (1970) 147–157.
[247] M. Devanathan, Z. Stachurski, The mechanism of hydrogen evolution on iron in acid solutions by determination of permeation rates, J. Electrochem. Soc. 111 (5) (1964) 619.
[248] F. Wei, K. Tsuzaki, Hydrogen trapping phenomena in martensitic steels, Gaseous hydrogen embrittlement of materials in energy technologies, Elsevier, 2012, pp. 493-525.
[249] H.-J. Christ, M. Decker, S. Zeitler, Hydrogen diffusion coefficients in the titanium alloys IMI 834, Ti 10-2-3, Ti 21 S, and alloy C, Metall. Mater. Trans. A 31 (6) (2000) 1507–1517.
[250] M. Shih, J. Miao, M. Mills, M. Ghazisaeidi, Stacking fault energy in concentrated alloys, Nat. Commun. 12 (1) (2021) 3590.
[251] G.A. Young, J.R. Scully, Hydrogen production, absorption and transport during environment assisted cracking of an Al-Zn-Mg-(Cu) alloy in humid air, International conference on hydrogen effects on material behavior and corrosion deformation interactions, 2002, pp. 22-26.
[252] B.O. Omiyale, T.O. Olugbade, T.E. Abioye, P.K. Farayibi, Wire arc additive manufacturing of aluminium alloys for aerospace and automotive applications: a review, Mater. Sci. Technol. 38 (7) (2022) 391–408.
[253] L. Wang, C.F. Dong, C. Man, Y.B. Hu, Q. Yu, X.G. Li, Effect of microstructure on corrosion behavior of high strength martensite steel-A literature review, Int. J. Miner. Metall. Mater. 28 (5) (2021) 754–773.
[254] S.L. Li, M.M. Liu, Y. Ren, Y.D. Wang, Hydrogen embrittlement behaviors of additive manufactured maraging steel investigated by in situ high-energy X-ray diffraction, Mater. Sci. Eng. a-Struct. Mater. Properties Microstruct. Process. 766 (2019) 6.
[255] X. Mei, Y. Yan, L. Qiao, Research on hydrogen embrittlement behavior of L-PBF 18Ni(300) maraging steel by experiments and numerical simulations, Acta Mater. 256 (2023), 119141.
[256] N. Zhou, S. Zhang, C. Ma, H. Zhang, C. Wu, J. Liu, F. Huang, Effect of heat treatment on the hydrogen embrittlement susceptibility of selective laser melted 18Ni-300 maraging steel, Mater. Sci. Eng. A 885 (2023), 145622.
[257] Z. Fu, B. Yang, K. Gan, D. Yan, Z. Li, G. Gou, H. Chen, Z. Wang, Improving the hydrogen embrittlement resistance of a selective laser melted high-entropy alloy via modifying the cellular structures, Corros. Sci. 190 (2021), 109695.

[258] H. Cheng, H. Luo, Z. Pan, X. Wang, Q. Zhao, Y. Fu, X. Li, Effects of laser powder bed fusion process parameters on microstructure and hydrogen embrittlement of high-entropy alloy, J. Mater. Sci. Technol. 155 (2023) 211–226.
[259] V. Macin, H.-J. Christ, Influence of hydride-induced microstructure modification on mechanical properties of metastable beta titanium alloy Ti 10V–2Fe-3Al, Int. J. Hydrogen Energy 40 (47) (2015) 16878–16891.
[260] E. Tal-Gutelmacher, D. Eliezer, Hydrogen cracking in titanium-based alloys, J. Alloy. Compd. 404–406 (2005) 621–625.
[261] V.N. Kudiiarov, E.B. Kashkarov, M.S. Syrtanov, A.M. Lider, Hydrogen sorption by Ni-coated titanium alloy VT1-0, Int. J. Hydrogen Energy 42 (15) (2017) 10604–10610.
[262] V.N. Kudiiarov, M.S. Syrtanov, Y.S. Bordulev, M.N. Babikhina, A.M. Lider, V. E. Gubin, T.L. Murashkina, The hydrogen sorption and desorption behavior in spherical powder of pure titanium used for additive manufacturing, Int. J. Hydrogen Energy 42 (22) (2017) 15283–15289.
[263] D. Hardie, S. Ouyang, Effect of hydrogen and strain rate upon the ductility of mill-annealed Ti6Al4V, Corros. Sci. 41 (1) (1999) 155–177.
[264] B.S. Yilbas, A. Coban, R. Kahraman, M.M. Khaled, Hydrogen embrittlement of Ti-6Al-4V alloy with surface modification by TiN coating, Int. J. Hydrogen Energy 23 (6) (1998) 483–489.
[265] R. Gaddam, M. Hörnqvist, M.L. Antti, R. Pederson, Influence of high-pressure gaseous hydrogen on the low-cycle fatigue and fatigue crack growth properties of a cast titanium alloy, Mater. Sci. Eng. A 612 (2014) 354–362.
[266] I.P. Mishin, G.P. Grabovetskaya, O.V. Zabudchenko, E.N. Stepanova, Influence of hydrogenation on evolution of submicrocrystalline structure of Ti–6Al–4V alloy upon exposure to temperature and stress, Russ. Phys. J. 57 (4) (2014) 423–428.
[267] E. Tal-Gutelmacher, D. Eliezer, The hydrogen embrittlement of titanium-based alloys, JOM 57 (9) (2005) 46–49.
[268] Y. Zong, K. Wu, Thermo hydrogen treatment for microstructure refinement and mechanical properties improvement of Ti-6Al-4V alloy, Mater. Sci. Eng. A 703 (2017) 430–437.
[269] Z. Liu, J. Li, Q. Ruan, K. Zhang, W. Ma, H. Dong, R. Wang, Probing the optimal thermohydrogen processing conditions of titanium alloy shavings via chemisorption method, Int. J. Hydrogen Energy 43 (45) (2018) 20783–20794.
[270] M.A. Murzinova, G.A. Salishchev, D.D. Afonichev, Formation of nanocrystalline structure in two-phase titanium alloy by combination of thermohydrogen processing with hot working, Int. J. Hydrogen Energy 27 (7) (2002) 775–782.
[271] Z. Cao, L. Ouyang, H. Wang, J. Liu, D. Sun, Q. Zhang, M. Zhu, Advanced high-pressure metal hydride fabricated via Ti–Cr–Mn alloys for hybrid tank, Int. J. Hydrogen Energy 40 (6) (2015) 2717–2728.
[272] J.-Y. Wang, R.-R. Jeng, J.-K. Nieh, S. Lee, S.-L. Lee, H.-Y. Bor, Comparing the hydrogen storage alloys—TiCrV and vanadium-rich TiCrMnV, Int. J. Hydrogen Energy 32 (16) (2007) 3959–3964.
[273] L.C. Campanelli, P.S.C.P. da Silva, A.M. Jorge, C. Bolfarini, Effect of hydrogen on the fatigue behavior of the near-β Ti-5Al-5Mo-5V-3Cr alloy, Scr. Mater. 132 (2017) 39–43.
[274] G.A. Lenning, C.M. Craighead, R.I. Jaffee, Constitution and mechanical properties of titanium-hydrogen alloys, JOM 6 (3) (1954) 367–376.
[275] L.E. Murr, Metallurgy of additive manufacturing: Examples from electron beam melting, Addit. Manuf. 5 (2015) 40–53.
[276] B. Dutta, F.H. Froes, The additive manufacturing (AM) of titanium alloys, Met. Powder Rep. 72 (2) (2017) 96–106.
[277] L. Parry, I.A. Ashcroft, R.D. Wildman, Understanding the effect of laser scan strategy on residual stress in selective laser melting through thermo-mechanical simulation, Addit. Manuf. 12 (2016) 1–15.
[278] A.R. Prasad, K. Ramji, G. Datta, An experimental study of wire EDM on Ti-6Al-4V alloy, Procedia Mater. Sci. 5 (2014) 2567–2576.
[279] R. Gupta, V.A. Kumar, C. Mathew, G.S. Rao, Strain hardening of titanium alloy Ti6Al4V sheets with prior heat treatment and cold working, Mater. Sci. Eng. A 662 (2016) 537–550.
[280] C. de Formanoir, A. Brulard, S. Vivès, G. Martin, F. Prima, S. Michotte, E. Rivière, A. Dolimont, S. Godet, A strategy to improve the work-hardening behavior of Ti–6Al–4V parts produced by additive manufacturing, Mater. Res. Lett. 5 (3) (2017) 201–208.
[281] R. Huang, M. Riddle, D. Graziano, J. Warren, S. Das, S. Nimbalkar, J. Cresko, E. Masanet, Energy and emissions saving potential of additive manufacturing: the case of lightweight aircraft components, J. Clean. Prod. 135 (2016) 1559–1570.
[282] E. Uhlmann, R. Kersting, T.B. Klein, M.F. Cruz, A.V. Borille, Additive manufacturing of titanium alloy for aircraft components, Procedia Cirp 35 (2015) 55–60.
[283] D. Eliezer, E. Tal-Gutelmacher, C.E. Cross, T. Boellinghaus, Hydrogen trapping in β-21S titanium alloy, Mater. Sci. Eng. A 421 (1) (2006) 200–207.
[284] A. Takasaki, Y. Furuya, K. Ojima, Y. Taneda, Hydride dissociation and hydrogen evolution behavior of electrochemically charged pure titanium, J. Alloy. Compd. 224 (2) (1995) 269–273.
[285] D. Banerjee, J.C. Williams, Perspectives on titanium science and technology, Acta Mater. 61 (3) (2013) 844–879.
[286] N.S. Pushilina, E.B. Kashkarov, M.S. Syrtanov, V.N. Kudiiarov, R.S. Laptev, M. Gustomyassov, Y. Kushnarev, V.V. Fedorov, Microstructure and properties of Ti-6.5Al-3.5Mo-1.5Zr-0.3Si parts produced by electron beam melting, J. Phys. Conf. Ser. 1115 (2018), 042057.
[287] M. Jamshidinia, F. Kong, R. Kovacevic, Numerical modeling of heat distribution in the electron beam melting® of Ti-6Al-4V, J. Manuf. Sci. Eng. 135 (6) (2013).
[288] K. Horke, A. Meyer, R.F. Singer, 24 - Metal injection molding (MIM) of nickel-base superalloys, in: D.F. Heaney (Ed.), Handbook of Metal Injection Molding (Second Edition), Woodhead Publishing, 2019, pp. 575–608.
[289] R.C. Reed, C.M.F. Rae, 22 - Physical Metallurgy of the Nickel-Based Superalloys, in: D.E. Laughlin, K. Hono (Eds.), Physical Metallurgy (Fifth Edition), Elsevier, Oxford, 2014, pp. 2215–2290.
[290] M.M. Kirka, P. Fernandez-Zelaia, Additive materials for high temperature applications, in: F.G. Caballero (Ed.), Encyclopedia of Materials: Metals and Alloys, Elsevier, Oxford, 2022, pp. 529–536.
[291] B. Graybill, M. Li, D. Malawey, C. Ma, J.-M. Alvarado-Orozco, E. Martinez-Franco, Additive manufacturing of nickel-based superalloys, ASME 2018 13th International Manufacturing Science and Engineering Conference, 2018.
[292] R.R. Dehoff, M. Kirka, W. Sames, H. Bilheux, A. Tremsin, L. Lowe, S. Babu, Site specific control of crystallographic grain orientation through electron beam additive manufacturing, Mater. Sci. Technol. 31 (8) (2015) 931–938.
[293] H. Gray, H. Nelson, R. Johnson, W. McPherson, F. Howard, J. Swisher, Potential structural material problems in a hydrogen energy system, Int. J. Hydrogen Energy 3 (1) (1978) 105–118.
[294] J.A. Lee, 17 - Hydrogen embrittlement of nickel, cobalt and iron-based superalloys, in: R.P. Gangloff, B.P. Somerday (Eds.), Gaseous Hydrogen Embrittlement of Materials in Energy Technologies, Woodhead Publishing, 2012, pp. 624–667.
[295] X. Lu, Y. Ma, D. Wang, On the hydrogen embrittlement behavior of nickel-based alloys: Alloys 718 and 725, Mater. Sci. Eng. A 792 (2020), 139785.
[296] Z. Zhang, K.L. Moore, G. McMahon, R. Morana, M. Preuss, On the role of precipitates in hydrogen trapping and hydrogen embrittlement of a nickel-based superalloy, Corros. Sci. 146 (2019) 58–69.
[297] G.R. Thellaputta, P.S. Chandra, C.S.P. Rao, Machinability of Nickel Based Superalloys: A Review, Mater. Today:. Proc. 4 (2) (2017) 3712–3721.
[298] O. Diegel, 10.02 - Additive manufacturing: an overview, in: S. Hashmi, G. F. Batalha, C.J. Van Tyne, B. Yilbas (Eds.), Comprehensive Materials Processing, Elsevier, Oxford, 2014, pp. 3–18.
[299] Q.B. Jia, D.D. Gu, Selective laser melting additive manufacturing of Inconel 718 superalloy parts: Densification, microstructure and properties, J. Alloy. Compd. 585 (2014) 713–721.
[300] H.E. Helmer, C. Körner, R.F. Singer, Additive manufacturing of nickel-based superalloy Inconel 718 by selective electron beam melting: Processing window and microstructure, J. Mater. Res. 29 (17) (2014) 1987–1996.
[301] E. Hosseini, V.A. Popovich, A review of mechanical properties of additively manufactured Inconel 718, Addit. Manuf. 30 (2019), 100877.
[302] A. Hirose, Y. Arita, Y. Nakanishi, K.F. Kobayashi, Decrease in hydrogen embrittlement sensitivity of Inconel 718 by laser surface softening, Mater. Sci. Eng. A 219 (1–2) (1996) 71–79.
[303] A. Yilmaz, The Portevin–Le Chatelier effect: a review of experimental findings, Sci. Technol. Adv. Mater. 12 (6) (2011), 063001.
[304] D. Deng, R.L. Peng, H. Brodin, J. Moverare, Microstructure and mechanical properties of Inconel 718 produced by selective laser melting: Sample orientation dependence and effects of post heat treatments, Mater. Sci. Eng. A 713 (2018) 294–306.
[305] G.H. Cao, T.Y. Sun, C.H. Wang, X. Li, M. Liu, Z.X. Zhang, P.F. Hu, A.M. Russell, R. Schneider, D. Gerthsen, Z.J. Zhou, C.P. Li, G.F. Chen, Investigations of γ′, γ″ and δ precipitates in heat-treated Inconel 718 alloy fabricated by selective laser melting, Mater Charact 136 (2018) 398–406.
[306] I.-J. Park, S.-M. Lee, H.-H. Jeon, Y.-K. Lee, The advantage of grain refinement in the hydrogen embrittlement of Fe–18Mn–0.6C twinning-induced plasticity steel, Corros. Sci. 93 (2015) 63–69.
[307] Z.H. Fu, G.Q. Gou, Z.Y. Zhu, C.P. Ma, W.Q. Yang, L.L. Zhang, Y.C. Hu, W. Gao, Stress corrosion cracking behavior of SUS301L-MT stainless steel laser-arc hybrid welded joints, Corros. Sci. 143 (2018) 23–30.
[308] H. Sarmiento Klapper, J. Klower, O. Gosheva, Hydrogen embrittlement: the game changing factor in the applicability of nickel alloys in oilfield technology, Philos Trans A Math Phys, Eng. Sci. 375 (2098) (2017).
[309] R.P. Singh, S.E. Zorrilla, S.K. Vidyarthi, R. Cocker, K. Cronin, Dairy Plant Design, Construction and Operation, in: P.L.H. McSweeney, J.P. McNamara (Eds.), Encyclopedia of Dairy Sciences (Third Edition), Academic Press, Oxford, 2022, pp. 239–252.
[310] C. San Marchi, 16 - Hydrogen embrittlement of stainless steels and their welds, in: R.P. Gangloff, B.P. Somerday (Eds.), Gaseous Hydrogen Embrittlement of Materials in Energy Technologies, Woodhead Publishing, 2012, pp. 592–623.
[311] C. Borchers, T. Michler, A. Pundt, Effect of hydrogen on the mechanical properties of stainless steels, Adv. Eng. Mater. 10 (1–2) (2008) 11–23.
[312] E.G. Astafurova, S.V. Astafurov, G.G. Maier, V.A. Moskvina, E.V. Melnikov, A. S. Fortuna, Hydrogen embrittlement of ultrafine-grained austenitic stainless steels, Rev. Adv. Mater. Sci. 54 (1) (2018) 25–45.
[313] S. Astafurov, E. Astafurova, Phase composition of austenitic stainless steels in additive manufacturing: A review, Metals (Basel) 11 (7) (2021) 1052.
[314] G. Ko, W. Kim, K. Kwon, T.-K. Lee, The corrosion of stainless steel made by additive manufacturing: a review, Metals 11 (3) (2021).
[315] A. Adeyemi, E.T. Akinlabi, R.M. Mahamood, Powder bed based laser additive manufacturing process of stainless steel: a review, Materials Today: Proceedings 5 (9, Part 3) (2018) 18510-18517.
[316] A. Hemmasian Ettefagh, S. Guo, J. Raush, Corrosion performance of additively manufactured stainless steel parts: A review, Addit. Manuf. 37 (2021), 101689.
[317] M. Laleh, A.E. Hughes, W. Xu, I. Gibson, M.Y. Tan, A critical review of corrosion characteristics of additively manufactured stainless steels, Int. Mater. Rev. 66 (8) (2021) 563–599.
[318] M. Alnajjar, F. Christien, K. Wolski, C. Bosch, Evidence of austenite by-passing in a stainless steel obtained from laser melting additive manufacturing, Addit. Manuf. 25 (2019) 187–195.

[319] H.K. Rafi, D. Pal, N. Patil, T.L. Starr, B.E. Stucker, Microstructure and mechanical behavior of 17–4 precipitation hardenable steel processed by selective laser melting, J. Mater. Eng. Perform. 23 (12) (2014) 4421–4428.
[320] F. Hengsbach, P. Koppa, K. Duschik, M.J. Holzweissig, M. Burns, J. Nellesen, W. Tillmann, T. Troster, K.P. Hoyer, M. Schaper, Duplex stainless steel fabricated by selective laser melting - Microstructural and mechanical properties, Mater. Des. 133 (2017) 136–142.
[321] K. Saeidi, L. Kevetkova, F. Lofaj, Z. Shen, Novel ferritic stainless steel formed by laser melting from duplex stainless steel powder with advanced mechanical properties and high ductility, Mater. Sci. Eng. A 665 (2016) 59–65.
[322] S. Prakash, 10 - Development of advanced alloys with improved resistance to corrosion and stress corrosion cracking (SCC) in power plants, in: A. Shirzadi, S. Jackson (Eds.), Structural Alloys for Power Plants, Woodhead Publishing, 2014, pp. 294–318.
[323] M. Louthan Jr, R. Derrick, Hydrogen transport in austenitic stainless steel, Corros. Sci. 15 (6–12) (1975) 565–577.
[324] T. Perng, M. Johnson, C. Altstetter, Influence of plastic deformation on hydrogen diffusion and permeation in stainless steels, Acta Metall. 37 (12) (1989) 3393–3397.
[325] C. Marchi, B. Somerday, S. Robinson, Permeability, solubility and diffusivity of hydrogen isotopes in stainless steels at high gas pressures, Int. J. Hydrogen Energy 32 (1) (2007) 100–116.
[326] G. Han, J. He, S. Fukuyama, K. Yokogawa, Effect of strain-induced martensite on hydrogen environment embrittlement of sensitized austenitic stainless steels at low temperatures, Acta Mater. 46 (13) (1998) 4559–4570.
[327] T. Michler, J. Naumann, Hydrogen environment embrittlement of austenitic stainless steels at low temperatures, Int. J. Hydrogen Energy 33 (8) (2008) 2111–2122.
[328] C. San Marchi, T. Michler, K. Nibur, B. Somerday, On the physical differences between tensile testing of type 304 and 316 austenitic stainless steels with internal hydrogen and in external hydrogen, Int. J. Hydrogen Energy 35 (18) (2010) 9736–9745.
[329] Y. Mine, K. Koga, O. Kraft, K. Takashima, Mechanical characterisation of hydrogen-induced quasi-cleavage in a metastable austenitic steel using micro-tensile testing, Scr. Mater. 113 (2016) 176–179.
[330] L. Zhang, Z. Li, J. Zheng, Y. Zhao, P. Xu, C. Zhou, X. Li, Effect of strain-induced martensite on hydrogen embrittlement of austenitic stainless steels investigated by combined tension and hydrogen release methods, Int. J. Hydrogen Energy 38 (19) (2013) 8208–8214.
[331] J. Brooks, A. West, A. Thompson, Effect of weld composition and microstructure on hydrogen assisted fracture of austenitic stainless steels, Metall. Trans. A 14 (1) (1983) 75–84.
[332] J.R. Buckley, D. Hardie, The effect of pre-straining and δ-ferrite on the embrittlement of 304L stainless steel by hydrogen, Corros. Sci. 34 (1) (1993) 93–107.
[333] C.H. Hung, A. Sutton, Y.Q. Li, Y.Y. Shen, H.L. Tsai, M.C. Leu, Enhanced mechanical properties for 304L stainless steel parts fabricated by laser-foil-printing additive manufacturing, J. Manuf. Process. 45 (2019) 438–446.
[334] G.B. Olson, M. Azrin, Transformation behavior of TRIP steels, Metall. Trans. A 9 (5) (1978) 713–721.
[335] J.-Y. Kang, H. Kim, K.-I. Kim, C.-H. Lee, H.N. Han, K.-H. Oh, T.-H. Lee, Effect of austenitic texture on tensile behavior of lean duplex stainless steel with transformation induced plasticity (TRIP), Mater. Sci. Eng. A 681 (2017) 114–120.
[336] Y.F. Shen, N. Jia, Y.D. Wang, X. Sun, L. Zuo, D. Raabe, Suppression of twinning and phase transformation in an ultrafine grained 2GPa strong metastable austenitic steel: Experiment and simulation, Acta Mater. 97 (2015) 305–315.
[337] K.A. Nibur, D.F. Bahr, B.P. Somerday, Hydrogen effects on dislocation activity in austenitic stainless steel, Acta Mater. 54 (10) (2006) 2677–2684.
[338] D.S.R. Krishna, Y. Sun, Effect of thermal oxidation conditions on tribological behaviour of titanium films on 316L stainless steel, Surf. Coat. Technol. 198 (1–3) (2005) 447–453.
[339] J.D. Majumdar, A. Kumar, L. Li, Direct laser cladding of SiC dispersed AISI 316L stainless steel, Tribol. Int. 42 (5) (2009) 750–753.
[340] M. Terada, D.M. Escriba, I. Costa, E. Materna-Morris, A.F. Padilha, Investigation on the intergranular corrosion resistance of the AISI 316L(N) stainless steel after long time creep testing at 600 °C, Mater Charact 59 (6) (2008) 663–668.
[341] K.M. Bertsch, S. Wang, A. Nagao, I.M. Robertson, Hydrogen-induced compatibility constraints across grain boundaries drive intergranular failure of Ni, Mater. Sci. Eng. A 760 (2019) 58–67.
[342] S. Wang, A. Nagao, K. Edalati, Z. Horita, I.M. Robertson, Influence of hydrogen on dislocation self-organization in Ni, Acta Mater. 135 (2017) 96–102.
[343] J.C.M. Farrar, Group C martensitic and ferritic stainless steels, in: J.C.M. Farrar (Ed.), The Alloy Tree, Woodhead Publishing, 2004, pp. 33–51.
[344] B. AlMangour, J.M. Yang, Improving the surface quality and mechanical properties by shot-peening of 17–4 stainless steel fabricated by additive manufacturing, Mater. Des. 110 (2016) 914–924.
[345] A. Kudzal, B. McWilliams, C. Hofmeister, F. Kellogg, J. Yu, J. Taggart-Scarff, J. Liang, Effect of scan pattern on the microstructure and mechanical properties of Powder Bed Fusion additive manufactured 17–4 stainless steel, Mater. Des. 133 (2017) 205–215.
[346] A. Yadollahi, J. Simsiriwong, S.M. Thompson, N. Shamsaei, Data demonstrating the effects of build orientation and heat treatment on fatigue behavior of selective laser melted 17–4PH stainless steel, Data Brief 7 (2016) 89–92.
[347] C.N. Hsiao, C.S. Chiou, J.R. Yang, Aging reactions in a 17–4PH stainless steel, Mater. Chem. Phys. 74 (2) (2002) 134–142.
[348] H.R. Habibi Bajguirani, The effect of ageing upon the microstructure and mechanical properties of type 15–5 PH stainless steel, Mater. Sci. Eng. A 338 (1) (2002) 142–159.
[349] U.K. Viswanathan, S. Banerjee, R. Krishnan, Effects of aging on the microstructure of 17–4PH stainless steel, Mater. Sci. Eng. A 104 (1988) 181–189.
[350] M. Murayama, K. Hono, Y. Katayama, Microstructural evolution in a 17–4PH stainless steel after aging at 400 °C, Metall. Mater. Trans. A 30 (2) (1999) 345–353.
[351] W.C. Chiang, C.C. Pu, B.L. Yu, J.K. Wu, Hydrogen susceptibility of 17–4PH stainless steel, Mater. Lett. 57 (16–17) (2003) 2485–2488.
[352] N. Guennouni, D. Maisonnette, C. Grosjean, D. Poquillon, C. Blanc, Susceptibility to pitting and environmentally assisted cracking of 17–4PH martensitic stainless steel produced by laser beam melting, Materials (Basel) 15 (20) (2022).
[353] A. Barroux, N. Ducommun, E. Nivet, L. Laffont, C. Blanc, Pitting corrosion of 17–4PH stainless steel manufactured by laser beam melting, Corros. Sci. 169 (2020), 108594.
[354] T.K. Shoemaker, Z.D. Harris, J.T. Burns, Comparing Stress Corrosion Cracking Behavior of Additively Manufactured and Wrought 17–4PH Stainless Steel, Corrosion 78 (6) (2022) 528–546.
[355] E.A. Jägle, P.-P. Choi, J. Van Humbeeck, D. Raabe, Precipitation and austenite reversion behavior of a maraging steel produced by selective laser melting, J. Mater. Res. 29 (17) (2014) 2072–2079.
[356] L.P.M. Santos, M. Béreš, I.N. Bastos, S.S.M. Tavares, H.F.G. Abreu, M.J. Gomes da Silva, Hydrogen embrittlement of ultra high strength 300 grade maraging steel, Corros. Sci. 101 (2015) 12–18.
[357] L.W. Tsay, H.L. Lu, C. Chen, The effect of grain size and aging on hydrogen embrittlement of a maraging steel, Corros. Sci. 50 (9) (2008) 2506–2511.
[358] G. Wang, Y. Yan, J. Li, J. Huang, L. Qiao, A.A. Volinsky, Microstructure effect on hydrogen-induced cracking in TM210 maraging steel, Mater. Sci. Eng. A 586 (2013) 142–148.
[359] M. Barsanti, M. Beghini, F. Frasconi, R. Ishak, B.D. Monelli, R. Valentini, Experimental study of hydrogen embrittlement in Maraging steels, Procedia Struct. Integrity 8 (2018) 501–508.
[360] K.G. Reddy, S. Arumugam, T.S. Lakshmanan, Hydrogen embrittlement of maraging steel, J. Mater. Sci. 27 (19) (1992) 5159–5162.
[361] M. Béreš, L. Wu, L.P.M. Santos, M. Masoumi, F.A.M. da Rocha Filho, C.C. da Silva, H.F.G. de Abreu, M.J. Gomes da Silva, Role of lattice strain and texture in hydrogen embrittlement of 18Ni (300) maraging steel, Int. J. Hydrogen Energy 42 (21) (2017) 14786–14793.
[362] E. Jelis, M. Hespos, S.L. Groeschler, R. Carpenter, L-PBF of 4340 Low Alloy Steel: Influence of Feedstock Powder, Layer Thickness, and Machine Maintenance, J. Mater. Eng. Perform. 28 (2) (2018) 693–700.
[363] E. Jelis, M. Clemente, S. Kerwien, N.M. Ravindra, M.R. Hespos, Metallurgical and mechanical evaluation of 4340 steel produced by direct metal laser sintering, JOM 67 (3) (2015) 582–589.
[364] R.G. Bonora, H.J.C. Voorwald, M.O.H. Cioffi, G.S. Junior, L.F.V. Santos, Fatigue in AISI 4340 steel thermal spray coating by HVOF for aeronautic application, Procedia Eng. 2 (1) (2010) 1617–1623.
[365] E. Jelis, M. Clemente, M. Hespos, S. Groeschler, E. Golden, R. Carpenter, Round robin study to evaluate consistency of 4340 steel specimens manufactured by different laser powder bed fusion machines, (2022).
[366] N. Eliaz, A. Shachar, B. Tal, D. Eliezer, Characteristics of hydrogen embrittlement, stress corrosion cracking and tempered martensite embrittlement in high-strength steels, Eng. Fail. Anal. 9 (2) (2002) 167–184.
[367] H.G. Nelson, Hydrogen embrittlement, Treatise on Materials Science & Technology, Elsevier, 1983, pp. 275-359.
[368] A. Turnbull, Hydrogen diffusion and trapping in metals, Gaseous Hydrogen Embrittlement of Materials in Energy Technologies, 2012, pp. 89-128.
[369] M.A. Liu, P.E.J. Rivera-Díaz-del-Castillo, J.I. Barraza-Fierro, H. Castaneda, A. Srivastava, Microstructural influence on hydrogen permeation and trapping in steels, Mater. Des. 167 (2019).
[370] J.M. Tartaglia, K.A. Lazzari, G.P. Hui, K.L. Hayrynen, A comparison of mechanical properties and hydrogen embrittlement resistance of austempered vs quenched and tempered 4340 steel, Metall. Mater. Trans. A 39 (3) (2008) 559–576.
[371] O. Dogan, M.F. Kapci, V. Esat, B. Bal, Experimental and molecular dynamics simulation-based investigations on hydrogen embrittlement behavior of chromium electroplated 4340 steel, J. Eng. Mater. Technol. 143 (4) (2021).
[372] J.C. Outeiro, 11 - Residual stresses in machining, in: V.V. Silberschmidt (Ed.), Mechanics of Materials in Modern Manufacturing Methods and Processing Techniques, Elsevier, 2020, pp. 297-360.
[373] F. Deirmina, N. Peghini, B. AlMangour, D. Grzesiak, M. Pellizzari, Heat treatment and properties of a hot work tool steel fabricated by additive manufacturing, Mater. Sci. Eng. A 753 (2019) 109–121.
[374] B. Ren, D.H. Lu, R. Zhou, Z.H. Li, J.R. Guan, Preparation and mechanical properties of selective laser melted H13 steel, J. Mater. Res. 34 (8) (2019) 1415–1425.
[375] G. Telasang, J. Dutta Majumdar, N. Wasekar, G. Padmanabham, I. Manna, Microstructure and mechanical properties of laser clad and post-cladding tempered aisi h13 tool steel, Metall. Mater. Trans. A 46 (5) (2015) 2309–2321.
[376] G. Telasang, J. Dutta Majumdar, G. Padmanabham, M. Tak, I. Manna, Effect of laser parameters on microstructure and hardness of laser clad and tempered AISI H13 tool steel, Surf. Coat. Technol. 258 (2014) 1108–1118.
[377] M. Wang, Y. Wu, Q. Wei, Y. Shi, Thermal fatigue properties of H13 hot-work tool steels processed by selective laser melting, Metals 10 (1) (2020) 116.

[378] J.J. Yan, D.L. Zheng, H.X. Li, X. Jia, J.F. Sun, Y.L. Li, M. Qian, M. Yan, Selective laser melting of H13: microstructure and residual stress, J. Mater. Sci. 52 (20) (2017) 12476–12485.
[379] R. Cottam, J. Wang, V. Luzin, Characterization of microstructure and residual stress in a 3D H13 tool steel component produced by additive manufacturing, J. Mater. Res. 29 (17) (2014) 1978–1986.
[380] K. Kempen, B. Vrancken, S. Buls, L. Thijs, J. Van Humbeeck, J.P. Kruth, Selective laser melting of crack-free high density M2 high speed steel parts by baseplate preheating, J. Manuf. Sci. E-Trans. Asme 136(6) (2014) 061026-061026-6.
[381] P.J. Maziasz, E.A. Payzant, M.E. Schlienger, K.M. McHugh, Residual stresses and microstructure of H13 steel formed by combining two different direct fabrication methods, Scr. Mater. 39 (10) (1998) 1471–1476.
[382] S. Jhavar, C.P. Paul, N.K. Jain, Causes of failure and repairing options for dies and molds: a review, Eng. Fail. Anal. 34 (2013) 519–535.
[383] E.P. George, D. Raabe, R.O. Ritchie, High-entropy alloys, Nat. Rev. Mater. 4 (8) (2019) 515–534.
[384] J.W. Yeh, S.K. Chen, S.J. Lin, J.Y. Gan, T.S. Chin, T.T. Shun, C.H. Tsau, S. Y. Chang, Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes, Adv. Eng. Mater. 6 (5) (2004) 299–303.
[385] Y. Qiu, S. Thomas, M.A. Gibson, H.L. Fraser, N. Birbilis, Corrosion of high entropy alloys, npj Mater. Degrad. 1 (1) (2017) 1–18.
[386] B. Gludovatz, A. Hohenwarter, D. Catoor, E.H. Chang, E.P. George, R.O. Ritchie, A fracture-resistant high-entropy alloy for cryogenic applications, Science 345 (6201) (2014) 1153–1158.
[387] O.N. Senkov, D.B. Miracle, K.J. Chaput, J.-P. Couzinie, Development and exploration of refractory high entropy alloys—A review, J. Mater. Res. 33 (19) (2018) 3092–3128.
[388] G. Laplanche, A. Kostka, O. Horst, G. Eggeler, E. George, Microstructure evolution and critical stress for twinning in the CrMnFeCoNi high-entropy alloy, Acta Mater. 118 (2016) 152–163.
[389] N. Guo, L. Wang, L. Luo, X. Li, Y. Su, J. Guo, H. Fu, Microstructure and mechanical properties of refractory MoNbHfZrTi high-entropy alloy, Mater. Des. 81 (2015) 87–94.
[390] O. Senkov, J. Scott, S. Senkova, D. Miracle, C. Woodward, Microstructure and room temperature properties of a high-entropy TaNbHfZrTi alloy, J. Alloy. Compd. 509 (20) (2011) 6043–6048.
[391] A. Takeuchi, K. Amiya, T. Wada, K. Yubuta, W. Zhang, High-entropy alloys with a hexagonal close-packed structure designed by equi-atomic alloy strategy and binary phase diagrams, JOM 66 (10) (2014) 1984–1992.
[392] Z. Wu, H. Bei, G.M. Pharr, E.P. George, Temperature dependence of the mechanical properties of equiatomic solid solution alloys with face-centered cubic crystal structures, Acta Mater. 81 (2014) 428–441.
[393] S. Chen, Y. Tong, P.K. Liaw, Additive manufacturing of high-entropy alloys: a review, Entropy 20 (12) (2018) 937.
[394] Y.M. Wang, T. Voisin, J.T. McKeown, J. Ye, N.P. Calta, Z. Li, Z. Zeng, Y. Zhang, W. Chen, T.T. Roehling, R.T. Ott, M.K. Santala, P.J. Depond, M.J. Matthews, A. V. Hamza, T. Zhu, Additively manufactured hierarchical stainless steels with high strength and ductility, Nat. Mater. 17 (1) (2018) 63–71.
[395] P. Kürnsteiner, M.B. Wilms, A. Weisheit, B. Gault, E.A. Jägle, D. Raabe, High-strength Damascus steel by additive manufacturing, Nature 582 (7813) (2020) 515–519.
[396] Z. Zhu, Q. Nguyen, F. Ng, X. An, X. Liao, P. Liaw, S. Nai, J. Wei, Hierarchical microstructure and strengthening mechanisms of a CoCrFeNiMn high entropy alloy additively manufactured by selective laser melting, Scripta Mater 154 (2018) 20-24.
[397] R. Li, P. Niu, T. Yuan, P. Cao, C. Chen, K. Zhou, Selective laser melting of an equiatomic CoCrFeMnNi high-entropy alloy: Processability, non-equilibrium microstructure and mechanical property, J. Alloy. Compd. 746 (2018) 125–134.
[398] Z. Xu, H. Zhang, W. Li, A. Mao, L. Wang, G. Song, Y. He, Microstructure and nanoindentation creep behavior of CoCrFeMnNi high-entropy alloy fabricated by selective laser melting, Addit. Manuf. 28 (2019) 766–771.
[399] S. Guan, D. Wan, K. Solberg, F. Berto, T. Welo, T.M. Yue, K. Chan, Additive manufacturing of fine-grained and dislocation-populated CrMnFeCoNi high entropy alloy by laser engineered net shaping, Mater. Sci. Eng. A 761 (2019), 138056.
[400] Y. Brif, M. Thomas, I. Todd, The use of high-entropy alloys in additive manufacturing, Scr. Mater. 99 (2015) 93–96.
[401] I. Kunce, M. Polanski, K. Karczewski, T. Plocinski, K. Kurzydlowski, Microstructural characterisation of high-entropy alloy AlCoCrFeNi fabricated by laser engineered net shaping, J. Alloy. Compd. 648 (2015) 751–758.
[402] H. Shiratori, T. Fujieda, K. Yamanaka, Y. Koizumi, K. Kuwabara, T. Kato, A. Chiba, Relationship between the microstructure and mechanical properties of an equiatomic AlCoCrFeNi high-entropy alloy fabricated by selective electron beam melting, Mater. Sci. Eng. A 656 (2016) 39–46.
[403] S. Guan, K. Solberg, D. Wan, F. Berto, T. Welo, T. Yue, K. Chan, Formation of fully equiaxed grain microstructure in additively manufactured AlCoCrFeNiTi0. 5 high entropy alloy, Mater. Des. 184 (2019), 108202.
[404] T. Borkar, B. Gwalani, D. Choudhuri, C.V. Mikler, C.J. Yannetta, X. Chen, R. V. Ramanujan, M.J. Styles, M.A. Gibson, R. Banerjee, A combinatorial assessment of AlxCrCuFeNi2 ($0 < x < 1.5$) complex concentrated alloys: microstructure, microhardness, and magnetic properties, Acta Mater. 116 (2016) 63–76.
[405] B. Gwalani, V. Soni, O.A. Waseem, S.A. Mantri, R. Banerjee, Laser additive manufacturing of compositionally graded AlCrFeMoVx (x= 0 to 1) high-entropy alloy system, Opt. Laser Technol. 113 (2019) 330–337.
[406] S. Guan, D. Wan, K. Solberg, F. Berto, T. Welo, T. Yue, K. Chan, Additively manufactured CrMnFeCoNi/AlCoCrFeNiTi0. 5 laminated high-entropy alloy with enhanced strength-plasticity synergy, Scr. Mater. 183 (2020) 133–138.
[407] Y. Zhao, D.-H. Lee, M.-Y. Seok, J.-A. Lee, M.P. Phaniraj, J.-Y. Suh, H.-Y. Ha, J.-Y. Kim, U. Ramamurty, J.-I. Jang, Resistance of CoCrFeMnNi high-entropy alloy to gaseous hydrogen embrittlement, Scr. Mater. 135 (2017) 54–58.
[408] K. Ichii, M. Koyama, C.C. Tasan, K. Tsuzaki, Comparative study of hydrogen embrittlement in stable and metastable high-entropy alloys, Scr. Mater. 150 (2018) 74–77.
[409] Z. Pu, Y. Chen, L.H. Dai, Strong resistance to hydrogen embrittlement of high-entropy alloy, Mater. Sci. Eng. A 736 (2018) 156–166.
[410] X. Zhou, W.A. Curtin, First principles study of the effect of hydrogen in austenitic stainless steels and high entropy alloys, Acta Mater. 200 (2020) 932–942.
[411] M. Koyama, K. Ichii, K. Tsuzaki, Grain refinement effect on hydrogen embrittlement resistance of an equiatomic CoCrFeMnNi high-entropy alloy, Int. J. Hydrogen Energy 44 (31) (2019) 17163–17167.
[412] H. Luo, Z. Li, D. Raabe, Hydrogen enhances strength and ductility of an equiatomic high-entropy alloy, Sci. Rep. 7 (1) (2017).
[413] M.S. Mehranpour, H. Shahmir, M. Nili-ahmadabadi, CoCrFeNiMn high entropy alloy microstructure and mechanical properties after severe cold shape rolling and annealing, Mater. Sci. Eng.: A 793 (2020) 139884.
[414] J. Wang, X.L. Zhou, J. Li, M. Brochu, Y.F. Zhao, Microstructures and properties of SLM-manufactured Cu-15Ni-8Sn alloy, Addit. Manuf. 31 (2020), 100921.
[415] D. Lin, L. Xu, H. Jing, Y. Han, L. Zhao, F. Minami, Effects of annealing on the structure and mechanical properties of FeCoCrNi high-entropy alloy fabricated via selective laser melting, Addit. Manuf. 32 (2020), 101058.
[416] K.M. Bertsch, G. Meric de Bellefon, B. Kuehl, D.J. Thoma, Origin of dislocation structures in an additively manufactured austenitic stainless steel 316L, Acta Mater. 199 (2020) 19–33.
[417] D. Kong, C. Dong, X. Ni, L. Zhang, H. Luo, R. Li, L. Wang, C. Man, X. Li, Superior resistance to hydrogen damage for selective laser melted 316L stainless steel in a proton exchange membrane fuel cell environment, Corros. Sci. 166 (2020), 108425.
[418] P. Ferro, R. Meneghello, G. Savio, F. Berto, A modified volumetric energy density–based approach for porosity assessment in additive manufacturing process design, Int. J. Adv. Manuf. Technol. 110 (7) (2020) 1911–1921.
[419] F. Otto, A. Dlouhý, C. Somsen, H. Bei, G. Eggeler, E.P. George, The influences of temperature and microstructure on the tensile properties of a CoCrFeMnNi high-entropy alloy, Acta Mater. 61 (15) (2013) 5743–5755.
[420] X. Chen, J. Kong, J. Li, S. Feng, H. Li, Q. Wang, Y. Liang, K. Dong, Y. Yang, High-strength AlCoCrFeNi2.1 eutectic high entropy alloy with ultrafine lamella structure via additive manufacturing, Mater. Sci. Eng. A 854 (2022), 143816.
[421] L. Huang, Y. Sun, N. Chen, H. Luan, G. Le, X. Liu, Y. Ji, Y. Lu, P.K. Liaw, X. Yang, Y. Zhou, J. Li, Simultaneously enhanced strength-ductility of AlCoCrFeNi2.1 eutectic high-entropy alloy via additive manufacturing, Mater. Sci. Eng. A 830 (2022), 142327.
[422] S. Wang, Y. Li, D. Zhang, Y. Yang, S. Marwana Manladan, Z. Luo, Microstructure and mechanical properties of high strength AlCoCrFeNi2.1 eutectic high entropy alloy prepared by selective laser melting (SLM), Mater. Lett. 310 (2022), 131511.
[423] T. Fujieda, H. Shiratori, K. Kuwabara, T. Kato, K. Yamanaka, Y. Koizumi, A. Chiba, First demonstration of promising selective electron beam melting method for utilizing high-entropy alloys as engineering materials, Mater. Lett. 159 (2015) 12–15.
[424] B. Dong, Z. Wang, Z. Pan, O. Muránsky, C. Shen, M. Reid, B. Wu, X. Chen, H. Li, On the development of pseudo-eutectic AlCoCrFeNi2.1 high entropy alloy using Powder-bed Arc Additive Manufacturing (PAAM) process, Mater. Sci. Eng. A 802 (2021), 140639.
[425] Q. Wang, Y. Lu, Q. Yu, Z. Zhang, The exceptional strong face-centered cubic phase and semi-coherent phase boundary in a eutectic dual-phase high entropy alloy AlCoCrFeNi, Sci. Rep. 8 (1) (2018) 14910.
[426] H. Luo, Z. Li, W. Lu, D. Ponge, D. Raabe, Hydrogen embrittlement of an interstitial equimolar high-entropy alloy, Corros. Sci. 136 (2018) 403–408.
[427] Y. Wang, X. Wang, J. Gong, L. Shen, W. Dong, Hydrogen embrittlement of catholically hydrogen-precharged 304L austenitic stainless steel: Effect of plastic pre-strain, Int. J. Hydrogen Energy 39 (25) (2014) 13909–13918.
[428] T. Michler, Y. Lee, R.P. Gangloff, J. Naumann, Influence of macro segregation on hydrogen environment embrittlement of SUS 316L stainless steel, Int. J. Hydrogen Energy 34 (7) (2009) 3201–3209.
[429] Y.H. Fan, H.L. Zhao, K.R. Weng, C. Ma, H.X. Yang, X.L. Dong, C.W. Guo, Y.G. Li, The role of delta ferrite in hydrogen embrittlement fracture of 17–4 PH stainless steel, Int. J. Hydrogen Energy 47 (79) (2022) 33883–33890.
[430] Y. Hong, C. Zhou, Y. Zheng, L. Zhang, J. Zheng, X. Chen, B. An, Formation of strain-induced martensite in selective laser melting austenitic stainless steel, Mater. Sci. Eng. A 740–741 (2019) 420–426.
[431] K. Saeidi, D.L. Zapata, F. Lofaj, L. Kvetkova, J. Olsen, Z. Shen, F. Akhtar, Ultra-high strength martensitic 420 stainless steel with high ductility, Addit. Manuf. 29 (2019), 100803.